Article Summaries

Abstraction Mechanisms for Virtual Avatars

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The following articles form the measurement foundation of the user study. Each entry includes the full bibliographic reference, DOI, a detailed content summary, and notes on the specific relevance to this thesis. Articles are grouped by topic.

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Uncanny Valley

MacDorman & Ishiguro (2006) — Uncanny Valley Empirical Programme

Full reference: MacDorman, K. F. & Ishiguro, H. (2006). The uncanny advantage of using androids in cognitive and social science research. In *Proceedings of the ICDL-2006 Workshop on Android Science*, 1–9.

Note: Conference proceedings; no DOI assigned. Available from author's website.

What the paper argues. MacDorman and Ishiguro argued that androids, robotic systems designed to closely resemble humans, are uniquely valuable scientific tools precisely because they can be positioned at any point along the human-likeness continuum with experimental control. By manipulating android appearance systematically, researchers can generate Uncanny Valley responses on demand and study the psychological and cognitive mechanisms that produce them. The paper proposed a set of semantic differential scales to operationalise eeriness and strangeness as measurable affective responses, converting Mori's qualitative curve into a quantifiable experimental variable. It also proposed that the Uncanny Valley effect is driven not merely by imperfection in appearance, but by violations of categorical perception: when an agent is ambiguous between the "human" and "non-human" categories, the cognitive mismatch produces a negative affective response.

Methodology. The paper combines a theoretical argument with the introduction of a measurement instrument. A pilot study is reported but sample size and full statistical details are not provided in the proceedings version.

Relevance to this thesis. MacDorman and Ishiguro's eeriness scales are the original source of the eeriness measurement approach used in this study. Their categorical perception account of the Uncanny Valley supports the prediction that NPR stylisation can reduce eeriness by shifting the stimulus away from the ambiguous boundary region between human and non-human categories, even if perceived human-likeness is not dramatically reduced.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #57** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  [check wording] THESIS: "The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly."
  SOURCE (PROCEDURE, P. 306; DISCUSSION, P. 308): "The same 31 images were presented again in random order, and the participants were asked, "Which of the following figures gives you an eerie feeling?" The selected images were then presented in random order and the participants were asked to rate how eerie each image was on a ten-point scale, ranging from slightly eerie to extremely eerie. ... human beings do seem to be highly sensitive to imperfections in near humanlike robots, both in their looks and movements"
  CHECK: The paper does introduce an eeriness rating separate from the human-likeness and familiarity scales, and it is motivated by the critique on p. 304 that Mori's familiarity axis 'appears to conflate novelty and valence or hedonic value'; the sensitivity to near-human imperfection is stated on p. 308. However, what is introduced is a single ad hoc ten-point eeriness item plus a forced-choice question, not a multi-item validated instrument comparable to Godspeed, and the plural 'instruments' overstates it. The validated eight-item eeriness

index is Ho and MacDorman (2010), not this 2006 paper; the Godspeed comparison itself appears nowhere here (Bartneck et al. postdates it).

Ho & MacDorman (2010) — Validated Eeriness Instrument

Full reference: Ho, C.-C. & MacDorman, K. F. (2010). Revisiting the uncanny valley theory: Developing and validating an alternative to the Godspeed indices. *Computers in Human Behavior*, 26(6), 1508–1518.

DOI: 10.1016/j.chb.2010.05.015

Core problem. Mori's (1970) uncanny valley graph has two axes: *bukimi* (eeriness) on the y-axis and degree of human likeness on the x-axis. To test the theory empirically, these must be measured by independent, validated instruments. The Godspeed questionnaire (Bartneck et al., 2009) proposed *anthropomorphism* and *likeability* as potential candidates. Ho and MacDorman tested this empirically and found it does not work. The paper has two phases.

Phase 1 — Testing the Godspeed indices. 384 participants rated 10 video clips (five computer-animated characters from *Final Fantasy*, *The Incredibles*, *The Polar Express*, and similar productions, plus five robots ranging from the non-humanoid Roomba to the android Aiko) on all 24 Godspeed items.

Reliability: Four of the five subscales were acceptable ($\alpha = .87$–$.92$). Perceived Safety failed ($\alpha = .60$).

Discriminant validity: This is where the Godspeed indices break down. The four cognitive-perceptual subscales were so strongly intercorrelated that they cannot claim to measure distinct concepts:

	Anthropomorphism	Animacy	Likeability	Perceived Intelligence
Animacy	.89***	—	—	—
Likeability	.73***	.74***	—	—
Perceived Intelligence	.67***	.72***	.71***	—

*** $p < .001$. Adapted from Ho & MacDorman (2010), Table 2.

Anthropomorphism and animacy correlate at $r = .89$; anthropomorphism and likeability at $r = .73$. These subscales are effectively measuring the same underlying construct — most likely interpersonal warmth, which is the dominant dimension of human social perception and accounts for 53% of the variance in everyday social judgments. Multidimensional scaling confirmed this: anthropomorphism and animacy items occupy a large overlapping region of the semantic space, with likeability items contained entirely within it. When robots and animated characters were plotted on anthropomorphism vs. likeability axes, all data fell along a single diagonal ($R^2 = .852$) — the axes are collinear, not orthogonal. The Godspeed indices cannot serve as the x- and y-axes of Mori's graph.

A second conceptual problem: *eeriness* is not the same as negative warmth (interpersonal coldness). A character can be cold and disagreeable without being eerie. The specific emotions that predict eeriness — fear, anxiety, disgust — are distinct from the emotional content of coldness. No Godspeed subscale captures this.

Phase 2 — Developing new indices. Ho and MacDorman developed four new indices through two rounds of testing (Round 1: $n = 19$; Round 2: $n = 253$), using the same 10 video clips. Items were semantic differential pairs selected to be decoupled from interpersonal warmth and from each other. A *warmth* index was developed alongside them as a decoupling check. The four new indices are:

Perceived Humanness (6 items, $\alpha = .92$, explains 68.96% variance):

Artificial – Natural	Human-made – Humanlike
Without Definite Lifespan – Mortal	Inanimate – Living
Mechanical Movement – Biological Movement	Synthetic – Real

Eeriness (8 items, $\alpha = .74$ overall, explains 62.04% variance). Two stable subfactors:

Eerie subfactor ($\alpha = .81$)	*Spine-tingling subfactor* ($\alpha = .81$)
Reassuring – Eerie	Unemotional – Hair-raising
Numbering – Freaky	Uninspiring – Spine-tingling
Ordinary – Supernatural	Boring – Shocking
Bland – Uncanny	Predictable – Thrilling

Attractiveness (5 items, $\alpha = .90$, explains 70.93% variance): Unattractive–Attractive, Ugly–Beautiful, Repulsive–Agreeable, Crude–Stylish, Messy–Sleek. Included because physical attractiveness influences social judgments independently of eeriness and must be controlled in uncanny valley research.

Warmth (5 items, $\alpha = .88$): Cold-hearted–Warm-hearted, Hostile–Friendly, Spiteful–Well-intentioned, Ill-tempered–Good-natured, Grumpy–Cheerful. Used as a decoupling check only, not as an outcome measure.

Validation results (Table 7 in the original paper):

	Attractiveness	Eeriness	Humanness	Warmth
Eeriness	−.03 (n.s.)	—	—	—
Humanness	.61***	.02 (n.s.)	—	—
Warmth	.62***	−.05 (n.s.)	.66***	—

*** $p < .001$. Adapted from Ho & MacDorman (2010), Table 7.

The critical result: **eeriness and humanness are not significantly correlated** ($r = .02$, $p = .514$). Eeriness is also decoupled from warmth ($r = -.05$, $p = .083$). The two axes of Mori's graph are now empirically independent. Multidimensional scaling confirmed that the three main indices occupy distinct, non-overlapping regions of the semantic space ($R^2 = .003$ for humanness vs. eeriness; compare to $R^2 = .852$ for Godspeed anthropomorphism vs. likeability in Phase 1).

One remaining limitation: perceived humanness retains a high correlation with warmth ($r = .66$). The authors acknowledge that human-likeness and warmth are practically difficult to fully decouple in people's mental models, and suggest refinement in future work.

CFA on the final 19-item set: $\chi^2 = 1229.29$, $df = 146$, CFI = .97, NFI = .97, GFI = .91, AGFI = .88, RMSEA = .075 (slightly above the .05 cutoff, but substantially better than Phase 1).

Relevance to this study. The **eeriness subscale** in Part B of the questionnaire is grounded in Ho and MacDorman's validated eeriness index. Items targeting feelings of strangeness,

creepiness, and being unsettled trace directly to the eerie subfactor. The paper's central finding — that eeriness and perceived humanness are orthogonal dimensions — provides the theoretical basis for combining the eeriness subscale (Part B) with the Godspeed anthropomorphism subscale (Part D) in this study. Because the two are independent, it is possible to test whether NPR stylisation shifts the C2 → C3 eeriness reduction through a change in perceived human-likeness (the mediation path in H5), or whether eeriness decreases for other reasons while anthropomorphism scores remain relatively stable. Without the Ho and MacDorman validation showing this independence, the mediation analysis in H5 would be structurally unsound.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #58** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "Their validation established that eeriness and human likeness are orthogonal."
  SOURCE (ABSTRACT, P. 1508; SECTION 5 CONCLUSION, P. 1517): "Phase 2 of this study developed alternative indices with nonsignificant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$). ... [Conclusion:] The apparent independence of the humanness and eeriness indices enables anthropomorphic characters to be plotted along nearly orthogonal axes, as implied by Mori's (1970) original graph of the uncanny valley."
  CHECK: The validation found a nonsignificant eeriness-humanness correlation ($r = .02$, $p = .514$, Section 3.3.5, p. 1515) and the authors themselves draw the orthogonality conclusion. Two small hedges the thesis drops: the paper writes 'nearly orthogonal', and the x-axis construct is the subjective 'perceived humanness' index rather than objective human likeness, with the authors noting 'decorrelation between indices was achieved for eeriness but only partly achieved for perceived humanness' (Section 3.1, p. 1513).

- **Entry #95** *(Ch.4 Methodology > User Study Overview)*
  THESIS: "Eeriness was taken from the instrument of Ho and MacDorman instead, since the Godspeed subscales do not separate cleanly from one another."
  SOURCE (SECTION 2.2 RESULTS, P. 1511): "Another serious problem was the significant and extremely high correlation between anthropomorphism, likeability, animacy, and perceived intelligence (Table 2). The correlations ranged from .67 for anthropomorphism and perceived intelligence to .89 for anthropomorphism and animacy. This suggests that those concepts had no discriminant validity. In other words, they were all measuring the same concept instead of measuring distinct concepts."
  CHECK: The paper's Phase 1 finding is exactly that the Godspeed indices fail to separate from one another, and Section 4 restates it: 'These findings indicate the Godspeed indices are not appropriate as distinct concepts for evaluating anthropomorphic agents' (p. 1516). The paper's own eeriness index was developed as the replacement, which matches the thesis sentence's substitution.

- **Entry #122** *(Ch.4 Methodology > User Study Design > Objective and Hypotheses)*
  THESIS: "Eeriness is a distinct negative affective response rather than general dislike."
  SOURCE (SECTION 1 PLOTTING EMOTIONAL RESPONSES TO HUMANLIKE CHARACTERS, P. 1508): "Mori (1970) refers to negative shinwakan as bukimi, which translates as eeriness. However, eeriness is not the negative anchor of warmth. A person can be cold and disagreeable without being eerie–at least not eerie in the way that an artificial human being is eerie. In addition, the set of negative emotions that predict eeriness (e.g., fear, anxiety, and disgust)

are more specific than coldness (Ho, MacDorman, & Pramono, 2008). Thus, shinwakan and bukimi appear to constitute distinct dimensions."
CHECK: The passage states directly that eeriness is not simply the negative pole of warmth or likeability and is predicted by a more specific set of negative emotions. Reinforced in Section 3, p. 1512: 'Eeriness, which is conceptually distinct from negative warmth (i.e., interpersonal coldness), would need to be included in any set of indices on the uncanny valley.'

- **Entry #128** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "It also contains no subscale corresponding specifically to eeriness."
  SOURCE (SECTION 1, P. 1509 (PARAGRAPH ON THE GODSPEED QUESTIONNAIRE)): "In addition, there is no index corresponding specifically to eeriness, a dimension that is arguably distinct from likeability but nevertheless important in determining whether a human-looking character has fallen into the uncanny valley."
  CHECK: Directly supports the sentence; the paper calls the Godspeed components 'indices' where the thesis says 'subscale', which is the same thing. The five Godspeed indices are enumerated in Section 2 as anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety, with no eeriness index among them.

- **Entry #139** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  [check wording] THESIS: "Cronbach's $\{}alpha$ was reported per subscale before the composites were formed, since averaging items presumes they measure one construct."
  SOURCE (SECTION 2.1.3 STATISTICAL ANALYSIS, P. 1510; SECTION 3.1 RESEARCH GOAL, P. 1513): "Cronbach's α was used to measure the reliability of each index. ... [Section 3.1:] (4) there should be at least three semantic differential scales per index to enable the estimation of reliability; and (5) the reliability of the indices should be acceptable (Cronbach's $\alpha \geq .70$)."
  CHECK: The paper supports the practice: it reports Cronbach's α separately for every index (Tables 1, 3, 4, 5) and applies the .70 cutoff, e.g. 'perceived safety had low reliability with a Cronbach's α of .60, which is below the standard .70 cutoff (Nunnally, 1978)' (Section 2.2, p. 1511). It does not, however, state the rationale the thesis sentence gives, that averaging items presumes they measure one construct; unidimensionality is handled in the paper by factor analysis, not by Cronbach's α ('items corresponding to a single, unidimensional concept should load on the same factor when applying exploratory factor analysis', Section 3.1, p. 1513). Note also that the thesis sentence describes its own analysis procedure rather than a claim from this paper. The α and $\geq$ glyphs are transcribed from the PDF's font-substituted extraction ('a' and 'P').

Mori (1970) — The Uncanny Valley

Full reference: Mori, M. (1970). The uncanny valley [*Bukimi no tani*]. *Energy*, 7(4), 33–35 [in Japanese]. Authorised English translation: MacDorman, K. F. & Kageki, N. (2012). The uncanny valley [from the field]. *IEEE Robotics & Automation Magazine*, 19(2), 98–100.

DOI (translation): `10.1109/MRA.2012.2192811`

Core argument. Affinity toward artificial figures does not increase monotonically as human likeness increases. Beyond a certain threshold of resemblance, affinity drops sharply into what Mori called the uncanny valley (*bukimi no tani*), before rising again toward a second and

higher peak. The original essay was published in a Japanese engineering magazine in 1970 and remained largely unknown to Western researchers until the 2012 IEEE Spectrum translation by MacDorman and Kageki became the canonical English source.

The two-curve graph. Mori presented a conceptual graph with two curves. The solid curve represents still (non-moving) figures; the dashed curve represents moving figures. Both share the valley shape but the moving curve descends far deeper into negative affect. Reference points on the safe slope include industrial robots and stuffed animals; Bunraku puppets and an Okina mask sit near the first peak. In the valley: a prosthetic hand (which appears realistic at first glance but reveals artificiality on closer inspection), an ordinary doll, a corpse, and a zombie. For the moving curve, the myoelectric prosthetic hand also descends into the valley because its movement is slightly unnatural.

Why movement amplifies eeriness. Mori observed that adding motion deepens the valley effect. He illustrated this with a smile: if the speed of a smile is reduced to half its natural rate, the expression becomes unsettling even if the still face is acceptable. The timing of social movement is part of the perceptual expectation against which the figure is evaluated, and mismatches in timing are as disturbing as mismatches in appearance.

Escape by design. Rather than attempting to cross the valley by progressively improving realism, Mori recommends targeting the first peak: figures with a deliberate and moderate level of human likeness that do not enter the ambiguous zone. He illustrates this with a wooden hand carved by a sculptor of Buddhist statues: it is clearly not a real hand, yet it is beautiful and harmonious because it makes no claim to be mistaken for one. Clarity of category identity — whether the figure is human or clearly not human — matters more than proximity to perfect realism.

An explanation of the uncanny. Mori speculates that the eerie sensation triggered in the valley may be a form of self-preservation instinct. The stimuli at the valley floor (corpses, zombies, severely damaged limbs) share features associated with death or severe biological abnormality. He frames this as an open question, acknowledging he had not yet worked through it fully. The 2012 translation's footnote 3 formalises this: eeriness is likely a response to proximal rather than distal sources of danger.

Relevance to this thesis. Mori's essay provides the foundational concept motivating the entire research programme. H1 (NPR stylisation produces lower eeriness than the photorealistic avatar) is a direct test of the valley's first slope: whether moving the Avaturn avatar from the near-human C2 toward the stylised C3 reduces eeriness. The escape-by-design principle also motivates the selection of NPR as an avenue rather than pursuing ever-increasing photorealism.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #32** *(Ch.3 Related Work > The Uncanny Valley and Avatar Realism)*
  THESIS: "Only then is comfort recovered."
  SOURCE (SEC. "VALLEY OF FAMILIARITY", PDF P. 1; AND SEC. "THE SIGNIFICANCE OF THE UNCANNY", PDF P. 2 (ENERGY 7(4), PP. 33-35)): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the "uncanny valley." ... In Figure 1, a healthy person is at the top of the second peak."
  CHECK: Mori's curve dips into the valley and only returns to positive values at the second peak, which he explicitly assigns to a healthy person; Figure 1 shows this recovery occurring

at 100% human likeness. The thesis word "comfort" stands in for Mori's translated term "familiarity" (shinwakan), which is a standard rendering.

- **Entry #33** *(Ch.3 Related Work > The Uncanny Valley and Avatar Realism)*
  [check wording] THESIS: "The eerie response arises only once the discrepancy registers, as with a prosthetic hand that turns unsettling the moment it begins to move."
  SOURCE (SEC. "VALLEY OF FAMILIARITY" AND SEC. "THE EFFECTS OF MOVEMENT", PDF P. 2): "But this kind of prosthetic hand is too real and when we notice it is prosthetic, we have a sense of strangeness. ... In mathematical terms, strangeness can be represented by negative familiarity, so the prosthetic hand is at the bottom of the valley. ... Conversely, if we add movement to a prosthetic hand, which is at the bottom of the uncanny valley, our sensation of strangeness grows quite large."
  CHECK: Mori supports the first half: the eerie response is triggered by noticing the discrepancy ("when we notice it is prosthetic"). He does not support the second half as phrased: the still prosthetic hand is already at the bottom of the valley, and movement magnifies an existing strangeness ("our sensation of strangeness grows quite large") rather than being the moment the hand "turns unsettling".

- **Entry #34** *(Ch.3 Related Work > The Uncanny Valley and Avatar Realism)*
  THESIS: "The solid curve describes affective response to still figures, while the dashed curve describes figures in motion."
  SOURCE (FIGURE 1 LEGEND, PDF P. 1; BODY SENTENCE IN SEC. "THE EFFECTS OF MOVEMENT", PDF P. 2): "[Fig. 1 key, upper left] "moving" (dashed line) / "still" (solid line); [y-axis] "familiarity"; [x-axis] "human likeness". [Body text] As shown in Figure 1, adding movement changes the shape of the uncanny valley by exaggerating the peaks and valley."
  CHECK: The figure key in this translation explicitly pairs a dashed line with "moving" and a solid line with "still", and the accompanying text confirms the dashed curve is the movement condition with exaggerated peaks and valley. The bracketed labels mark figure elements; the quoted words are the verbatim legend and axis labels, which pdftotext cannot extract since the figure is an image.

- **Entry #47** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity."
  SOURCE (SEC. "VALLEY OF FAMILIARITY", PDF PP. 1-2): "There are mathematical functions of the form y = f(x) for which the value of y increases (or decreases) continuously with the value of x. ... Climbing a mountain is an example of a function that does not increase continuously ... I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. ... So in this case, the appearance is quite human like, but the familiarity is negative. This is the uncanny valley."
  CHECK: Mori frames the valley precisely as a departure from a continuously increasing function, with familiarity going negative at high human likeness and recovering at the second peak (healthy person), which is the structure the thesis sentence describes.

- **Entry #93** *(Ch.4 Methodology > Research Questions)*
  THESIS: "Uncanny valley theory places a real human speaker at the peak of the affinity curve, since maximal human likeness corresponds to maximal comfort."

SOURCE (SEC. "THE SIGNIFICANCE OF THE UNCANNY", PDF P. 2; AND SEC. "ESCAPE BY DESIGN", PDF P. 2): "In Figure 1, a healthy person is at the top of the second peak. ... Although the second peak is higher, there is a far greater risk of falling into the uncanny valley."
CHECK: Mori places the healthy (real) human at the top of the second peak, and states that this peak is the higher of the two, so maximal human likeness corresponds to maximal familiarity. Mori writes about a healthy person generally, not a speaker, but the thesis's "real human speaker" is its own stimulus rather than an attributed claim.

- **Entry #120** *(Ch.4 Methodology > User Study Design > Objective and Hypotheses)*
  [check wording] THESIS: "Photorealistic avatars occupy the uncanny valley region where negative affect peaks, while stylised abstraction moves the representation towards the stable ascending slope at lower fidelity."
  SOURCE (SEC. "VALLEY OF FAMILIARITY", PDF P. 2; AND SEC. "ESCAPE BY DESIGN", PDF P. 2): "But this kind of prosthetic hand is too real and when we notice it is prosthetic, we have a sense of strangeness. ... so the prosthetic hand is at the bottom of the valley. ... So I recommend designers take the first peak as the goal in building robots rather than the second. Although the second peak is higher, there is a far greater risk of falling into the uncanny valley. We predict that it is possible to produce a safe familiarity by a nonhumanlike design."
  CHECK: The second half is well supported: Mori recommends targeting the lower first peak because a nonhumanlike design yields "safe familiarity". The first half generalises beyond the source: Mori's valley occupants are prosthetic hands, corpses and zombies, he never discusses digital avatars or photorealistic rendering, and in his account the valley is entered when a near-human artefact reveals a discrepancy, not by high visual fidelity as such, since the fully realistic healthy person sits at the peak.

- **Entry #124** *(Ch.4 Methodology > User Study Design > Objective and Hypotheses)*
  THESIS: "Uncanny valley theory places a real human speaker at the peak of the affinity curve, where human likeness is maximal and comfort is highest."
  SOURCE (SEC. "THE SIGNIFICANCE OF THE UNCANNY", PDF P. 2; AND SEC. "ESCAPE BY DESIGN", PDF P. 2): "In Figure 1, a healthy person is at the top of the second peak. ... Although the second peak is higher, there is a far greater risk of falling into the uncanny valley."
  CHECK: Same support as entry 93: the healthy person occupies the highest point of the curve at maximal human likeness. Note this sentence duplicates entry 93 almost verbatim in the thesis, which may be worth checking for redundancy.

McDonnell et al. (2012) — Render me Real?

Full reference: McDonnell, R., Breidt, M., & Bülthoff, H. H. (2012). Render me real? Investigating the effect of render style on the perception of animated virtual humans. *ACM Transactions on Graphics*, 31(4), Article 91.

DOI: 10.1145/2185520.2185587

Research question. Does rendering style affect how trustworthy, appealing, familiar, and reassuring virtual humans are perceived to be? Does motion change these judgements? Does

rendering style unconsciously bias viewers when judging whether a virtual character is lying or telling the truth?

Stimulus creation. Two virtual replicas were created of two motion-captured actors (one male, one female) using 3D scanning and next-generation game character rigs. Motion was captured with a 13-camera Vicon optical system with 52 body markers and 36 face markers, synchronised with an eye-tracker at 250Hz. The same motion capture performances were rendered in 11 styles, from the highly abstract ToonPencil (black outlines on a white background, no shading) to HumanHQ2 and HumanWrinkle (ray-traced area lights, accurate reflections, dynamic wrinkle maps). The intermediate styles cover the toon-to-realistic continuum: ToonFlat, ToonShaded, ToonBare (Phong, no textures), ToonCG (high-quality games-industry cartoon), HumanIll (desaturated yellowed skin, deliberately unappealing), HumanBasic (Phong with human textures, no shadows), and HumanSSS (subsurface scattering).

Experiment 1 — Explicit ratings (still and moving). Participants rated each character on six 7-point scales: realistic vs. abstract, appealing vs. unappealing, reassuring vs. eerie, familiar vs. unfamiliar, friendly vs. unfriendly, and trustworthy vs. untrustworthy.

- **Realism:** the 11 styles fell into six distinct perceptual groups, confirming the stimuli covered the full realism continuum.
- **Appeal:** both the most abstract styles (ToonPencil, ToonFlat, ToonShaded, ToonCG) and the most realistic (HumanHQ1, HumanHQ2) were rated as most appealing. A statistically significant drop occurred for mid-range styles, directly replicating the valley shape Mori predicted. HumanIll was the least appealing of all.
- **Reassurance:** HumanIll was rated most eerie; ToonFlat and ToonShaded were rated more reassuring than HumanHQ2, suggesting that highly stylised rendering can be more comfortable than nearly photorealistic rendering.
- **Trustworthy:** ToonPencil was rated most trustworthy. HumanIll was least trustworthy. Most realistic and most abstract styles received statistically equivalent trustworthiness; mid-range styles received the lowest trust ratings.
- **Friendly:** HumanIll and HumanBasic were most unfriendly; ToonPencil, ToonFlat, ToonShaded, and ToonCG were most friendly.
- **Motion:** no significant main effect of motion was found for most scales. HumanIll specifically became significantly less appealing when moving ($p < .009$), confirming that motion amplifies the valley effect for unappealing styles.

Experiment 2 — Motion anomalies. Large motion artifacts (half-face animation removed, static eyes) were applied to five styles. Toon characters tolerated large motion anomalies far better than realistic ones: the half-face artifact reduced pleasantness far more on HumanWrinkle than on ToonCG. This confirms that realism raises the expected standard and thereby amplifies the cost of any visible imperfection.

Experiment 3 — Implicit lie detection. Participants detected lies from 60 silent video clips at near-chance accuracy. Rendering style did *not* significantly bias deception detection. The audio track was the most reliable cue. This null result is notable: render style influences explicit social impressions (appeal, trust) without contaminating the implicit judgment of honesty.

Relevance to this thesis. McDonnell et al. provide direct empirical precedent for a non-linear

relationship between rendering style and trust-related perceptions. Their finding that both abstract and photorealistic styles receive higher trustworthiness ratings than mid-range styles is the clearest published evidence that NPR stylisation does not reduce trust relative to photorealistic rendering. This directly motivates H2. The finding that motion amplifies the valley for unappealing styles also supports the choice to use video stimuli.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #38** *(Ch.3 Related Work > The Uncanny Valley and Avatar Realism)*
  THESIS: "However, appeal does not follow from abstraction alone, since one of their abstract styles ranked among the least appealing."
  SOURCE (SECTION 7 DISCUSSION, P. 91:10 (FIRST TWO SENTENCES); SECTION 5.1.3 RESULTS, APPEAL, P. 91:5 (FINAL CLAUSE)): "However, we cannot say that all abstract cartoon characters are considered appealing. One of our render styles (ToonBare) was considered quite unappealing and evoked negative reactions from participants across most of the scales tested. We believe that this is related to the lack of familiarity of this style, as participants gave it a low rating on the familiarity scale. ... HumanBasic and ToonBare were the next least appealing ($p < 0.04$ in all cases)"
  CHECK: The Discussion states explicitly that not all abstract cartoon styles were appealing, and the Appeal results place ToonBare among the least appealing styles (second-least, after HumanIll). This matches the thesis sentence's claim that one abstract style ranked among the least appealing.

- **Entry #90** *(Ch.3 Related Work > Research Gap)*
  [check wording] THESIS: "First, NPR stylisation is a principled strategy for avoiding the uncanny valley, with empirical support from McDonnell et al.."
  SOURCE (SECTION 1 INTRODUCTION, P. 91:2 (FIRST PASSAGE); SECTION 7 DISCUSSION, P. 91:10 (SECOND AND THIRD PASSAGES)): "We found that highly realistic and highly abstract styles were considered appealing, whether still or moving. However, render styles that were in the middle of the range between abstract and realistic were considered unappealing, and the most unappealing character was considered even more so when motion was applied. ... These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. ... Based on the results of our experiments, we can reassure developers that selecting realistic virtual characters is not as risky as often discussed (provided that the motion does not contain artifacts)."
  CHECK: The paper does supply empirical support that highly abstract styles are appealing and that mid-range styles fall into an appeal trough, so stylisation avoids the trough. However, it explicitly counters the framing of stylisation as the principled route out of the uncanny valley: the most realistic style was rated as appealing as the cartoon styles, the authors reassure developers that realistic characters are 'not as risky as often discussed', and they note that not all abstract styles are appealing (ToonBare). Citing this paper for NPR as a principled avoidance strategy therefore takes only one half of a two-sided result.

- **Entry #100** *(Ch.4 Methodology > NPR Technique Design)*
  THESIS: "Characters rendered at an intermediate level, between clearly stylised and fully realistic, have been rated least appealing."
  SOURCE (SECTION 5.1.3 RESULTS, APPEAL, P. 91:5): "Characters that were rated halfway

between abstract and realistic in the Realism ratings, were considered the least appealing, with HumanIll being rated as significantly less appealing than all other styles ($p < 0.0005$ in all cases). ... These findings imply that characters that appear highly abstract or highly realistic are considered equally appealing. Furthermore, that a drop in appeal occurs for characters that are neither abstract nor realistic, and therefore may be difficult to categorize."
CHECK: Direct match: the paper states that styles rated halfway between abstract and realistic were considered the least appealing, and repeats the point in the Discussion ('The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our abstract to realistic scale').

- **Entry #136** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  [check wording] THESIS: "The repeated-measures ANOVA is the established test for rating studies that compare rendering styles within participants."
  SOURCE (SECTION 5.1.3 RESULTS, P. 91:5 (FIRST PASSAGE); SECTION 5.1 EXP 1: STILL VS. MOVING, P. 91:4 (SECOND PASSAGE)): "For each of the rating scales, a two-way repeated measures ANalysis Of VAriance (ANOVA) was conducted on the data from both blocks with within-subjects factor render style (10) and between-subjects factor movement (2 - still or moving). ... In psychology, rating scales are commonly used to gather information on subjective conditions such as attitudes and emotions of participants, which could not be collected directly."
  CHECK: The paper demonstrates the practice, running two-way repeated-measures ANOVAs with render style as the within-subjects factor across all six rating scales and again in Exp 2 and Section 6.2. It nowhere claims that repeated-measures ANOVA is 'the established test' for such studies, and it makes no methodological survey statement of that kind. The paper is only a single instance of the practice, so it supports the sentence by example rather than by assertion. Note also that the paper's own Exp 1 factor 'movement' was between-groups, not within-participants.

McDonnell et al. (2009) — Body Shape and the Perception of Emotion

Full reference: McDonnell, R., Jörg, S., McHugh, J., Newell, F. N., & O'Sullivan, C. (2009). Investigating the role of body shape on the perception of emotion. *ACM Transactions on Applied Perception*, 6(3), Article 14.

DOI: `10.1145/1577755.1577757`

Physical PDF: `McDonnell09_HappyZombies.pdf` in Articles folder. Verified.

Research question. Are real and virtual characters moving with the same biological motion perceived to be exhibiting the same emotions? Specifically, does the visual representation of the character's body (photorealistic, stylised, cartoon, or nonhuman) change how emotional content is recognised and experienced?

Stimuli. A video of a real actor performing six basic emotions (sadness, happiness, surprise, fear, anger, disgust) served as the motion source. The same motion-captured performance was applied to five virtual body types:

- *High-resolution virtual male* — a detailed realistic mesh counterpart of the actor

- *Low-resolution virtual male* — the same mesh with reduced polygon count
- *Cartoon character* — an anthropomorphic but clearly non-human toon figure (blue, simplified proportions)
- *Wooden mannequin* — a geometrically neutral, non-expressive body
- *Zombie-like character* — a visually disturbing near-human figure placed deliberately in the uncanny zone

Faces and hands were blurred in all conditions to remove confounding facial and gestural cues. A second experiment used point-light displays (luminance dots at joints, no body form) to test whether motion alone conveys emotional information.

Procedure. Participants rated each performance across all character types on a list of 41 complex emotion descriptors. The key measure was whether the dominant perceived emotion matched the actor's intended emotion. A point-of-light experiment then isolated the contribution of body form from motion.

Key findings.

- **Body representation had little or no effect on emotion recognition.** Participants correctly identified the intended emotion at comparable rates across all body types: real video, high-resolution virtual human, low-resolution virtual human, cartoon character, wooden mannequin, and zombie. The authors conclude that "captured body motion is as effective at depicting emotional body language regardless of character appearance."
- **Emotion perception is highly robust and largely independent of body shape,** so long as any form at all is present. No significant difference emerged between the zombie rendering and the photorealistic condition for recognition accuracy.
- **Emotion intensity was lower for point-light displays.** When motion alone was presented without body form, emotions were perceived as less intense than in all form-present conditions. Body geometry contributes to perceived emotional strength even when it does not determine which emotion is identified.

Interpretation. Biological motion carries the emotional signal independently of rendering style or mesh complexity. The visual body can vary from photorealistic to cartoon to zombie without disrupting emotion recognition. Only the complete removal of body form (point-light) noticeably reduces perceived emotional intensity.

Relevance to this thesis. This paper establishes that NPR stylisation does not impair the emotional communicative function of a virtual character, even when the stylisation is as extreme as a cartoon or zombie body. This supports the design assumption underlying the C3 (NPR) condition: applying an NPR shader to the Avaturn avatar is not expected to reduce the advisor's communicative clarity. Any differences in trust ratings between C2 and C3 are therefore unlikely to be driven by reduced comprehension of the avatar's body language. The paper also provides an empirical parallel to Mori's prediction: even a zombie-like body, which sits in the uncanny valley for visual appearance, conveys emotion as clearly as a photorealistic avatar, suggesting that emotional legibility and eeriness operate on separate perceptual channels.

Dill et al. (2012) — Evaluation of the Uncanny Valley in CG Characters

Full reference: Dill, V., Flach, L. M., Hocevar, R., Lykawka, C., Musse, S. R., & Pinho, M. S. (2012). Evaluation of the uncanny valley in CG characters. In Y. Nakano, M. Neff, A. Paiva, & M. Walker (Eds.), *Intelligent Virtual Agents: 12th International Conference, IVA 2012*, Lecture Notes in Artificial Intelligence, vol. 7502, pp. 511–513. Springer, Heidelberg.

Note: Poster paper (3 pages). Proceedings DOI: `10.1007/978-3-642-33197-8`

Research question. Does the uncanny valley effect, originally proposed by Mori (1970) for robots and androids, extend to contemporary CG characters from movies and games? The paper also asks whether adding motion to these characters changes the shape of the uncanny valley curve.

Methodology. The authors selected ten representative characters from recent and well-known movies and games spanning a range of human-likeness. A two-stage questionnaire was administered to participants. In the first stage, participants rated still images of the characters; in the second stage, they rated video clips of the same characters performing the same scenes. Ratings were plotted with human-likeness on the horizontal axis and familiarity level on the vertical axis, directly replicating Mori's original graph format. Participants were also asked which parts of the face caused the most discomfort where any was experienced. The exact sample size is not reported in this poster contribution.

Results. The obtained data produced a chart closely following the original uncanny valley curve proposed by Mori. Characters in the near-human range (C.11, C.16, C.1, and C.14) showed the characteristic negative response: they were rated as highly human-like but received low familiarity scores, consistent with the "almost human but unnatural" pattern. The curves for still images and for video were both coherent with Mori's original. The video curve showed more pronounced valley effects for near-human characters than the still-image curve, consistent with Mori's prediction that movement amplifies the uncanny response. When asked which facial features caused discomfort, participants consistently identified two regions: the **eyes** and the **mouth**.

Limitations. This is a three-page poster paper; no statistical tests are reported and the sample size is not stated. The ten characters were selected from existing productions rather than systematically placed along a controlled realism continuum, meaning the spacing of realism levels is uneven. No demographic data (age, gender, familiarity with animated films or games) were collected or controlled.

Relevance to this thesis. This paper is one of the few empirical validations of the uncanny valley effect for CG characters from contemporary digital media, rather than for robots or androids. It confirms that the phenomenon is not confined to physical hardware and extends to the type of content this thesis works with. The identification of eyes and mouth as the primary sources of facial discomfort is directly relevant to avatar design: these are the most expressive and detail-dependent features of a face, and NPR stylisation that abstracts skin texture and smooths micro-detail in these regions is expected to reduce precisely this discomfort. The replication of the valley for both still images and video supports the thesis's video-based study design, and the motion amplification result provides additional justification for using dynamic stimuli. The paper provides direct empirical grounding for H1 and for the overall motivation that photorealistic avatars risk entering the uncanny zone in a way that NPR stylisation can mitigate.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #152** *(Ch.6 Evaluation > Discussion > What the Stylisation Left Unchanged)*
  THESIS: "The features that the participants of Dill et al. blamed for their discomfort, the eyes and the mouth, were largely preserved."
  SOURCE (SECTION 2 (OBTAINED RESULTS), PRINTED P. 512 (PDF P. 527 OF THE IVA 2012 LNAI 7502 PROCEEDINGS VOLUME)): "The questionnaire also asked to the participants which parts of the face the participant felt more discomfort, if it was the case. Analysing the answers, we could highlight two specific parts of the face: eyes and mouth."
  CHECK: The paper explicitly reports that participants named the eyes and the mouth as the facial parts causing the most discomfort, which is exactly the attributed part of the thesis sentence. The trailing clause 'were largely preserved' describes the thesis's own rendering result and carries no attribution to Dill et al.

Seymour et al. (2021) — Have We Crossed the Uncanny Valley?

Full reference: Seymour, M., Yuan, L., Dennis, A. R., & Riemer, K. (2021). Have we crossed the uncanny valley? Understanding affinity, trustworthiness, and preference for realistic digital humans in immersive environments. *Journal of the Association for Information Systems*, 22(3).

DOI: `10.17705/1jais.00491`

Research questions. Are current human-realistic avatars capable of crossing the uncanny valley? Do users show greater affinity, trustworthiness ratings, and preference for a human-realistic avatar than a caricature avatar? Does an immersive 3D VR display amplify these differences compared to a 2D screen?

Study design. A mixed-methods field study was conducted at the SIGGRAPH 2017 conference over four days and 18 sessions as part of the MEETMIKE event series. The host avatar was Digital MIKE, a highly human-realistic digital human created to resemble the real host Mike Seymour. Real-time facial animation was driven by Cubic Motion deep-learning face-tracking and rendered in Unreal Engine 4 at 90 stereo frames per second. Guest avatars were 18 unique caricature avatars, one per session, generated by Loom.ai and representing the actual guest expert participants. Audience members observed the live interview either via VR headset (HTC VIVE, $n = 72$) or via a traditional 2D projection screen ($n \approx 550$). Data collection was concurrent and mixed-method: qualitative observation, one-on-one structured interviews, and a quantitative survey collecting affinity, trustworthiness, and preference measures.

Hypotheses and results.

- **H1** (greater affinity for human-realistic avatar) — Supported. Both VR and 2D groups rated Digital MIKE significantly higher on affinity than the caricature guest avatars. Effect sizes were medium to large.
- **H2** (greater trustworthiness for human-realistic avatar) — Supported. Digital MIKE received higher trustworthiness ratings.
- **H3** (greater preference for human-realistic avatar as virtual agent) — Supported.
- **H4** (VR amplifies the effect) — Partially supported. VR participants showed a stronger

preference for the human-realistic avatar and stronger affinity for it than 2D screen observers, consistent with the prediction that immersive display environments heighten the perceptual differences between realism levels.

Qualitative findings. Interview participants acknowledged the different realism levels of the two avatars but evaluated both positively. The caricature avatars did not produce overt negative reactions: participants engaged socially with them. This is consistent with Mori's two-slope model: caricature avatars near the first peak produce positive affect, while human-realistic avatars near the second peak also produce positive affect. The paper's contribution is not showing a valley response to caricatures, but showing that the second peak is now reachable in real-time rendering.

Relevance to this thesis. This paper establishes that even current human-realistic real-time avatars, though rated more trustworthy and preferred over caricature alternatives, still exist below the real-human benchmark in affinity terms. H3 in this study (both avatar conditions produce lower trust than the real person) is consistent with this finding. The C2 vs. C3 comparison mirrors Seymour et al.'s human-realistic vs. caricature comparison, though in this study the NPR shader is applied to the same base avatar rather than using an entirely distinct caricature character.

User Study — Trust and Avatar Perception

McKnight & Chervany (2001) — Trust Conceptual Typology

Full reference: McKnight, D. H. & Chervany, N. L. (2001). What trust means in e-commerce customer relationships: An interdisciplinary conceptual typology. *International Journal of Electronic Commerce*, 6(2), 35–59.

DOI: 10.1080/10864415.2001.11044235

Background and motivation. Trust research across disciplines had produced dozens of incompatible definitions. The word "trust" carried on average 17 distinct meanings across major unabridged dictionaries, more than conceptually adjacent terms such as "cooperation" or "confidence" (mean of 4.7 definitions). Psychologists viewed trust as a personal trait, sociologists viewed it as a social structure, and economists viewed it as a mechanism for rational choice. This disciplinary fragmentation meant that empirical results could not be meaningfully compared or accumulated. McKnight and Chervany argued that before empirical research on trust can progress, the field needs a clear, parsimonious, interdisciplinary conceptual typology, one that identifies what the principal types of trust are, how they differ from one another, and how they relate to each other causally.

Method: how the typology was constructed. The authors reviewed approximately 80 articles and books on trust across three disciplinary clusters: psychology and social psychology (23 sources), sociology, economics, and political science (19 sources), and management and communications (23 sources). From these they identified 65 sources that provided usable trust definitions. Searching stopped when conceptual saturation was reached, that is, when no new definition types emerged. By analysing these definitions, the authors found that all existing definitions fell into two independent cross-cutting dimensions. The first was the *conceptual type* of the construct: whether trust was defined as a disposition, an attitude, a belief, an intention, or a behavior. The second was the *referent* of the construct: whether trust was directed at people in general, at a specific situation or structural arrangement, or at a specific other person. They also identified 16 specific trust-related characteristics mentioned across all definitions, which collapsed into five second-order categories (Table 1 in the original paper):

Second-Order Category	Count	% of Total
Benevolence (good will, caring, responsiveness)	38	38.8
Integrity (honest, reliable, dependable)	26	26.5
Competence (capable, expert, dynamic)	20	20.4
Predictability	6	6.1
Other	8	8.2
Total	98	100.0

Mapping all 65 definitions onto a five-by-six grid crossing these dimensions produced the final typology: four high-level trust constructs and ten measurable subconstructs.

The four trust constructs. From the analysis of 65 definitions, McKnight and Chervany derive the following comprehensive overarching definition of trust, which covers all three domain levels simultaneously:

Overarching definition of trust (McKnight & Chervany, 2001, p. 10):

"To willingly become vulnerable to the trustee — another person, institution, or people generally — having taken into consideration the characteristics of the trustee, with a feeling of relative security in a situation of risk."

This definition applies across all trust types. The **direct object** (who or what one trusts) is what distinguishes dispositional, institutional, and interpersonal trust from each other. All forms of trust share two implicit background conditions: a feeling of relative security, and an actual situation of risk or uncertainty. Trust is meaningful only because dependence carries the possibility of negative consequences.

1. Disposition to Trust *(from trait psychology and economics)*

Definition: the extent to which one displays a consistent tendency to be willing to depend on general others across a broad spectrum of situations and persons.

This is a stable, generalised personality tendency that is cross-situational and cross-personal. It does not mean one believes specific people are trustworthy; it means one has a propensity to be willing to depend on others generally. Disposition to trust matters most in novel situations where no specific evidence about the other party yet exists. It colours how new situations and new actors are interpreted. It has two subconstructs:

- *Faith in Humanity:* one assumes that other people in general are usually competent, benevolent, honest, and predictable. This is a belief about human nature. Someone with high Faith in Humanity is less judgmental upfront and more tolerant of others' mistakes. Because it concerns beliefs about people's attributes, it is more likely to predict Trusting Beliefs about a specific party directly.
- *Trusting Stance:* regardless of what one assumes about people's attributes, one assumes that better outcomes will result from treating people as though they are well-meaning and reliable. This is a personal strategy rather than a belief, derived from the calculative economics tradition. Someone with a high Trusting Stance would say "I trust until given a reason not to." Because it is a strategy rather than a belief about people, it connects more directly to Trusting Intentions than to Trusting Beliefs.

2. Institution-based Trust *(from sociology: Zucker, Luhmann, Garfinkel)*

Definition: one believes that favorable conditions are in place that are conducive to situational success in an endeavor or aspect of one's life.

This is trust in the surrounding situation or structural arrangement rather than in any specific person. It is situation-specific but cross-personal: it applies to this type of situation regardless of who the specific other party is. It has two subconstructs:

- *Structural Assurance:* one believes that protective structures — guarantees, contracts, regulations, promises, legal recourse, processes, or procedures — are in place that are conducive to success.
- *Situational Normality:* one believes that the situation in a venture is normal, favorable, or conducive to success. Trust breaks down when things "go weird," when the situation seems

inexplicable, improper, or abnormal. When a web consumer believes the online shopping interaction is normal and that the vendor's role is appropriate and expected, situational normality supports trust in that vendor.

3. Trusting Beliefs *(from social psychology — interpersonal trust, belief component)*

Definition: one believes that the other party has one or more characteristics beneficial to one.

This is directed at a specific other person or entity and is cross-situational: it applies across various interactions with that party. It is a cognitive and affective belief about the other party's attributes, not merely an expectation. The paper identifies four subconstructs, building on Mayer, Davis, and Schoorman (1995):

- *Competence belief:* one believes the other party has the ability or power to do what one needs done. In an advisory context, the trustor believes the advisor can provide relevant and correct guidance. This addresses the fear that the other party may be willing but simply unable.
- *Benevolence belief:* one believes the other party cares about one and is motivated to act in one's interest. This is about the specific relationship between trustor and trustee: benevolence does not mean the trustee is universally kind to everyone, but that they specifically look out for this trustor. A benevolent advisor would not exploit informational asymmetry to further their own interests at the trustor's expense.
- *Integrity belief:* one believes the other party makes good-faith agreements, tells the truth, acts ethically, and fulfills promises. Integrity is about the moral character of the trustee rather than the dyadic relationship. It is value-laden: a party with integrity adheres to principles the trustor considers acceptable. An advisor with high integrity would be honest about trade-offs even when honesty is costly to themselves or complicated for the trustor to hear.
- *Predictability belief:* one believes the other party's actions — whether good or bad — are consistent enough that one can forecast them in a given situation. Unlike integrity, predictability is value-neutral: the party is believed to be consistent in both virtues and flaws. Predictability interacts with the other three beliefs because it answers whether the benevolence, competence, and integrity one perceives are stable over time. The authors include predictability in their typology (unlike Mayer et al., who excluded it) because consistency of behavior is a meaningful separate dimension in real trust relationships.

4. Trusting Intentions *(from social psychology — interpersonal trust, intention component)*

Definition: one is willing to depend on, or intends to depend on, the other party even though one cannot control that party.

This follows from Trusting Beliefs via the theory of reasoned action (Fishbein and Ajzen, 1975): beliefs produce intentions. It is also person-specific and cross-situational. Three elements are essential: a readiness to depend or rely, person-specificity, and the absence of control (one is willingly vulnerable). It has two subconstructs:

- *Willingness to Depend:* one is volitionally prepared to make oneself vulnerable to the other party by relying on them in a situation. This is about desire or volition.

- *Subjective Probability of Depending:* the extent to which one forecasts or predicts that one will depend on the other party. This is stronger than willingness: it is a committed forecast or intent. While willingness expresses desire, subjective probability expresses a verifiable commitment.

The paper also defines mirror-image *distrusting intentions* subconstructs for contrast: *no willingness to depend* and *subjective probability of not depending*. Distrust feelings are described as usually strong and emotionally charged, qualitatively different from the cooler cognitive character of trust.

The causal model. The four constructs are arranged in a left-to-right causal model. Figure 1 reproduces the structure of Figure 1 from the original paper. Solid arrows indicate primary causal paths; thinner arrows indicate weaker or mediated links. Disposition to Trust also feeds directly into Trusting Beliefs about specific parties, especially in novel situations where no situational cues yet exist. Trust-Related Behaviors sit outside the formal typology because behaviors like purchasing, cooperating, and information sharing already carry their own labels; they are downstream consequences of Trusting Beliefs and Trusting Intentions, not trust constructs in themselves.

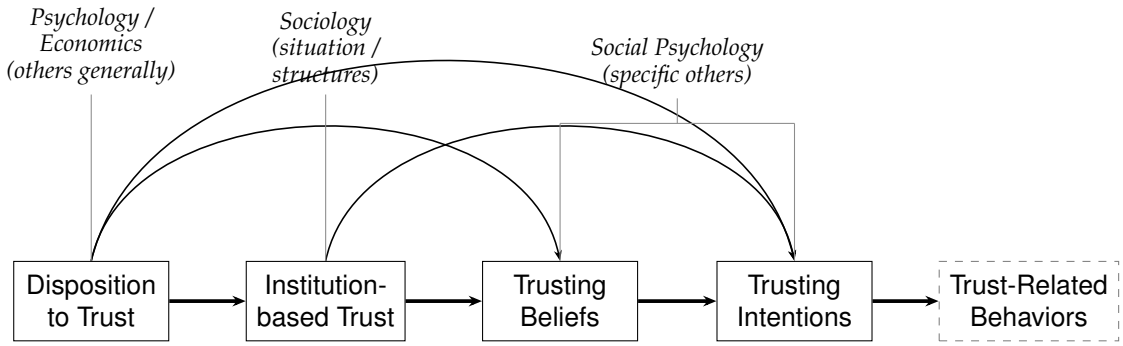


Figure 1: Interdisciplinary model of trust constructs (adapted from McKnight & Chervany, 2001, Figure 1). Thicker arrows are primary causal paths. Thinner arrows are weaker or mediated links. The dotted box lies outside the formal trust typology.

The "grammar" of trust (Figure 2 in the original paper) makes the differentiating logic explicit: the trustor is the subject, trust is the verb, and the *direct object* determines the type of trust construct involved.

Subject (Trustor)	Verb	Object (Trustee)	Type
The e-commerce consumer	trusts	*the E-vendor*	Interpersonal
The e-commerce consumer	trusts	*the web itself*	Institutional
The e-commerce consumer	trusts	*others generally*	Dispositional

Adapted from McKnight & Chervany (2001), Figure 2.

Contextual orientation of constructs. The four constructs differ along two contextual dimensions: whether they are situation-specific or cross-situational, and whether they are person-specific or cross-personal. It makes explicit why collapsing trust into a single score is analytically unsound: each construct has a different contextual scope, and an intervention that affects one need not affect another.

	Situational		Personal	
Trust Type	Situation-specific	Cross-situational	Person-specific	Cross-personal
Disposition to Trust		✓		✓
Institution-based Trust	✓			✓
Trusting Beliefs		✓	✓	
Trusting Intentions		✓	✓	

Adapted from McKnight & Chervany (2001), Table 3.

Methodology. The paper is a purely conceptual synthesis. It contains no empirical study, no experiment, and no participant data. Its contribution is the taxonomic framework itself: the definitions, the hierarchy of constructs and subconstructs, and the proposed causal model. The authors report that 93 student raters validated the categorisation of trust characteristics, with 71 percent agreement with the authors' assignments. The authors also note that scales for each subconstruct were developed and tested separately, with results to be reported elsewhere.

Limitations acknowledged by the authors. Two limitations are identified. First, crossing disciplinary lines risks losing nuance from original domain-specific definitions; the authors address this by grounding each subconstruct carefully in its originating literature. Second, some subconstructs may not be empirically discriminant in all conditions: particularly benevolence and integrity tend to factor together when the trustor knows little about the trustee. The dispositional and institution-based subconstructs are described as consistently distinguishable.

Relevance to this thesis. The McKnight and Chervany typology provides the theoretical foundation for the three-part trust battery used in this study (Parts E, F, and G of the questionnaire). The three Trusting Beliefs subconstructs retained here — competence, benevolence, and integrity — correspond directly to their framework, as adopted by Canales et al. (2024) for the avatar trust context. Predictability is not measured as a separate subscale in this study because, as McKnight and Chervany themselves note, it tends to merge with competence and integrity when the trustor and trustee are unfamiliar, which is the case for all participants in a first-impression video study.

The scenario design maps the three constructs onto distinct advisor interactions. Scenario 1 (Career Choice) targets *competence*: the participant must judge whether the advisor has the knowledge and judgment to navigate a complex career trade-off. Scenario 2 (Travel Route) targets *benevolence* alongside competence: the advisor must demonstrate that they are looking out for the participant's specific interests, not giving generic advice. Scenario 3 (Learning Resource) targets *integrity*: the advisor is honest about a trade-off that could undermine their recommendation (the free option has equivalent content), demonstrating transparent reasoning over self-serving simplification.

Why Disposition to Trust and Institution-based Trust are not measured. The remaining two constructs in the McKnight and Chervany typology are excluded from this study for principled reasons that follow directly from the model. Disposition to Trust is a stable personality trait: it is cross-situational and cross-personal, meaning it captures how much a given participant tends to trust people in general, regardless of who they are or what context they appear in. It does not vary based on which avatar a participant just watched. Measuring it three times across three video blocks would yield the same score each time; it cannot function as a within-subjects dependent variable. In a more comprehensive study design it would enter as a between-subjects

covariate, measured once per participant. Institution-based Trust is similarly invariant across conditions because every participant in this study encounters the same institutional context, that is, the same survey platform, the same study framing, and the same university ethics-approved environment. None of this changes between C1, C2, and C3. Moreover, the research question is not about the trustworthiness of the study environment but about the trustworthiness of a specific advisor as perceived through their visual rendering. That is precisely the interpersonal domain. Both excluded constructs are antecedents in the McKnight and Chervany causal model: they feed into Trusting Beliefs rather than being outcomes of them. Measuring Trusting Beliefs is therefore the theoretically correct choice for a study whose independent variable is the appearance of a specific other party.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #44** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "McKnight and Chervany subsequently carried that distinction into a typology separating trusting beliefs from trusting intentions in technology-mediated contexts."
  SOURCE (CONCEPTUAL DEFINITIONS OF TRUST CONSTRUCTS, PP. 13-14 (SEE ALSO THE ABSTRACT, P. 2: "THE CONCEPTUAL-LEVEL CONSTRUCTS CONSIST OF DISPOSITION TO TRUST ..., INSTITUTION-BASED TRUST ..., AND TRUSTING BELIEFS AND TRUSTING INTENTIONS")): "Trusting Beliefs means one believes that the other party has one or more characteristics beneficial to one. ... Trusting Intentions means one is willing to depend on, or intends to depend on, the other party even though one cannot control that party."
  CHECK: The paper builds an interdisciplinary typology in which Trusting Beliefs (beliefs about the trustee's characteristics) and Trusting Intentions (willingness to depend) are separate constructs, matching the thesis sentence. The setting is technology-mediated: the running context is e-commerce consumer to e-vendor relationships (p. 13, "In the Internet context, the people involved include consumers and E-vendors").

- **Entry #130** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Parts E, F, G — Trust battery (7 point Likert). Nine items operationalised the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in ."
  SOURCE (A TYPOLOGY OF RELATED TRUST CONSTRUCTS, P. 11; CONCEPTUAL DEFINITIONS OF TRUST SUBCONSTRUCTS, P. 17): "Trusting Beliefs includes Competence, Benevolence, Integrity, and Predictability beliefs, corresponding to the first four rows of Table 2. ... Trusting beliefs subconstructs defined here consist of four types, building on Mayer et al. (1995), though we recognize that other types exist."
  CHECK: The source explicitly derives its trusting-beliefs subconstructs from Mayer et al. (1995) and names competence, benevolence and integrity among them, so the thesis framing is accurate. Two caveats worth noting: McKnight and Chervany label the first belief Competence rather than Mayer's Ability, and they add a fourth belief, Predictability (p. 17, "unlike Mayer et al., we feel predictability is important"), which a three-subscale nine-item battery does not cover. The paper is conceptual and supplies no measurement items itself, only the statement that the subconstructs are "measurable via scales" (p. 11).

Mayer, Davis & Schoorman (1995) — Integrative Model of Trust

Full reference: Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3), 709–734.

DOI: 10.5465/amr.1995.9508080335

Core argument. Prior trust research suffered from conceptual fragmentation: disciplines used definitions that were not interchangeable, making findings hard to compare. Mayer et al. synthesised these into one integrative model. Trust is defined as *"the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party"* (p. 712). Three elements are essential: (1) willingness to be vulnerable; (2) the absence of monitoring and control; (3) a positive expectation about the other party's behaviour. Trust is therefore a disposition toward risk, not a response to certainty.

The three trustworthiness factors — the ABI model. Trust in a specific other party arises from that party's perceived trustworthiness, decomposed into three independent factors:

- **Ability:** A cluster of skills and competencies that enable a party to have influence within a specific domain. Ability is domain-specific: a trustee may be highly competent in one area and not in another. In an advisory context, ability concerns whether the advisor has the knowledge and judgment to provide useful guidance on the particular topic at hand.
- **Benevolence:** The extent to which the trustee is believed to want to do good for the trustor, apart from any self-interested motive. Benevolence is relational and specific to the trustor-trustee dyad — it is not a general trait of kindness but a belief that this trustee specifically cares about this trustor's wellbeing. An advisor with high benevolence would not exploit an informational asymmetry to push the trustor toward an option that serves the advisor's own interests.
- **Integrity:** The trustor's perception that the trustee adheres to a set of principles the trustor finds acceptable. This includes honesty, consistency between stated values and observed actions, and credibility in communication. A trustee with high integrity discloses trade-offs honestly even when doing so is complicated for the trustor to hear.

The three factors are conceptually independent and can take different values for the same trustee: a party can be highly able but low in benevolence (competent but self-serving), or high in benevolence but low in ability (caring but ineffective). Collapsing them into a single trust score obscures these distinctions and masks differential effects.

Full causal model. Perceived trustworthiness (ability + benevolence + integrity) produces trust (willingness to be vulnerable), which produces risk-taking behaviour in the relationship. This behavioural outcome — acting on advice, following a recommendation, selecting a trustee for a consequential task — sits outside the definition of trust itself but is its downstream consequence. Two moderators are included: (1) the trustor's *propensity to trust*, a stable individual trait that determines how much weight is placed on perceived trustworthiness before much evidence has accumulated; and (2) the *perceived risk* of the situation, which moderates whether trust translates into actual risk-taking behaviour at all.

What trust is not. The authors explicitly distinguish trust from cooperation (which can occur under monitoring), from confidence (which implies certainty, not vulnerability), and from

predictability (one can predict an untrustworthy person's behaviour without trusting them). A scale that asks whether participants can predict a trustee's behaviour is not measuring trust.

Relevance to this study. The Mayer et al. ABI model is the theoretical foundation for the trust battery in Parts E, F, and G of the questionnaire, as adopted by Canales et al. (2024) for the avatar trust context. The three subscales correspond directly: ability (Part E: competence and relevant knowledge), benevolence (Part F: caring and acting in the participant's interest), and integrity (Part G: honesty and transparent reasoning). The forced-choice advisor selection (Part P1) operationalises the risk-taking behavioural outcome that Mayer et al. place downstream of trust in their model. Including both the self-report battery and the behavioural measure allows this study to test whether the self-report and behavioural layers dissociate across rendering conditions.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #43** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "They further separated trustworthiness as an attribute of the trustee from trust as an attitude held by the trustor."
  SOURCE (CHARACTERISTICS OF THE TRUSTEE: THE CONCEPT OF TRUSTWORTHINESS, P. 717): "In the remainder of this section, three characteristics of the trustee that determine trustworthiness are examined. Although they are not trust per se, these variables help build the foundation for the development of trust."
  CHECK: The separation is explicit here and reinforced on p. 724 ('The distinction between a trustor's characteristics and trustee's characteristics is important.') and p. 729 ('trust is conceptually easy to differentiate from perceived ability, benevolence, and integrity of the trustee'). Minor wording caveat: Mayer et al. define trust as 'the willingness of a party to be vulnerable' (p. 712) rather than as an attitude, though they once write 'no risk involved with holding such an attitude' (p. 726).

- **Entry #129** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Parts E, F, G — Trust battery (7 point Likert). Nine items operationalised the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in ."
  SOURCE (THE FACTORS OF TRUSTWORTHINESS, P. 717): "Even though a number of factors have been proposed, three characteristics of a trustee appear often in the literature: ability, benevolence, and integrity. As a set, these three appear to explain a major portion of trustworthiness."
  CHECK: Mayer et al. propose the ability-benevolence-integrity model as cited. They supply no measurement items, so the nine-item 7-point battery is the thesis's own operationalisation; the thesis label 'Competence' corresponds to Mayer's 'ability' (mapping stated on p. 724).

- **Entry #133** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "They expect it to explain variance in trust when included alongside a fuller set of variables."
  SOURCE (CHARACTERISTICS OF THE TRUSTOR: THE PROPENSITY TO TRUST, P. 716): "Propensity should contribute to the explanation of variance in trust if used as a part of a more complete set of variables."
  CHECK: Verbatim match for the expectation attributed to the authors, with 'it' in the thesis

sentence referring to propensity to trust.

Bartneck et al. (2009) — Godspeed Questionnaire Series

Full reference: Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics*, 1(1), 71–81.

DOI: 10.1007/s12369-008-0001-3

Background and motivation. Robot developers in HRI had been creating their own ad hoc questionnaires without validation, making results from different studies impossible to compare. Bartneck et al. conducted a literature review of existing instruments for five key perceptual concepts, then standardised and validated them into a single coherent battery, which they named *Godspeed*. All five subscales use **5-point semantic differential** format: participants rate their impression of the robot by choosing between two opposing adjective anchors (e.g., *Fake* ↔ *Natural*). This format was chosen over Likert items to reduce acquiescence bias and to maintain consistency across subscales. Validation used a within-subjects study with three robot appearance conditions (human, android, masked android); Cronbach's α was computed per condition, with 0.7 as the minimum acceptable threshold following Nunnally (1978).

The five Godspeed subscales.

Godspeed I — Anthropomorphism *5 items*

Captures the degree to which participants attribute human form, characteristics, or behavior to the observed entity. Connected directly to the uncanny valley: a robot that scores high on anthropomorphism but fails to match the expectations that score creates will trigger stronger eeriness responses.

Negative anchor	Scale	Positive anchor
Fake	1 2 3 4 5	Natural
Machinelike	1 2 3 4 5	Humanlike
Unconscious	1 2 3 4 5	Conscious
Artificial	1 2 3 4 5	Lifelike
Moving rigidly	1 2 3 4 5	Moving elegantly

Cronbach's α: 0.878 (Study 1); 0.929 (human condition), 0.923 (android), 0.856 (masked android) in the within-subjects study. All above threshold.

Godspeed II — Animacy *6 items*

Captures perceived aliveness and responsiveness. Grounded in the Piagetian framework: things that move of their own accord, react to stimuli, and interact with their environment are perceived as animate.

Negative anchor	Scale	Positive anchor
Dead	1 2 3 4 5	Alive
Stagnant	1 2 3 4 5	Lively
Mechanical	1 2 3 4 5	Organic
Artificial	1 2 3 4 5	Lifelike
Inert	1 2 3 4 5	Interactive
Apathetic	1 2 3 4 5	Responsive

Cronbach's α: 0.702 (Study 1); above 0.7 across all conditions in the within-subjects study. The authors note that *Artificial/Lifelike* appears in both Anthropomorphism and Animacy; correlation analysis is recommended when both are used together.

Godspeed III — Likeability *5 items*

Captures positive affective appeal and social warmth. Research shows that first impressions form within seconds to minutes: people decide within 30 seconds whether a blind date will succeed, and interviewers know within 1–2 minutes whether a job applicant is a strong candidate. Computers and robots are treated as social actors by the same mechanisms.

Negative anchor	Scale	Positive anchor
Dislike	1 2 3 4 5	Like
Unfriendly	1 2 3 4 5	Friendly
Unkind	1 2 3 4 5	Kind
Unpleasant	1 2 3 4 5	Pleasant
Awful	1 2 3 4 5	Nice

Cronbach's α: 0.865 (Study 1); 0.923 (human), 0.878 (android), 0.842 (masked android). All well above threshold.

Godspeed IV — Perceived Intelligence *5 items*

Captures the observer's subjective judgment of the entity's intellectual capacity, explicitly distinct from actual computational intelligence. Perceived intelligence depends on behavioral cues: an entity that displays contextually appropriate, non-random behavior is perceived as more intelligent, regardless of whether that behavior is genuinely reasoned or scripted.

Negative anchor	Scale	Positive anchor
Incompetent	1 2 3 4 5	Competent
Ignorant	1 2 3 4 5	Knowledgeable
Irresponsible	1 2 3 4 5	Responsible
Unintelligent	1 2 3 4 5	Intelligent
Foolish	1 2 3 4 5	Sensible

Cronbach's α: 0.750, 0.769, 0.763 across three independent studies. All above threshold.

Godspeed V — Perceived Safety *3 items*

Captures the observer's subjective sense of comfort and absence of anxiety during interaction. Unlike the four previous subscales, the instruction asks about the participant's own affective state rather than their impression of the robot. Developed by Kulic and Croft from physiological

sensor data and validated against heart rate and skin conductance measurements.

Negative anchor	Scale	Positive anchor
Anxious	1 2 3 4 5	Relaxed
Agitated	1 2 3 4 5	Calm
Quiescent	1 2 3 4 5	Surprised

Cronbach's α: 0.91 in the validation study. Strong correlation with physiological arousal measures established concurrent validity.

Which Godspeed subscales are used in this study and which are not.

Used:

- **Godspeed I (Anthropomorphism) → Part D of the questionnaire.** This subscale operationalises perceived human-likeness, which is the perceptual axis along which the three conditions differ. C1 (real person) is expected to score highest; C3 (NPR avatar) is expected to score lower than C2 (realistic avatar) because NPR rendering removes photorealistic texture detail and softens skin gradients. Including anthropomorphism ratings allows testing whether differences in eeriness across conditions are mediated by shifts in perceived human-likeness (H5), or whether eeriness changes independently of humanness scores.
- **Godspeed III (Likeability) → Part C of the questionnaire.** This subscale is the primary affective indicator in this study. Likeability captures warmth, social appeal, and positive first impressions: the affective dimension most directly disrupted by eeriness. If NPR abstraction successfully reduces the uncanny valley response, one mechanism by which it does so is by making the avatar more likeable. Likeability also partially predicts trust: an advisor who is perceived as warm and pleasant is more likely to be listened to and followed. Separating likeability from the trust subscales (Parts E–G) allows the analysis to distinguish affective response from cognitive trust assessment.

Not used, and why:

- **Godspeed II (Animacy) — excluded.** Animacy measures perceived aliveness and biological responsiveness. In this study, all three conditions use the same video recordings with the same movement quality per scenario block. The animation is held constant across C2 and C3 (the critical comparison pair), and C1 is a real person with naturally organic motion. There is no design manipulation targeting animacy perception, so including it would add items without contributing to the core hypotheses. Animacy also overlaps substantially with anthropomorphism at the item level (*Artificial/Lifelike* appears in both), which would require additional factor analysis to disentangle.
- **Godspeed IV (Perceived Intelligence) — excluded.** Perceived intelligence is covered with greater specificity by the McKnight and Chervany trust subscales in Parts E–G. The Competence trust items (Part E: *This advisor seems knowledgeable about this topic*, *This advisor gave me relevant advice*) are a direct and theory-grounded measure of the same construct as Godspeed IV, but framed in terms of the advisory relationship rather than as a general impression of cognitive ability. Adding Godspeed IV alongside the trust battery would create redundancy and lengthen the questionnaire without adding explanatory power.
- **Godspeed V (Perceived Safety) — excluded.** Perceived safety measures anxiety and

discomfort during physical proximity to a robot. This construct has no meaningful referent in a video-only study where the avatar is not physically co-present and poses no motion hazard. The original scale was validated specifically for industrial robot motion scenarios where the robot's speed and trajectory could create genuine physical threat. Applying it to a seated video advisor would produce floor-effect ratings without variance, making the scale analytically useless in this context.

General limitation noted by the authors. Results from Godspeed subscales should be interpreted as comparative tools, not absolute values. Ground truth for constructs like anthropomorphism or likeability is inaccessible: cultural background, prior experience with robots, and individual personality all influence ratings. The questionnaires provide a snapshot that allows conditions to be compared within a study, but absolute scores should not be generalised across different participant populations or stimulus types.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #56** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  [check wording] THESIS: "The Godspeed Questionnaire Series, developed by Bartneck et al., offers one such option, providing validated semantic differential scales for five perceptual dimensions."
  SOURCE (ABSTRACT, P. 71; AND SECT. 1 INTRODUCTION (VALIDITY/RELIABILITY DISCUSSION), P. 73): "A literature review has been performed on the measurements of five key concepts in HRI: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety. The results have been distilled into five consistent questionnaires using semantic differential scales. We report reliability and validity indicators based on several empirical studies that used these questionnaires. ... Psychologists are invited to further develop the questionnaires by adding new concepts, and to conduct further validations where it appears necessary. || Given the short history of research in HRI it would appear difficult to achieve construct validity. The same holds true for criterion validity. ... The proposed questionnaires can therefore be considered to have content validity."
  CHECK: The paper plainly supports 'Bartneck et al.', 'five' concepts (anthropomorphism, animacy, likeability, perceived intelligence, perceived safety), 'semantic differential scales' and the Godspeed name (Sect. 7: 'This series will be called "Godspeed"'). The word 'validated' overstates the source: the authors claim only content validity plus internal consistency reliability (Cronbach's Alpha), explicitly state construct and criterion validity are difficult to achieve in HRI, and invite others 'to conduct further validations where it appears necessary'. For perceived safety they only write that 'the validity of the questionnaire may be assumed'.

- **Entry #94** *(Ch.4 Methodology > User Study Overview)*
  THESIS: "The Godspeed questionnaire supplied the likeability items."
  SOURCE (SECT. 4 LIKEABILITY, PP. 75-76; ITEM LIST ALSO IN APPENDIX, 'GODSPEED III: LIKEABILITY', P. 79): "Jennifer Monahan [41] complemented her "liking" question with 5-point semantic differential scales: nice/awful, friendly/unfriendly, kind/unkind, and pleasant/unpleasant, because these judgments tend to demonstrate considerable variance in common with "liking" judgments [42]. ... We decided to only include the five items, since it is always possible to exclude items in cases where they would not contribute to the reliability and validity of the questionnaire."
  CHECK: The paper presents the likeability items as one of the five Godspeed questionnaires

(named 'Godspeed' in Sect. 7 Conclusions), so it is the correct source for likeability items.

- **Entry #127** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Part B used the five-item Godspeed likeability subscale, from Bartneck et al.."
  SOURCE (APPENDIX, P. 79 (GODSPEED III); SECT. 4 LIKEABILITY, P. 76): "GODSPEED III: LIKEABILITY. Please rate your impression of the robot on these scales: Dislike 1 2 3 4 5 Like; Unfriendly 1 2 3 4 5 Friendly; Unkind 1 2 3 4 5 Kind; Unpleasant 1 2 3 4 5 Pleasant; Awful 1 2 3 4 5 Nice || We decided to only include the five items ... Two studies used this new likeability questionnaire. The first reports a Cronbach's Alpha of 0.865 [21] ... 0.923 for the human condition, 0.878 for the android condition, and 0.842 for the masked android condition."
  CHECK: The Appendix confirms exactly five likeability items on a 5-point semantic differential scale, matching the thesis description of a five-item Godspeed likeability subscale.

- **Entry #138** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Cronbach's $\{}alpha$ was reported per subscale before the composites were formed, since averaging items presumes they measure one construct."
  SOURCE (SECT. 7 CONCLUSIONS, P. 78; AND SECT. 1 INTRODUCTION (RELIABILITY DISCUSSION), P. 73): "Before calculating the mean scores for anthropomorphism, animacy, likeability, or perceived intelligence it is good practice to perform a reliability test and report the resulting Cronbach's Alpha. || Internal consistency ... is a measurement for how well the different items measure the same concept, and it is of particular importance to the questionnaires proposed because they are designed to be homogenous in content. Internal consistency involves the calculation of a statistic known as Cronbach's Alpha."
  CHECK: The source explicitly recommends running a reliability test and reporting Cronbach's Alpha before computing subscale means, and defines internal consistency as how well items measure the same concept, which is the rationale the thesis sentence gives. Minor detail: the paper's list of subscales in that recommendation omits perceived safety.

Canales, Roble & Neff (2024) — Trust and Avatar Stylisation

Full reference: Canales, R., Roble, D., & Neff, M. (2024). The impact of avatar stylization on trust. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 418–428.

DOI: `10.1109/VR58804.2024.00063`

Study design. 66 participants in VR each interviewed three virtual doctors in a 3 (identity) × 3 (stylisation level) within-subjects design. Three doctor identities (Asian female AF, South Asian male AM, Caucasian male CM) were each rendered at three stylisation levels: photorealistic (Real), semi-realistic (Mid), and caricatured. All responses were pre-recorded using high-quality offline motion capture (Vicon for body, Faceware for face) and played back in Unreal Engine 5 on a Meta Quest 2. The scenario was high-stakes medical: participants imagined they had been diagnosed with a potentially cancerous kidney lump and needed a second opinion. Each doctor answered three preset questions, one designed to address each trust component from the Mayer et al. (1995) ABI model.

Questionnaire and behavioural measure. After each doctor interview, participants completed a trust survey on 7-point Likert scales. The survey covered the three Mayer et al. trust components:

- **Ability** — two items assessing perceived competence and relevant skills
- **Benevolence** — two items assessing perceived care and alignment with patient interests
- **Integrity** — two items assessing honesty and adherence to acceptable principles
- **Composite trust** — one overall trust item

All items were 7-point Likert scales. Scale labels were withheld from participants to avoid priming. Dialogues were pre-tested on Mechanical Turk to be statistically equivalent in believability across all three doctor identities. After all three interviews, participants chose their most preferred doctor for a follow-up second opinion and their least preferred — the primary behavioural trust indicator.

Key results. Style had no significant main effect on any self-report trust dimension (ability: $p = .12$; benevolence partial: $p = .034$; integrity: $p = .16$; composite trust: $p = .13$). Realistic avatars were not trusted more than stylised ones. However, stylisation significantly affected *behavioural* doctor selection ($p = .044$): Mid-style was chosen as most preferred doctor by 48% of participants, compared to approximately 26% each for Real and Caricature. The strongest effect was *identity*: CM was consistently rated lower on all trust dimensions and selected as least preferred most often, driven by motion and body language artifacts from the retargeting pipeline rather than visual style.

Relevance to this study. This is the closest prior work to the current study. Three design decisions are adopted directly: (1) the video-based advisor paradigm with pre-recorded scripted responses; (2) the Mayer et al. ABI trust battery on 7-point Likert scales (Parts E, F, G of the questionnaire); (3) the forced-choice advisor selection as a behavioural trust indicator (Part P1). The finding that self-reported trust was equivalent across stylisation levels is the empirical basis for H2. The dissociation between self-report (no style effect) and behavioural selection (Mid preferred) is the reason this study includes both measures: a self-report battery alone would have missed the style preference entirely.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #49** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "The three avatar stylisation levels used by Canales et al.: realistic (AF-Real), semi-realistic (AM-Mid) and caricatured (CM-Caricature)."
  SOURCE (ABSTRACT, P. 1; §4.1 AVATARS, P. 4; §5.1 DESIGN, P. 5): "Participants meet three different doctors with three different style levels: realistic, caricatured, and an in-between "Mid" level. ... [§4.1, p. 4] they created concept art for a caricatured version of each person along with a semi-realistic model intended to lie between the caricature and realistic model. ... [§5.1, p. 5] each participant sees only three avatars, featuring one of each identity and each style (e.g. AM-Caricature, AF-Real and CM-Mid)."
  CHECK: The three style levels (Real, Mid, Caricature) and the three identities (AF, AM, CM) are as described, and the paper itself calls the Mid level 'semi-realistic'. The specific identity-style pairing in the caption (AF-Real, AM-Mid, CM-Caricature) is one of the six between-subjects combinations, not a fixed property of the design, so the caption should not be read as implying each style was tied to one identity.

- **Entry #51** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "However, the semi-realistic condition attracted the highest rate of behavioural

selection."
SOURCE (§6.2 DOCTOR SELECTION, P. 7; §9 DISCUSSION, P. 8): "We found that doctor selection by Style was unlikely to occur by chance (p = 0.044), with the probability of Mid being the style of the most preferred doctor significantly higher than chance (p < 0.05). ... [§9, p. 8] the strongest result related to style shows that people were more likely to select the doctor portrayed in the Mid-style, doing this 48% of the time, compared to 26% for the other two categories."
CHECK: The Mid (semi-realistic) style was selected as the preferred doctor 48% of the time versus 26% for each of the other two styles, significantly above chance. The paper cautions that the test 'just passed our alpha cutoff (p = 0.044)' and that statistical power was limited.

- **Entry #89** *(Ch.3 Related Work > NPR for Virtual Humans)*
  [check wording] THESIS: "Canales et al. applied stylisation to custom assets in an open pipeline."
  SOURCE (§4.1 AVATARS, P. 4): "The two stylized versions of each identity were hand modeled by a team of artists. The scanned high realism models and photos of each person were provided to the art team. ... The art team also provided albedo, normal, and roughness maps for each stylized avatar, which were applied in Unreal using the default material with tuned parameters."
  CHECK: The 'custom assets' half is fully supported: geometry was hand sculpted, wrapped in Wrap3D, retuned in Maya, and custom albedo/normal/roughness maps were authored and applied in Unreal. However, the paper never characterises its pipeline as 'open', and the stylisation was achieved through sculpting and texturing inside the MetaHuman/Unreal toolchain rather than through shader-level access, so 'open pipeline' is the thesis's own framing, not a claim in the source.

- **Entry #91** *(Ch.3 Related Work > Research Gap)*
  [check wording] THESIS: "Second, stylised avatars have not been found to carry a trust penalty relative to photorealistic avatars in controlled evaluations."
  SOURCE (§9 DISCUSSION, P. 8): "With regard to Style, we do not see evidence that people are more likely to trust more realistic avatars nor less likely to trust our most stylized caricatures. This runs counter to some previous work that suggests a preference for realistic avatars [18, 25], but is consistent with other work that found no such preference [5, 29]"
  CHECK: Canales et al. found no trust penalty for stylisation in their own study, which supports the gist. However the thesis sentence generalises to 'controlled evaluations' in the plural, and this paper explicitly states its result 'runs counter to some previous work that suggests a preference for realistic avatars' (Jo et al. [18], McDonnell et al. [25]). A single-study citation therefore over-generalises; the sentence should either be hedged or carry additional citations.

- **Entry #92** *(Ch.3 Related Work > Research Gap)*
  THESIS: "It compared coarse style categories rather than a specific NPR shader pipeline and did not measure eeriness as a dimension separate from overall stylisation."
  SOURCE (§4.1 AVATARS, P. 4; §5.3 PROCEDURE, P. 5): "The two stylized versions of each identity were hand modeled by a team of artists. ... they created concept art for a caricatured version of each person along with a semi-realistic model intended to lie between the caricature and realistic model. ... [§5.3, p. 5] Prompts were customized from [23] and designed to

measure Ability, Benevolence and Integrity, the three constituents of the trust model used in this study, and then a composite measure of Trust."
CHECK: Both halves hold. Stylisation was a coarse three-level artist-authored sculpt and texture manipulation, not a shader pipeline. The post-doctor survey instrument (Supplemental §11.1) contains only Ability, Benevolence, Integrity and Trust items, with no eeriness or uncanniness scale; eeriness appears only as unprompted remarks in the structured interviews ('only four describing them as uncanny or creepy', §7, p. 7).

- **Entry #121** *(Ch.4 Methodology > User Study Design > Objective and Hypotheses)*
  THESIS: "Prior work varied stylisation level within a fixed advisor role and scenario and found no significant effect on self-reported trust."
  SOURCE (§10 CONCLUSION, P. 9): "This study used a high stakes medical scenario in order to explore how people's tendency to trust was impacted by avatar stylization. Stylization had no significant impact on trust ratings. The Realistic avatars received lower ratings for benevolence on average; however, the differences were not significant."
  CHECK: The advisor role (doctor) and scenario (kidney lump, second opinion) were fixed for every participant while style varied, and no significant effect on self-reported trust was found. One caveat worth knowing: §6.1 (p. 6) reports that Benevolence was 'the only category that show a significant main effect for Style (chi2(2) = 6.78, p = .034)', although post-hoc tests found no significant style pairs. Avatar identity was also varied simultaneously in the same design.

- **Entry #126** *(Ch.4 Methodology > User Study Design > Conditions)*
  THESIS: "Canales et al. crossed avatar stylisation level with speaker identity in their study and each of the three advisors was a different person delivering different dialogue."
  SOURCE (§5.1 DESIGN, P. 5; §1 INTRODUCTION, P. 1; §3.2 DIALOGUE TESTING, P. 3): "the full experiment uses a 3 (avatar identity) x 3 (avatar style) experimental design, but each participant sees only three avatars, featuring one of each identity and each style ... [§1, p. 1] Each doctor is a different person (varied identity) and each is represented at a different level of stylization ... [§3.2, p. 3] the avatar's voice must match the avatar, which means that a different actor must perform each avatar's dialogue. During the scenario, each avatar must also provide different answers to ensure believability."
  CHECK: The crossed 3x3 identity by style design, the three distinct source persons/actors, and the distinct dialogue per advisor are all stated explicitly. The paper notes the dialogues were pretested to be statistically equivalent on ability, benevolence and integrity, so 'different dialogue' means different text, not different trust content.

- **Entry #131** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "P1 was the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al.."
  SOURCE (§5.3 PROCEDURE, P. 5): "After the survey, the next doctor appeared, and the process was repeated twice more so that all three doctors were interviewed and rated. Participants then selected a single doctor for a second opinion, and also selected the doctor they would least prefer."
  CHECK: The post-study forced-choice advisor selection paradigm is exactly as described: after rating all three advisors, participants chose one for the second opinion. The paper additionally collected a least-preferred choice, which the thesis's P1 adaptation apparently

does not.

- **Entry #141** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Comparable work has inferred an absence of stylisation effects on trust from non-significant results alone."
  SOURCE (§10 CONCLUSION, P. 9): "Stylization had no significant impact on trust ratings. The Realistic avatars received lower ratings for benevolence on average; however, the differences were not significant. This suggests that realistic avatars are not required and it may be possible to use a range of avatar visual styles for applications where trust is critical."
  CHECK: The paper draws a positive design conclusion ('realistic avatars are not required') directly from non-significant results. No equivalence test, TOST procedure, or Bayesian null-evidence analysis appears anywhere in the paper, so the inference does rest on non-significance alone.

- **Entry #149** *(Ch.6 Evaluation > User Study Results > Post-Study Rankings)*
  THESIS: "Canales et al. found a divergence between explicit and behavioural measures in their comparable study."
  SOURCE (§10 CONCLUSION, P. 9; §9 DISCUSSION, P. 9): "People were more likely to select the Mid stylized doctor, despite this preference not being reflected in the subjective trust surveys. ... [§9, p. 9] It is also important to note that this style preference was not reflected in the trust survey ratings."
  CHECK: The dissociation between the explicit self-report trust ratings (no style effect) and the behavioural forced-choice selection (Mid preferred) is stated in both the Discussion and the Conclusion.

- **Entry #150** *(Ch.6 Evaluation > User Study Results > Secondary Measures)*
  THESIS: "Canales et al. similarly found no significant relationship between propensity and trust ratings in their comparable manipulation."
  SOURCE (§6 RESULTS, P. 5): "There was also no significant correlation between people's propensity to trust and the average of their actual trust survey ratings as described in Sec.6.1 (Pearson's R of 0.062, p=.62), so it appears that personal trust propensity was not an important factor in the trust results."
  CHECK: Directly stated. The paper also reports no significant difference in propensity across the nine assigned avatars (p=.52), confirming propensity did not confound the style manipulation.

Alipour, Hartmann & Alimardani (2025) — Systematic Review

Full reference: Alipour, A., Hartmann, T., & Alimardani, M. (2025). Would you rely on an eerie agent? A systematic review of the impact of the Uncanny Valley Effect on trust in human-agent interaction. *arXiv preprint arXiv:2505.05543*.

DOI / arXiv: `arXiv:2505.05543` https://arxiv.org/abs/2505.05543

Core question. How does the Uncanny Valley Effect (UVE) impact human trust in agents? This is the first systematic review to map the intersection of UVE research and trust across the empirical literature as a whole.

Method. PRISMA 2020 systematic review. The search covered seven databases (Scopus, Web of Science, PubMed, APA PsycInfo, IEEE Xplore, ProQuest, WorldCat) using the string (*"uncanny valley" OR "uncanny effect"*) AND (*trust* OR believab* OR persuas* OR acceptab* OR acceptance OR competen* OR credib**). The initial pool of 641 records was reduced to 311 after removing duplicates and excluded formats. After two-stage screening (title/abstract then full text), 53 studies were retained. Inter-rater reliability was Fleiss' Kappa $= 0.61$, indicating substantial agreement. Data were extracted using predefined codes covering agent type, modality, study design, UVE measures, trust measures, and key findings.

Key observation on interaction type. 42 out of 53 studies (79%) used indirect encounters: participants passively viewed static images, videos, or transcripts rather than engaging in live interaction with an agent. Agent types spanned robots ($n = 25$), virtual agents ($n = 21$), chatbots ($n = 2$), and voice-only systems ($n = 2$).

Novel trust measurement framework. The paper's central theoretical contribution is a three-category taxonomy of how trust has been assessed across the literature:

Category	Studies	What it captures
Direct trust measures	24	Explicit self-reported trust or trustworthiness ratings
Trust-related constructs	11	Competence, acceptability, credibility, believability, persuasion
Trust-related intentions	18	Behavioural intentions: advice following, willingness to interact, purchase decisions

Of the 53 studies, 46 relied exclusively on subjective self-report. Only 7 used objective behavioural measures such as trust games or decision tasks.

UVE measurement. The best-practice measures for capturing the UVE are eeriness ($n = 38$ studies) and creepiness ($n = 4$ studies), which showed the largest effect sizes in the meta-analysis by Diel et al. (2021). Broader affective responses such as discomfort, threat, disgust, or anxiety were used in 12 studies. Positive affinity measures (familiarity, likability) appeared in 3 studies.

Main findings. Among the 41 studies that empirically tested the UVE–trust relationship:

- **17 studies reported a direct effect.** In every case, the direction was the same: UVE diminished trust or trust-related outcomes. The most rigorous example is Mathur and Reichling (2016), who ran a monetary investment game with robot face stimuli and showed that trust followed the same valley-shaped curve as liking: robots falling into the uncanny valley received less money from participants.
- **24 studies (more than half) reported a conditional or indirect effect.** The UVE–trust relationship was moderated or mediated by additional factors. Prominent moderators included: context of use and task type ($n = 6$; e.g., the trust cost of uncanniness was weaker in high-expertise tutoring roles than in low-expertise hotel service roles); age ($n = 3$; Tu et al. found that younger and middle-aged adults showed a trust dip for near-human agents while older adults did not); familiarity with the agent ($n = 3$); and personality traits and individual neural differences ($n = 3$).
- **10 studies found no effect.** The UVE was successfully induced but did not translate into reduced trust. Kim (2022) confirmed that hyper-realistic avatars were rated significantly

higher on uncanniness and freakiness than cartoonish avatars, yet trustworthiness scores were statistically equivalent across both types.

Competence as a buffer. A cross-cutting finding across several no-effect and moderation studies is that perceived competence can counteract the negative influence of eeriness on trust. MacDorman (2019) found that eeriness did not reduce persuasion when the computer-animated agent was perceived as competent. Dai and MacDorman (2018, 2021) reported that warmth and competence predicted advice adherence independently of the agent's realism level. Patel and MacDorman (2015) found that advice quality exerted a stronger influence on compliance than the agent's appearance or rated trustworthiness.

Three categories of moderating factors. The review synthesises all moderation findings into three groups:

- **User-related:** age, personality traits, individual neural response differences, initial impressions, familiarity and satisfaction with the agent, disconfirmed expectations, perceived identity threat, user self-justification, prior interpersonal trust levels.
- **Agent-related:** design mismatches (e.g., realistic visual appearance with artificial voice), competence, warmth, autonomy, emotional expressiveness, curated imperfections, overall trustworthiness signal.
- **Environment-related:** context of use, task type (social vs. mechanical vs. empathy-requiring), interaction frequency and repetition.

Limitations of the existing literature (Table 4 of the paper).

Validity	Issue	Studies
External	Indirect encounters only; no live interaction	41
External	No sample size justification via power analysis	46
External	Narrow or specialist sample	7
Internal	Limited range of stimuli (often only 2 levels)	19
Internal	Uncontrolled confounding variables	23
Internal	Failed or weak manipulation checks	5
Construct	Measurement robustness issues (single-item scales)	20
Construct	Sole reliance on subjective self-report	46

Recommendations for future research (Table 5 of the paper).

1. Move toward dynamic, real-time interaction paradigms; add longitudinal designs to track trust formation across multiple sessions.
2. Combine self-report surveys with objective behavioural measures: trust games, forced-choice tasks, physiological signals (HRV, EEG, pupil size).
3. Manipulate human-likeness across at least 3–5 levels to reveal the non-linear UVE pattern; measure both eeriness and positive responses (familiarity, likability).
4. Justify sample sizes through power analysis; ensure diversity across gender, age, and

cultural background.

5. Model the relationship structurally: treat UVE as predictor and trust as outcome, with candidate mediators including emotional discomfort, perceived competence, warmth, anthropomorphism, and threat perception.

Relevance to this study.

- **Validates the video-based design:** 42 out of 53 reviewed studies used indirect exposure. Video stimuli sit squarely within the established empirical norm and are explicitly classified as an appropriate stimulus type by the authors.
- **Validates combining self-report with a behavioural indicator:** the paper explicitly identifies reliance on self-report alone as a major limitation and recommends pairing surveys with behavioural measures. Part P1 (forced-choice advisor selection) implements exactly this recommendation.
- **Validates eeriness as the primary UVE instrument (Part B):** eeriness is identified as the best-practice measure with the largest effect sizes across the prior meta-analysis (Diel et al., 2021), used in 38 of the 53 reviewed studies.
- **Validates the dual positive/negative measurement approach (Parts B and C):** the paper recommends measuring both negative (eeriness) and positive (likeability, familiarity) dimensions; this is implemented via the Ho and MacDorman eeriness subscale (Part B) and the Godspeed Likeability subscale (Part C).
- **Grounds hypothesis H4** (eeriness negatively predicts trust across conditions): the majority of direct-effect studies confirm this direction, supporting H4 as a correlational prediction.
- **Explains the plausibility of H2** (no significant trust difference between C2 and C3): the 10 no-effect studies and the competence-as-buffer literature both support the expectation that equivalent advice quality can produce equivalent trust across rendering styles, even where eeriness differs.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #50** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  THESIS: "Together these span all three measurement approaches in the Alipour et al. taxonomy."
  SOURCE (SECTION 4.3.2 TRUST MEASUREMENT, P. 9 (SEE ALSO SECTION 4.4.2, P. 11: 'THE THREE TYPES OF TRUST MEASUREMENT APPROACHES THAT WE PREVIOUSLY IDENTIFIED')):
  "Studies were categorized into three groups based on the type of trust measurement that they employed (see Table 3). The first group (n = 24) consisted of studies that directly and explicitly measured trust or trustworthiness empirically (e.g., based on self-report measures) ... The second group (n = 11) included studies that assessed trust-related constructs, such as competence, acceptability, credibility, believability, and persuasiveness. The third group (n = 18) contained studies that assessed trust-related intentions, such as adherence to an agent's advice, willingness to interact, or conformity to an agent's recommendation in a decision-making task (e.g., purchase or investment decisions)."
  CHECK: The paper does define exactly three trust measurement approaches (direct trust/trustworthiness, trust-related constructs, trust-related intentions), so the taxonomy referred to is correctly described. The mapping of the Canales measures onto the three groups is the thesis's own

inference, but it is a reasonable one: competence is explicitly listed as a trust-related construct, and a follow-up consultation choice matches 'adherence to an agent's advice, willingness to interact, or conformity to an agent's recommendation in a decision-making task'.

- **Entry #123** *(Ch.4 Methodology > User Study Design > Objective and Hypotheses)*
  [check wording] THESIS: "Conditional or indirect effects were more common at 24 and 10 studies found none."
  SOURCE (SECTION 4.4.1 GENERAL FINDINGS, PP. 10-11): "A total of 41 studies provided empirical evidence on the relationship between the UVE and trust or trust-related constructs/intentions. ... Among the 41 coded studies, 19 were categorized as showing a direct effect. ... We coded 24 studies as showing a conditional/indirect effect. ... Among the 41 studies that empirically tested the relationship between the UVE and trust or trust-related constructs/intentions, 10 were coded as showing no effect."
  CHECK: Three of the four numbers are exact (41 empirically testing, 24 conditional/indirect, 10 no effect), but the thesis says 17 studies reported a direct effect whereas the paper states 19. The paper's own subgroup counts confirm 19 (11 in the direct-trust group, p. 12; 3 in the trust-related-constructs group, p. 13; 5 in the trust-related-intentions group, p. 14). Change 17 to 19. The claim that conditional or indirect effects were more common still holds (24 > 19). Note also that these categories overlap: 12 studies were dual coded under both direct and conditional/indirect effects (Fig. 8 caption, p. 11), so the counts do not partition the 41.

- **Entry #132** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "It captured implicit trust judgment that explicit Likert ratings may not fully reflect and this addresses the methodological gap identified by Alipour et al.."
  SOURCE (SECTION 5.2 METHODOLOGICAL LIMITATIONS OF THE EXISTING LITERATURE, P. 16; AND SECTION 5.3 RESEARCH GAPS AND RECOMMENDATIONS FOR FUTURE RESEARCH, P. 17 (ALSO TABLE 4, GAP: 'OVERRELIANCE ON SELF-REPORT MEASURES', RECOMMENDATION: 'COMBINE SUBJECTIVE SURVEYS WITH OBJECTIVE BEHAVIORAL OUTCOMES')): "Finally, almost all studies relied heavily on self-report measures, which are vulnerable to biases such as social desirability and participants' misinterpretation of constructs. Moreover, self-report measures may fail to capture automatic or unconscious reactions, which are especially relevant in HAI research. For example, participants may consciously report trust while unconsciously exhibiting hesitation or discomfort. These points highlight the need for integrating behavioral or physiological measures in future research to more reliably capture UVE-related responses. ... We also recommend complementing single-time subjective self-report measures with additional methodological approaches, such as implicit trust tasks, decision-making scenarios or trust games, behavioral indicators (e.g., advice following, task delegation), and physiological signals"
  CHECK: The paper explicitly names overreliance on self-report as a methodological gap, states that self-report may miss automatic or unconscious reactions, and recommends implicit trust tasks, decision-making scenarios and behavioural indicators such as advice following. A forced-choice advisor selection is the kind of measure the review counts as objective: it classifies a choice-task decision preference among its seven objective trust measures (Section 4.3.2, p. 9).

- **Entry #148** *(Ch.6 Evaluation > User Study Results > Hypothesis Tests)*
  THESIS: "This result matches the direct negative relationship between eeriness and trust that

Alipour et al. report in about half of the studies examining the association."
SOURCE (SECTION 5.1 DOES THE UVE AFFECT TRUST?, P. 15; SECOND SENTENCE FROM SECTION 4.4.2.1 STUDIES DIRECTLY MEASURING TRUST, P. 12): "Overall, the findings suggest that while the UVE affects trust, this relationship is often not straightforward. Only about half of the studies that empirically tested the impact of the UVE on trust or trust-related constructs or intentions reported a direct effect. ... Of these, 11 provided a direct effect of the impact of the UVE on trust. All of these studies reported that the UVE was induced, and that the elicited negative affinity response directly diminished trust."
CHECK: 'About half' matches the paper's own wording and its counts (19 of 41 empirically testing studies, i.e. 46 percent). The negative direction is explicit for the studies that measured trust directly, where the elicited negative affinity response is said to have diminished trust. One nuance: the source writes 'Only about half', framing the figure as a limitation on the evidence, and codes the effect as UVE to trust rather than eeriness to trust specifically, although eeriness or creepiness is the UVE measure in 38 of the 53 studies (Table 2, p. 9).

Nowak & Biocca (2003) — Agency, Anthropomorphism, and Social Presence

Full reference: Nowak, K. L. & Biocca, F. (2003). The effect of the agency and anthropomorphism on users' sense of telepresence, copresence, and social presence in virtual environments. *Presence: Teleoperators and Virtual Environments*, 12(5), 481–494.

DOI: 10.1162/105474603322761289

Key distinctions. The paper operationalises three presence dimensions as independent dependent variables. *Telepresence* is the sense of being in the virtual environment itself. *Copresence* is the sense of sharing that space with a specific other person (assessed from both the participant's own perspective and as an attributed property of the partner). *Social presence* is a medium-level property: the perceived capability of the communication channel to convey social cues, irrespective of the specific partner.

Study design. A 2×3 between-subjects experiment with 134 undergraduates. The two independent variables were: (1) *agency* — whether the interaction partner was described as a human-controlled avatar or a computer-controlled agent; and (2) *visual anthropomorphism* — high-anthropomorphic (a detailed 3D human face), low-anthropomorphic (a simplified cartoon face with exaggerated eyes), or no image (text-only interaction). Participants spent 15 minutes in a virtual meeting room with the confederate partner, ostensibly preparing for a scavenger hunt.

Scales. Telepresence: 8 items, $\alpha = .88$. Perceived other copresence: 15 items, $\alpha = .90$. Self-reported copresence: 11 items, $\alpha = .78$. Social presence: 9 items on sliding scales.

Results.

- **Agency had no significant effect** on any presence measure (telepresence, copresence, or social presence). Whether participants believed their partner was human or a bot did not change how connected or present they felt. This challenges the assumption that knowing an agent is human is a prerequisite for meaningful social engagement.
- **Presence of any virtual image** (vs. no image) marginally increased telepresence ($p = .12$). Having a visual representation of a partner, even a minimal one, helps participants feel situated in the virtual space.

- **Counterintuitive finding:** low-anthropomorphic images produced significantly *more* copresence ($F = 6.52$, $p = .01$) and social presence ($F = 4.90$, $p = .03$) than high-anthropomorphic images or no image. The high-anthropomorphic condition produced the *least* copresence among the image conditions.

Explanation offered. High-anthropomorphic visual representations establish expectations of rich, human-level social responsiveness. When those expectations are not met by the limited behaviour of the scripted virtual confederate, participants experience a gap between expectation and reality, which reduces their sense of connection. Low-anthropomorphic representations do not trigger such high expectations, so the same limited interaction produces a satisfying social experience. This expectation-mismatch mechanism is closely related to the uncanny valley: a highly realistic avatar that fails to meet human-level behavioural expectations is penalised more severely than a less realistic one.

Relevance to this thesis. Nowak and Biocca's counterintuitive finding — that lower anthropomorphism can produce higher social presence — provides theoretical support for the prediction that NPR stylisation (C3) need not reduce trust or social engagement relative to the photorealistic avatar (C2). If the realistic C2 avatar establishes higher perceptual expectations that the recorded video cannot fully satisfy, the stylised C3 may perform comparably or better. The paper also operationalises copresence and social presence in ways relevant to interpreting trust ratings across conditions.

Nowak & Rauh (2005) — Avatar Appearance and Online Social Perceptions

Full reference: Nowak, K. L. & Rauh, C. (2005). The influence of the avatar on online perceptions of anthropomorphism, androgyny, credibility, homophily, and attraction. *Journal of Computer-Mediated Communication*, 11(1).

DOI: `10.1111/j.1083-6101.2005.tb00291.x`

Research question. How do avatar appearance choices affect how an online interaction partner is perceived in terms of anthropomorphism, credibility, homophily (perceived similarity), and social attraction?

Method. Participants ($n = 177$) were shown 30 different avatar images that varied systematically in human likeness (from abstract geometric icons to photorealistic faces) and gender cues (masculine, feminine, androgynous). Images were presented as representations of potential online interaction partners. Participants completed multi-item validated scales for each dependent variable.

Key findings.

- Avatar human-likeness significantly predicted anthropomorphism ratings: more human-like avatars were perceived as having more human-like thoughts and motivations.
- The relationship between human-likeness and credibility was not linear. Highly anthropomorphic but androgynous avatars received the most favourable credibility, homophily, and attraction ratings.
- Androgynous avatars (those without strong gender cues) received higher credibility ratings than strongly gendered avatars, suggesting that demographic ambiguity reduces the

opportunity for stereotype-based discounting of the partner's credibility.

- Participants showed greater social attraction to avatars perceived as most similar to themselves (homophily effect), with similarity assessed both in gender terms and in perceived humanness.
- More abstract avatars did not automatically produce lower social perceptions: the relationship depended on the specific perceptual dimension, confirming that avatar appearance effects are multidimensional and not reducible to a simple "more human-like is better" principle.

Relevance to this thesis. Nowak and Rauh establish that avatar appearance choices reliably affect how an online partner is socially perceived across dimensions closely related to trust: credibility, attraction, and perceived similarity. The finding that social perception is sensitive to anthropomorphism level in a nonlinear way supports the expectation that NPR stylisation will affect perceived credibility of the advisory avatar, while not necessarily in a straightforwardly negative direction.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #48** *(Ch.3 Related Work > Trust and Credibility in Virtual Agents)*
  [check wording] THESIS: "Avatar design choices therefore function as meaningful social signals that shape participants' willingness to engage and trust."
  SOURCE (CONCLUSION, PRINTED PAGE 174): "Avatars that were more anthropomorphic were perceived to be more attractive and credible, and people were more likely to choose to be represented by them. The strongest predictor of these variables, however, was the degree of masculinity or femininity (lack of androgyny) of an avatar."
  CHECK: The "meaningful social signals" half is well supported (Discussion, p. 171: "the avatar's characteristics were more influential than the users' individual difference variables"), but the study measured credibility, not trust: credibility is defined as "the degree to which a person is perceived as believable, trustworthy, and competent" and is described as "one of the primary predictors for certain interaction outcomes such as attitude change and trust" (p. 158), i.e. a predictor of trust rather than trust itself, operationalised by the items "intelligent," "informed," and "reliable." "Willingness to engage" also overstates the behavioural measure, which asked "how likely the participants would be to choose that image to represent them" (self-representation), and the Limitations section (p. 173) concedes that "we did not examine how, or if, people's responses to these images would actually influence their responses to a person who chose to be represented by a particular avatar."

Weidner et al. (2023) — Systematic Review of Avatars in AR and VR

Full reference: Weidner, F., Boettcher, G., Arevalo Arboleda, S., Diao, C., Sinani, L., Kunert, C., Gerhardt, C., Broll, W., & Raake, A. (2023). A systematic review on the visualization of avatars and agents in AR & VR displayed using head-mounted displays. *IEEE Transactions on Visualization and Computer Graphics*, 29.

DOI: 10.1109/TVCG.2023.3247072

Scope. PRISMA systematic review of 72 empirical papers published between 2015 and 2022, studying avatars or agents rendered in AR or VR using HMDs. Sources: ACM Digital Library,

IEEE Xplore, SpringerLink, Frontiers, and MIT Press. Inclusion criteria required: HMD use during the study, at least two conditions comparing different avatar visualisation choices, and empirical human participant data. 233 papers screened; 72 included.

Rendering style taxonomy. The 72 included papers used the following style categories (with overlap): abstract ($n = 21$), stickman ($n = 10$), cartoon ($n = 16$), stylized ($n = 20$), robot ($n = 17$), generic realistic ($n = 56$), personalized realistic ($n = 10$), scanned realistic ($n = 3$), video ($n = 5$), point cloud ($n = 6$), translucent ($n = 2$), and hybrid ($n = 3$). The majority used full-body visualization ($n = 50$, 69.4%).

Task domain taxonomy. Five task domains: (1) physical activity (walking and non-walking); (2) hand interaction (direct and indirect manipulation); (3) communication (dynamic bidirectional and static unidirectional); (4) game-like scenarios; (5) education and training.

Key high-level findings.

- Full-body visualization consistently outperforms partial-body configurations (head-only, head-and-hands) across nearly all measures: embodiment, task performance, presence, and social presence.
- Generic realistic avatars significantly outperform other styles for bidirectional communication tasks. Realistic appearance supports richer nonverbal communication through gaze, facial expression, and gesture fidelity.
- For game-like and educational tasks, visual fidelity contributes far less: abstract or cartoon avatars perform comparably to realistic ones in task performance and learning outcomes.
- Personalized realistic and scanned realistic avatars tend to produce the strongest effects on embodiment and body ownership, though at higher technical cost.
- Facial animation and lip synchronisation are under-studied and under-reported: only 22 papers manipulated or controlled for animation quality, despite its known influence on the uncanny valley.
- AR and VR findings transfer reasonably well to each other: 47 papers tested in VR only, 12 in AR only, and 13 in both; the basic style effects were broadly consistent across display type.

Practical guidelines. The review synthesises seven practitioner guidelines: (1) use generic realistic full-body as a default for communication-oriented applications; (2) give users agency over visible body parts; (3) prioritise realistic styles where interpersonal trust, presence, or social norm adherence matter; (4) for engaging game content or easy training tasks, realistic rendering is not essential; (5) realistic hands improve hand interaction performance; (6) limit visual complexity in learning contexts to avoid distraction; (7) personalized realistic and video avatars can outperform generic realistic when individual identity matters.

Relevance to this thesis. This review provides the widest-scope summary of how rendering style affects avatar-mediated behaviour in immersive environments. Its finding that rendering style significantly affects social perception and communication behaviour across dozens of independent studies corroborates the central motivation for this thesis's user study: that NPR stylisation is a meaningful independent variable with measurable perceptual effects. The review also confirms that avatar visualization research has focused primarily on first-person self-avatars and physical interaction tasks; the present study extends this to third-person advisory

contexts and trust formation from scripted video stimuli, filling a gap Weidner et al. identify as understudied.

Kuffner dos Anjos & Pereira (2024) — Realism and Self-Embodied Avatars

Full reference: Kuffner dos Anjos, R. & Madeiras Pereira, J. (2024). Effects of realism and representation on self-embodied avatars in immersive virtual environments. *arXiv preprint*, arXiv:2405.02672.

Preprint: `arXiv:2405.02672`

Research question. How do avatar realism level and camera perspective (first-person vs. third-person) interact to affect sense of embodiment, task performance, and social presence in VR navigation tasks?

Study design. 24 participants performed four navigation tasks (obstacle avoidance by walking, by jumping, by crouching, and object catching) in a 4m×4m physical room tracked by Kinect depth sensors. Three avatar representations were compared: (1) *Abstract* — spheres for joints, cylinders for limbs, no texture; (2) *Mesh* — a realistic 3D humanoid mesh driven by Kinect skeleton tracking; (3) *Point Cloud* — a real-time depth-sensor reconstruction of the participant's actual body. Each was presented in both first-person perspective (1PP) and third-person perspective (3PP), giving 12 conditions, administered in counterbalanced order via Latin Square.

Embodiment questionnaire. Participants rated four embodiment sub-components after each condition: sense of agency (motor control), body ownership (the virtual body felt like their own), self-location (perceived position matched the avatar), and body control (general coordination sense). All items on a 6-point Likert scale.

Key results.

- **Perspective outweighs representation.** 1PP produced faster task completion times than 3PP for all three avatar types in most tasks. The clearest effect was on the walking task: 1PP with Mesh was significantly faster than all 3PP conditions ($Z = -3.352$, $p = .001$).
- **Mesh in 1PP showed the strongest embodiment:** agency ($Z = -2.687$, $p = .007$), body ownership ($Z = -2.775$, $p = .006$), and self-location ($Z = -3.574$, $p < .001$) were all significantly higher than Abstract or Point Cloud in 1PP.
- **Mesh in 3PP was the worst performing condition** on the obstacle-crouching task ($Z = -3.254$, $p = .001$). The authors attribute this to the uncanny valley: the realistic appearance sets higher behavioural expectations that the imperfect Kinect animation fails to meet when viewed from outside the body.
- **Abstract in 3PP produced higher body ownership** than Mesh in 3PP, and users showed greater spatial confidence (fewer errors in distance estimation) with the abstract avatar.
- **Point Cloud in 3PP showed the highest agency score** ($Z = -2.812$, $p < .01$) but caused visual occlusion problems and occasional rendering artefacts.

Design guidelines.

- Avoid mesh avatars in third-person perspective: the uncanny valley is amplified in 3PP

because the user can observe the avatar's full body motion.

- Use Abstract or Point Cloud representations for third-person spatial navigation tasks.
- 1PP with Mesh is the optimal choice for embodiment-critical applications.

Relevance to this thesis. This paper extends uncanny valley research to the self-embodiment domain, demonstrating that imperfect animation on a realistic mesh amplifies negative perception when the avatar is observed from outside the body. This is directly analogous to why the photorealistic C2 advisory avatar, observed by participants as a video, may produce higher eeriness than the stylised C3: subtle imperfections in facial animation or expression timing are perceptually more costly against a photorealistic appearance than against a stylised one.

Alimardani, de Roode, Vaitonytė & Louwerse (2024) — Appearance, Voice, and Trust in Virtual Agent Collaboration

Full reference: Alimardani, M., de Roode, R., Vaitonytė, J., & Louwerse, M. M. (2024). Effect of a virtual agent's appearance and voice on uncanny valley and trust in human-agent collaboration. In *Proceedings of the 24th ACM International Conference on Intelligent Virtual Agents (IVA '24)*, Article 5. ACM.

DOI: 10.1145/3652988.3673970

Research question. Does a mismatch between a virtual agent's visual appearance (humanlike vs. robotic) and voice (humanlike vs. synthesised robotic) affect uncanny valley perception, anthropomorphism attribution, and trusting behaviour in a collaborative task?

Study design. A 2 × 2 between-subjects factorial experiment. Independent variables: (1) *agent appearance* — humanlike (photorealistic 3D virtual human face) or robotic (clearly mechanical robot face); (2) *agent voice* — humanlike (natural sounding synthesis) or robotic (clearly synthetic speech). Four resulting conditions: Humanlike/Humanlike (matched), Humanlike/Robotic (mismatch), Robotic/Humanlike (mismatch), Robotic/Robotic (matched). $n = 80$ participants (20 per condition) performed an emotion recognition task in which the virtual agent presented images and offered recommendations. After each trial the participant could accept or override the agent's recommendation, so trusting behaviour was operationalised as the proportion of agent recommendations adopted across trials.

Dependent measures. Anthropomorphism (Godspeed Anthropomorphism subscale), uncanny valley perception (6-item eeriness scale from Ho & MacDorman, 2010), and trusting behaviour (advice-following rate).

Key results.

- **Anthropomorphism:** A significant appearance × voice interaction. The mismatch Robotic/Humanlike condition elevated anthropomorphism ratings above the matched Robotic/Robotic condition — the human voice "upgraded" perceived humanness beyond what the robotic appearance alone suggested.
- **Eeriness:** The Humanlike/Robotic mismatch (realistic face + robotic voice) produced the highest eeriness ratings, significantly above all matched conditions. Mismatching modalities amplified the uncanny valley response.

- **Trusting behaviour: No significant main or interaction effect** of appearance or voice on advice-following rate. All four conditions produced statistically equivalent trust behaviour. This is the critical null result.

Interpretation. A clean dissociation: manipulating the agent's appearance and voice changes anthropomorphism and eeriness perception, but does not change whether participants follow its advice. The authors interpret this through the Mayer et al. (1995) ABI lens: when the agent's demonstrated competence (consistent and reliable recommendations) is held constant, perceptual shifts in appearance do not alter the trust placed in that competence.

Relevance to this thesis. This paper provides the strongest single-study support for H2 (NPR stylisation does not significantly reduce trust). The design isolates visual appearance while holding task competence constant, which mirrors the C2 vs. C3 comparison where both conditions deliver identical scripted advice. If even a dramatic mismatch combining a humanlike face with a robotic voice does not reduce trusting behaviour, a shift from photorealistic to NPR-stylised rendering of the same advisor is similarly unlikely to produce a significant trust reduction.

Dubosc, Gorisse, Lefrou, Richir & Christmann (2025) — Avatar Stylisation and Facial Expression Intensity

Full reference: Dubosc, C., Gorisse, G., Lefrou, T., Richir, S., & Christmann, O. (2025). Effect of avatar stylization and facial expression intensity in virtual interactions. *Virtual Reality*, Springer Nature. Published online 8 October 2025.

DOI: `10.1007/s10055-025-01238-6`

Research question. Does the avatar's stylisation level affect co-presence, eeriness, and trustworthiness in a virtual face-to-face interaction? Does the intensity of facial expressions modulate these effects?

Study design. 40 participants in a VR face-to-face interaction. A professional actress performed scripted dialogue; her motion-capture performance was retargeted simultaneously onto three avatar versions: *stylised* (cartoon-like, simplified geometry, high colour saturation), *semi-realistic* (intermediate stylisation, visible skin texture), and *realistic* (photorealistic, subsurface scattering, full texture detail). Three expression intensity levels were applied: *reduced* (movements dampened to 50%), *unaltered* (original), and *augmented* (amplified to 150%). Stylisation was the within-subjects factor; expression intensity was between-subjects. Mixed 3×3 design.

Outcome measures. Co-presence (7-point scale adapted from Nowak & Biocca, 2003), eeriness (adapted from Ho & MacDorman, 2010), and trustworthiness (5-point rating).

Key results.

- **Co-presence:** Significant main effect of stylisation ($F = 7.83$, $p < .001$, $\eta_p^2 = .18$). Stylised produced significantly greater co-presence than realistic ($p = .002$). No effect of expression intensity. No interaction.
- **Eeriness:** Significant main effect of stylisation ($F = 11.24$, $p < .001$, $\eta_p^2 = .23$). Stylised and semi-realistic produced significantly lower eeriness than realistic ($p < .001$ and $p = .004$ respectively). No significant difference between stylised and semi-realistic. No expression intensity effect. No interaction.

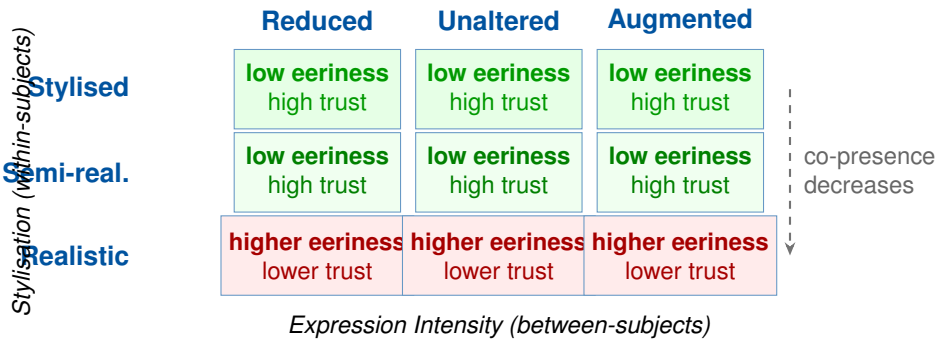


Figure 2: Experimental design of Dubosc et al. (2025). Stylisation is the within-subjects factor (rows); expression intensity is between-subjects (columns). The main effect of stylisation on eeriness and trustworthiness was significant across all expression intensity conditions. No interaction was found.

- **Trustworthiness:** Significant main effect of stylisation ($F = 6.94$, $p < .001$, $\eta_p^2 = .16$). Stylised and semi-realistic were rated significantly *more* trustworthy than realistic ($p = .001$ and $p = .009$ respectively). No interaction.

Interpretation. The photorealistic avatar triggers a subtle uncanny valley response even among participants who do not consciously label it as eerie, and this reduces social confidence in it. Stylised avatars sitting clearly outside the uncanny zone do not trigger this response and are consequently rated as more socially reliable. The absence of any interaction between stylisation and expression intensity confirms that the expression content of scripted content does not confound the stylisation effect.

Relevance to this thesis. This is the most directly relevant new paper to the current study. Both the C2 vs. C3 comparison and the Dubosc et al. realistic vs. stylised comparison manipulate rendering style while holding performance content constant. The finding that stylised avatars produce lower eeriness *and* higher trustworthiness than photorealistic avatars is strong prior evidence that H1 will be supported (C3 lower eeriness than C2) and that H2 may in fact yield null or even a positive direction. The no-interaction result further validates the assumption that holding the adviser script constant across conditions is sufficient to isolate the rendering effect.

Tao, Liu, Qiu & Li (2025) — Network Meta-Analysis of Avatar Realism and Perceptual Experience

Full reference: Tao, Z., Liu, Y., Qiu, J., & Li, S. (2025). Impact of virtual avatar appearance realism on perceptual interaction experience: a network meta-analysis. *Frontiers in Psychology*, 16, Article 1624975. Published 3 December 2025.

DOI: `10.3389/fpsyg.2025.1624975`

Research question. What is the quantitative effect of virtual avatar appearance realism on three perceptual outcomes — attractiveness, trustworthiness, and eeriness — as synthesised across the empirical literature?

Method. Network meta-analysis (NMA). A standard meta-analysis pools studies comparing the same two conditions. NMA extends this using indirect comparisons: if Study A compares

Low vs. Medium and Study B compares Medium vs. High, NMA can estimate the Low vs. High effect without a study that directly compares them, enabling simultaneous ranking across all levels. Search period: April 2015 to April 2025, four databases. Inclusion: studies that manipulated avatar appearance realism as an independent variable and measured at least one target outcome. **13 studies with 2,343 participants included.** Realism was coded into three levels: *low* (abstract, cartoon, stylised), *medium* (semi-realistic, realistic-but-not-photorealistic), and *high* (photorealistic, scanned).

Key findings at a glance.

Outcome	Low realism	Medium realism	High realism
Attractiveness	lowest	middle	**highest**
Trustworthiness	lowest	middle	**highest**
Eeriness	low	**peak (valley!)**	medium

Bold indicates the extreme value on that dimension.

- **Attractiveness:** High realism significantly more attractive than both medium and low (SUCRA: High = 83%, Medium = 46%, Low = 21%). Linear increase with realism.
- **Trustworthiness:** High realism significantly more trustworthy than both medium and low (SUCRA: High = 79%, Medium = 44%, Low = 27%). Medium and low not significantly different from each other.
- **Eeriness:** Medium realism produced the strongest eeriness, significantly higher than both high and low (SUCRA for high eeriness: Medium = 82%, Low = 19%, High = 49%). This is the quantitative confirmation of the uncanny valley: semi-realistic avatars are the most eerie, while both abstract and photorealistic avatars are less eerie.

Interpretation. The first NMA-level quantitative confirmation that the uncanny valley is specific to medium realism levels, not an artefact of individual study design. High-realism avatars achieve the second peak (lower eeriness than medium, highest attractiveness and trustworthiness). Low-realism avatars avoid the valley by remaining far from the ambiguous near-human zone.

Relevance to this thesis. The C2 avatar (Avaturn photorealistic) sits near the high-realism zone; the C3 avatar (NPR-stylised) moves toward the low-realism zone. The NMA predicts that this shift reduces eeriness because both high and low realism score lower than medium. However, the NMA also shows that trustworthiness follows realism monotonically: low-realism avatars score the lowest on trustworthiness. This provides the countervailing pressure motivating H2: the two hypotheses together test whether the eeriness reduction from C2 to C3 can be achieved without a significant trustworthiness cost that the meta-analysis would predict.

Cihodaru-Ştefanache & Podina (2025) — Uncanny Valley in Embodied Conversational Agents

Full reference: Cihodaru-Ştefanache, Ş. & Podina, I. R. (2025). The uncanny valley effect in embodied conversational agents: a critical systematic review of attractiveness, anthropomorphism, and uncanniness. *Frontiers in Psychology*, 16, Article 1625984.

DOI: `10.3389/fpsyg.2025.1625984`

Scope. PRISMA 2020 systematic review investigating how ECA (embodied conversational agent) design features and user characteristics interact to produce uncanny valley responses, with three target outcomes: attractiveness, anthropomorphism, and uncanniness. ECAs are defined as screen-based or VR virtual humans capable of dialogue and non-verbal communication.

Method. Search across ACM Digital Library, IEEE Xplore, Scopus, ProQuest, and Web of Science. Initial pool: 21,897 records. After deduplication, title/abstract screening, and full-text review: **29 studies included.** Methodological quality was assessed using a standardised checklist; most included studies were rated as weak to moderate quality.

ECA design factors identified as uncanny valley predictors.

- **Physical appearance:** skin texture realism, hair rendering quality, eye contact behaviour, and blink rate were the most commonly studied features. All showed significant effects on uncanniness when manipulated.
- **Non-verbal communication:** gesture fluency, lip synchronisation accuracy, and facial expression timing all affected the UVE. Asynchronies between speech and lip movement, or between emotional content and expression onset, consistently amplified eeriness.
- **Voice-appearance consistency:** the most consistent cross-study finding. Mismatches between visual appearance realism and voice quality amplify the UVE. An ECA with a highly realistic face paired with a synthetic voice, or vice versa, received higher uncanniness ratings than matched agents.
- **Stylisation:** studies comparing stylised versus realistic ECAs consistently found that stylised agents produced lower uncanniness scores. Deliberate low-realism stylisation is the most frequently recommended avoidance strategy.

User characteristics identified as moderators.

- Age: younger adults showed more pronounced UVE responses than older adults.
- Technology familiarity: more experience with virtual agents attenuated UVE responses.
- Individual disgust sensitivity predicted eeriness ratings independently of stimulus properties.

Checklist for Avoiding the UVE in ECAs (three-domain practical output of the review):

1. *Physical appearance:* use consistent realism across all visual features; avoid partial stylisation (e.g. realistic eyes with cartoon hair); consider deliberate low-realism stylisation when uncanny valley avoidance is prioritised over visual fidelity.
2. *Non-verbal communication:* ensure accurate lip synchronisation; match facial expression timing and intensity to speech prosody; maintain a natural blink rate (average 15–20 blinks per minute).

3. *Multimodal consistency:* match visual realism level to voice quality; avoid appearance-voice modality mismatches.

Relevance to this thesis. The recommendation to use consistent stylisation across all visual features validates the approach of applying a single NPR shader uniformly to the Avaturn avatar (C3) rather than selectively stylising only certain body regions. The voice-appearance consistency finding is also relevant to interpreting C2 vs. C3 differences: both conditions use the same recorded voice, meaning C2 (realistic appearance + human voice) is fully consistent while C3 (stylised appearance + human voice) introduces a moderate mismatch. This mismatch could amplify eeriness for C3 relative to what pure appearance-level stylisation would produce, which should be acknowledged as a limitation in the discussion of H1.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #153** *(Ch.6 Evaluation > Discussion > What the Stylisation Left Unchanged)*
  [check wording] THESIS: "Cihodaru-\{}c{S}tefanache and Podina report that the perceptual mismatch hypothesis has the strongest empirical support among competing explanations of the effect."
  SOURCE (SECTION 1 (INTRODUCTION), PRINTED P. 02 (PDF P. 2)): "Understanding UVE remains challenging due to the existence of multiple competing hypotheses that try to explain our perception of anthropomorphism in ECAs. These range from the morbidity and movement hypotheses to the category ambiguity (Cheetham and Jancke, 2013; Katsyri et al., 2015; Pollick, 2010). Among these, the perceptual mismatch hypothesis has received strong empirical support."
  CHECK: The authors do single the perceptual mismatch hypothesis out from the competing hypotheses, but they write that it 'has received strong empirical support', not the strongest; no ranking of the hypotheses by weight of evidence appears anywhere in the paper. The thesis converts a positive into a superlative, so 'strong empirical support among competing explanations' would match the source, while 'the strongest' overstates it. Searched the full text for 'mismatch', 'strongest', 'most support' and 'competing hypothes'; the only other substantive mentions are the Discussion's own framework (mismatch detection between perceived anthropomorphism and user expectations, p. 15) and the Katsyri et al. (2015) reference entry.

Wallraven, Breidt, Cunningham & Bülthoff (2008) — Evaluating the Perceptual Realism of Animated Facial Expressions

Full reference: Wallraven, C., Breidt, M., Cunningham, D. W., & Bülthoff, H. H. (2008). Evaluating the perceptual realism of animated facial expressions. *ACM Transactions on Applied Perception*, 4(4), Article 23, 20 pages.

DOI: `10.1145/1278760.1278764`

Background and motivation. Creating believable animated CG faces remains one of the most challenging tasks in computer graphics. The challenge of evaluating quality is equally hard: most prior work relied on explicit judgments such as "does this look real?", which capture only a narrow slice of perceptual experience. Wallraven et al. propose an integrated psychophysical framework that replaces explicit realism judgments with task-dependent perceptual measures,

benchmarked against real video of the same actor performing the same expressions.

Method: three experiments.

- **Experiment 1 (animation technique comparison).** Six animation conditions were compared: a full hybrid avatar (motion capture timing + high-resolution 3D scan shape), avatar without rigid head motion, linear morph, cluster animation, 4D dynamic scan, and static 4D peak frame. All expressed seven emotions (confusion, disgust, fear, happy, sad, surprise, thinking). Participants ($n = 12$) rated each stimulus on four perceptual measures: recognition accuracy, reaction time, perceived intensity, sincerity, and typicality.
- **Experiment 2 (information channels).** Using the benchmarked avatar as baseline, the three information channels of shape (surface normals), texture (albedo), and motion were each independently degraded using Gaussian blur across four blur levels. The experiment assessed how each channel contributes to the perceptual measures.
- **Experiment 3 (eyes and teeth).** The avatar lacked eyes in the first two experiments. Eyes (two spherical eye meshes with constrained gaze-at-camera motion) and teeth were added, and perceptual measures were re-collected for both static and dynamic conditions.

Key findings.

- **Avatar achieves comparable recognition to video.** The full avatar condition reached 65.5% recognition accuracy, matching real-world video benchmarks for most expressions (with exceptions: "confusion" and "thinking").
- **Rigid head motion is critical.** Removing head motion alone (AvaNoR condition) dropped recognition to 44.2% — a loss of 21 percentage points. Adding head motion back restored performance and significantly reduced reaction time.
- **Motion dominates over shape and texture.** In Experiment 2, degrading shape or texture had no significant effect on recognition as long as motion was intact. When motion was removed (static peak frame), shape degradation severely reduced recognition. Temporal information can compensate for degraded static channels, but not the reverse.
- **Intensity and sincerity require motion.** Blurring either shape or motion significantly reduced perceived intensity and sincerity ratings, even when recognition was unaffected. This demonstrates that recognition and quality are not the same perceptual outcome.
- **Eyes significantly improve recognition and quality.** Adding eyes produced a gain of approximately 5% in recognition accuracy ($p < 0.001$) and significantly increased intensity and sincerity for most expressions. The benefit was larger for dynamic sequences than static frames.

Key citable sentences (original text and location).

- **Section 2.2, p. 23:4:** "Having access to an indirect, perceptual measure is especially important in the field of human–machine communication, which requires, for example, avatars for virtual kiosks or sales tasks [...] For these applications it is crucial that the facial information displayed by the avatar should not only be recognizable, but also believable and sincere."
- **Section 3, p. 23:5:** "Computer-animated expressions should be recognized as quickly as human expressions are. Any human–machine interaction will be strongly and negatively affected if it takes a person too long to decipher the computer animated expressions."

- **Section 3.2.3 Summary, p. 23:11:** "Both types of temporal information ensure that the resulting facial expressions are recognized reliably and quickly, as well as providing important cues for judging their intensity, sincerity, and typicality."
- **Section 3.3.3 Summary, p. 23:14:** "Interestingly, blurring the shape and texture channels had no effect on recognition rates, as long as the motion channel was present, although it did increase reaction time in both conditions."
- **Section 3.4, p. 23:14:** "One of the biggest drawbacks, in terms of realism and perceptual plausibility, of the avatar as it was used in the two previous experiments is its lack of eyes and teeth. It is well known, for example, that humans are very sensitive to eye motion."
- **Section 4 (Summary), p. 23:18:** "Whereas the ultimate goal of computer graphics usually has been to achieve maximum physical realism, perceptual measures seem better suited to capture success or failure of animated facial expressions."
- **Section 4 (Summary), p. 23:19:** "Taken together, our experimental results shed light on how humans use different information channels to evaluate various aspects of facial expressions — results that are not only of interest for perception research but also for computer animation (including the use of avatars in virtual reality and computer-supported collaborative work)."

Relevance to this thesis. This paper motivates two key design choices in the user study. First, the argument that recognition alone is insufficient to measure avatar quality directly supports the multi-component questionnaire design used here, which collects eeriness, likeability, trust, and sincerity ratings alongside the primary behavioural measure. A CG avatar that is recognised correctly may still trigger negative responses along other perceptual dimensions, as Wallraven et al. demonstrate for intensity and sincerity. Second, the finding that motion is the dominant information channel for both recognition and quality ratings supports the choice of video stimuli rather than static images: all three conditions (C1 real, C2 photorealistic avatar, C3 NPR avatar) present dynamic facial motion, ensuring the dominant channel is preserved across the comparison. The concept of "perceptual equivalence" — that the goal is to match the human benchmark on perceptual measures rather than on physical fidelity — is directly applicable to framing the comparison between C2 and C3: the question is not whether C3 looks objectively less detailed, but whether it is perceptually equivalent to C2 on the measures of trust and eeriness that the study targets.

Exact sentence cited in thesis. The thesis claim that reducing shape detail degrades expressive readability cites `wallraven2008` using the following sentence from §4 Summary (p. 23:19): *"In addition, in Experiment 2, we found clear effects of shape blurring on both intensity and sincerity judgments. This demonstrates that, even though reduced shape information does not affect recognition, it does influence the perceived quality of the expression."*

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #101** *(Ch.4 Methodology > NPR Technique Design)*
  THESIS: "By contrast, excessive stylisation can reduce visual detail and compromise expressive readability."
  SOURCE (SECTION 5, CONCLUSION AND OUTLOOK, PRINTED PAGE ARTICLE 16 / 22): "In addition, one should, in general, avoid high degrees of abstraction for stylization (this is true both for real-world data and for animated sequences) as these adversely influence the perceived intensity and sincerity."
  CHECK: The paper states directly that stylization techniques remove detail (Section 2.1: they "usually remove some detail from the original image"; Section 3.2.2.1 on cartoon stylization: "It does, however, remove small details, as well as low-contrast details"), and that high abstraction degrades expressive readability: Experiment 3 (Article 16 / 17) found "illustrative and especially brush-stroke stylization seem to provide far less adequate cues for robust recognition at the coarsest level," and Section 4.3.3 (Article 16 / 20) reports that "higher abstraction (or coarser resolution) results in a decrease in both perceived intensity and sincerity." The thesis sentence is appropriately hedged with "can," matching the paper's finding that a significant recognition drop appeared only at the coarsest levels.

Latoschik et al. (2017) — The Effect of Avatar Realism in Immersive Social VR

Full reference: Latoschik, M. E., Roth, D., Gall, D., Achenbach, J., Waltemate, T., & Botsch, M. (2017). The Effect of Avatar Realism in Immersive Social Virtual Realities. In *Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology (VRST '17)*, Gothenburg, Sweden, November 8–10, 2017. ACM, 10 pages.
DOI: `10.1145/3139131.3139156`

What the paper does. The paper investigates how the visual realism of an avatar affects embodiment and social interaction in immersive virtual reality (VR). It compares an abstract, gender-neutral wooden mannequin avatar against high fidelity avatars reconstructed from photogrammetry scans of real people. Both representations were assigned alternately to the participant's own body and to the body of a virtual interaction partner, so that the effect of one's own appearance and the effect of the other's appearance could be separated within a single session. The authors state that no prior approach had examined the self-effects of one's own avatar appearance together with the effects of the other's avatar appearance in a virtual social encounter.

Method. Twenty participants (11 female, 9 male, aged 19–24, $M = 20.25$, $SD = 1.21$) took part in a 2 (self avatar) $\times$ 2 (other avatar) within-subjects design with randomised condition order. Participants wore an Oculus Rift head-mounted display, were tracked with an OptiTrack optical system, and first inspected their own avatar in a virtual mirror for 120 s to induce the illusion of virtual body ownership (IVBO), then waved back at a virtual partner appearing in a virtual window. Measures taken after every condition were IVBO (Acceptance, Control, Change), the uncanny valley instrument of Ho et al. with its Humanness, Eeriness and Attractiveness factors, co-presence, rapport, telepresence, social presence, and a three-item trust scale. Data were analysed with two-factorial repeated-measures ANOVAs.

Key findings. The scanned realistic avatars were rated significantly more human-like than the

wooden mannequin when used for the interaction partner ($F_{1,19} = 10.91$, $p = .004$, $\eta_p^2 = .365$; $M = 2.97$ versus $M = 2.23$). Eeriness was slightly higher for the realistic avatars ($M = 2.703$ versus $M = 2.556$) and Attractiveness slightly higher for the mannequin ($M = 3.220$ versus $M = 3.160$), but neither difference reached significance, which the authors read as only an indication of a potential uncanny valley. Realistic avatars produced significantly higher IVBO Acceptance when worn by the participant ($F_{1,19} = 6.19$, $p = .022$, $\eta_p^2 = .246$), and a marginally significant effect of the other's avatar on the Change factor ($F_{1,19} = 4.21$, $p = .054$, $\eta_p^2 = .181$) suggested that the partner's appearance shifts self-perception. No significant results were obtained for presence, trust, or rapport.

Relevance to this thesis. The study is a direct precedent for contrasting an abstracted avatar against a photorealistic one while holding animation and environment constant, and it uses the same uncanny valley factor structure (Humanness, Eeriness, Attractiveness) that this thesis adopts through the Ho and MacDorman instrument. Its null result for trust across realism levels supports treating trust equivalence, rather than a trust advantage, as the expected outcome when only rendering style is varied. It also documents that a large difference in rated human likeness need not produce a matching difference in eeriness, which is the effect this thesis tests between its photorealistic and non-photorealistic (NPR) Avaturn conditions.

Passages this thesis relies on. Full source quotes for these claims are in `citation_check.pdf`.

- **Entry #87** *(Ch.3 Related Work > NPR for Virtual Humans)*
  THESIS: "The realistic avatars were rated significantly more human-like and evoked stronger virtual body ownership, alongside some indication of an uncanny valley response."

- **Entry #137** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "The repeated-measures ANOVA is the established test for rating studies that compare rendering styles within participants."

Gao et al. (2025) — Trust in Virtual Agents: Exploring the Role of Stylization and Voice

Full reference: Gao, Y., Dai, Y., Zhang, G., Guo, H., Mostajeran, F., Zheng, B., & Yu, T. (2025). Trust in Virtual Agents: Exploring the Role of Stylization and Voice. *IEEE Transactions on Visualization and Computer Graphics*.

What the paper does. The paper examines how the visual style of a virtual agent and the origin of its voice jointly affect user trust, perceived realism, empathy, and social presence in VR. The authors build a data-driven pipeline that reconstructs a stylised agent from a TED talk video, converts the reconstructed face into a MetaHuman model, and drives it with audio-generated facial animation and gesture. Three appearance styles are compared: *Real* (the original video of the speaker), *Stylization* (a MetaHuman refined to resemble that speaker), and *No-Stylization* (a generic preset MetaHuman). The study is framed as a test of whether such data-driven stylisation can retain the trust afforded by a real human speaker.

Method. Twenty-seven participants (9 female, mean age 25.4, $SD = 4.28$) took part in a 3 $\times$ 3 $\times$ 2 mixed design. Agent Identity (Young, Middle-aged, Elder) was between subjects, while Style (Real, Stylization, No-Stylization) and Voice (Authentic, Synthesized) were within subjects, giving six stimuli per participant in a Latin square order. Participants watched 40–60 s

speeches from a front-row seat in a virtual theatre using a Meta Quest 3, answered a true/false attention check after each speech, and then completed questionnaires. Trust was measured with an adaptation of the Mayer et al. model split into Ability, Benevolence, Integrity, Trust and Propensity, alongside perceived realism (Appearance, Voice, Movement, Overall), empathy, social presence, and satisfaction, all on 7-point Likert scales. Galvanic skin response (GSR) was recorded at 50 Hz as a physiological trust indicator, and analysis used repeated-measures ANOVA or aligned rank transform ANOVA with Tukey-corrected post-hoc comparisons.

Key findings. Style had a significant main effect on every trust item, whereas voice was significant only for Ability ($F_{1,24} = 5.741$, $p = 0.025$). On the overall Trust item ($F_{2,120} = 5.912$, $p = 0.004$) the Real style was rated significantly more trustworthy than No-Stylization ($p = 0.002$), but was *not* significantly different from Stylization ($p = 0.216$), even though Real scored significantly higher than both other styles on Ability, Benevolence and Integrity. Elder agents were trusted more than Young agents under both the No-Stylization ($p < 0.001$) and Real ($p = 0.013$) styles. Trust correlated moderately with perceived overall realism ($R = 0.3732$), empathy ($R = 0.3783$) and appearance realism ($R = 0.3718$), weakly with social presence ($R = 0.2944$), and most weakly with perceived voice realism ($R = 0.2322$). Mean GSR was lowest in the Real condition, which the authors align with its highest trust ratings.

Relevance to this thesis. This is the closest published paradigm to the present study: a real recording of a human speaker is compared against rendered versions of the same speaker delivering the same speech, with trust decomposed into competence, benevolence and integrity as in this thesis. The finding that a stylised agent was statistically indistinguishable from the real recording on overall trust, while a generic non-matching agent was not, gives direct empirical grounding for expecting equivalence rather than a deficit between the real speaker and a stylised avatar. Its moderate correlations between trust and perceived appearance realism also motivate collecting an appearance influence item alongside the trust scales.

Natarajan and Gombolay (2020) — Effects of Anthropomorphism and Accountability on Trust in Human Robot Interaction

Full reference: Natarajan, M., & Gombolay, M. (2020). Effects of Anthropomorphism and Accountability on Trust in Human Robot Interaction. In *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction (HRI '20)*, March 23–26, 2020, Cambridge, United Kingdom. ACM, New York, NY, USA, 10 pages.
DOI: `10.1145/3319502.3374839`

What the paper does. The paper studies how trust in and dependence on a robot that provides decision support vary with the robot's perceived anthropomorphism, its behaviour, and its physical or virtual presence. The authors state that theirs is the first integrated study to manipulate several robot attributes at once and the first to examine trust across multiple robots in both embodied and virtual form. A second aim is to test whether holding the human accountable for accepting bad advice reduces errors caused by over-compliance. Trust is treated alongside two objective behavioural measures, inappropriate compliance and inappropriate reliance.

Method. Seventy-five participants were recruited (analysis used 72; age 18–58, mean 25.298, $SD = 8.475$, 51.47% female) in a 4 (robot type) $\times$ 4 (behaviour) $\times$ 2 (robot presence) $\times$ 2 (coalition-building preface) mixed design, yielding 720 data points. Robot type (Pepper, Nao,

Kuri, Sawyer) and the coalition-building preface were between subjects, while behaviour and presence were within subjects. Participants played a time-constrained arithmetic quiz in which an agent offered a hint before each of 10 questions and they chose to accept or reject it. Correct agents always gave good advice; Apologetic, Accountable and Indifferent agents gave correct advice only 50% of the time and differed only in the feedback they gave after an error. Trust was measured on a 7-point Likert questionnaire after each agent and perceived anthropomorphism on the 9-point Godspeed anthropomorphism differential, with analysis by repeated-measures ANOVA on a linear mixed effects model.

Key findings. Perceived anthropomorphism was a highly significant positive predictor of trust ($p < 0.001$, $F_{1,720} = 17.6276$, coefficient 0.29335), while robot type itself was not significant, meaning that perceived human likeness rather than the robot's form factor drove trust. Agent behaviour was also significant ($p < 0.001$, $F_{4,670} = 73.5384$): median trust was highest for the two correct agents (55 and 54), then apologetic (33), indifferent (25.5) and accountable (24), with all correct-versus-incorrect pairs differing at $p < 0.001$ and no significant difference between indifferent and accountable. Physical presence affected neither trust nor perceived anthropomorphism. Inappropriate compliance depended significantly on the previously encountered behaviour ($p < 0.001$, $F_{4,670} = 4.860626$), with acceptance of bad advice lowest after an accountable agent, and the coalition-building preface raised trust only for the accountable agent ($\chi^2 = 7.6376$, $p = 0.00571$).

Relevance to this thesis. The result that perceived anthropomorphism predicts trust while the specific agent embodiment does not supports measuring perceived human likeness as its own construct rather than assuming it follows from the rendering platform, which is why the Godspeed anthropomorphism subscale is administered separately from the trust battery here. The study also demonstrates the value of pairing self-reported trust with a behavioural acceptance measure, the same logic behind the forced-choice advisor selection used in this thesis. Its advice-giving task, where an agent recommends one option under time pressure, is structurally close to the advisor vignettes used as stimuli.

Rahman et al. (2024) — Navigating Avatar-Based Marketing From Value Aspects

Full reference: Rahman, M. S., Sabbir, M. M., Zhang, J., Taufique, K. M. R., & Rahman, M. M. (2024). Navigating Avatar-Based Marketing From Value Aspects: A Systematic Literature Review and Forward-Looking Research Agenda. *International Journal of Consumer Studies*, 48, e13096.
DOI: `10.1111/ijcs.13096`

What the paper does. The paper is a systematic literature review of avatar-based marketing (ABM) organised through the theory–context–characteristics–methods (TCCM) framework. It synthesises how avatar attributes shape consumer-perceived value, drawing on the Theory of Consumption Values to classify functional, social, emotional, epistemic and conditional value. The authors argue that the existing literature on avatar attributes and consumer values is fragmented, and that earlier reviews have not addressed the value dimensions that drive purchase decisions. The review closes with a research agenda covering theory, avatar design, deployment context and method.

Method. The authors followed the SPAR-4-SLR protocol across its assembling, arranging and assessing stages. Web of Science, Scopus and Google Scholar were searched for the period 2005 to 10 September 2023 with a search string combining avatar and marketing terms with chatbot, embodied conversational agent, virtual influencer and virtual assistant terms, returning 1237, 1428 and 12,541 documents respectively. Subject-area, language and document-type filters reduced this to 478 published and 67 in-press items, and a full-text review by two authors removed duplicates and papers not concerned with ABM, leaving 57 publications. These were coded in Microsoft Excel through content analysis against the TCCM categories.

Key findings. Anthropomorphism in this literature is consistently decomposed into form realism, meaning appearance and spatial or dynamic likeness to a human, and behavioural realism, meaning the ability to interact naturally, and both are reported to improve customer engagement and loyalty. The review notes that a higher degree of form embodiment, covering features such as gender, voice and hair, has been found to increase perceived credibility and source trust. Anthropomorphism is described as fostering trust when consumers adopt a new technology, but the authors set the Uncanny Valley effect against the Computers Are Social Actors paradigm, observing that an almost authentic yet imperfect depiction of a real person evokes eerie feelings and can reduce emotional connection and trust. Of the 57 articles, 50 were quantitative, and the authors report that many lack an explicit conceptual framework and that most rely on classic human–computer interaction theories rather than marketing-side theory.

Relevance to this thesis. The review supplies the applied framing for the advisor scenario used here: avatars are already deployed as recommenders and service assistants, so the trust consequences of their visual appearance carry practical weight beyond the laboratory. Its form realism versus behavioural realism distinction is the terminology this thesis relies on when arguing that only appearance is manipulated between the photorealistic and NPR conditions while animation is held constant. It also documents that the uncanny valley is invoked rarely and mostly as a caution in this literature, which locates the gap that a controlled comparison of a stylised avatar against a photorealistic one is intended to fill.

NPR — Nonphotorealistic Rendering

Gooch et al. (1998) — A Nonphotorealistic Lighting Model for Technical Illustration

Full reference: Gooch, A., Gooch, B., Shirley, P., & Cohen, E. (1998). A non-photorealistic lighting model for automatic technical illustration. In *Proceedings of SIGGRAPH 1998* (pp. 447–452). ACM.

DOI: `10.1145/280814.280950`

Problem. Standard Phong shading maps surface illumination to a grayscale or colour gradient that fills the entire dynamic range from black (shadow) to white (highlight). Technical illustrators solve a different problem: they need every surface region to be clearly readable, so they confine shading to the midtone range and reserve pure black for outlines and pure white for specular highlights. This produces drawings where form is unambiguous regardless of the lighting angle, which is rarely achievable with Phong shading.

The tone-transfer model. Gooch et al. propose a shading function that replaces the Phong

diffuse term with a cool-to-warm tone shift:

$$I = \frac{1 + \hat{l} \cdot \hat{n}}{2} \cdot k_{\text{warm}} + \frac{1 - \hat{l} \cdot \hat{n}}{2} \cdot k_{\text{cool}}$$

where $\hat{l}$ is the unit vector toward the light and $\hat{n}$ is the surface normal. The warm colour $k_{\text{warm}} = k_{\text{yellow}} + \beta k_d$ and the cool colour $k_{\text{cool}} = k_{\text{blue}} + \alpha k_d$ blend the object's diffuse colour k_d with yellow and blue tint colours. The parameters $\alpha, \beta \in [0, 1]$ control how much the object's own colour contributes versus the pure tint. This construction keeps all shading values in the midtone range: faces pointing toward the light receive a warm midtone, those pointing away receive a cool midtone, and neither reaches the extremes of black or white. Black is reserved for drawn outlines; white for Phong specular highlights.

GPU implementation approximation. The paper describes a real-time approximation using two Phong lights placed symmetrically about the view direction: one with positive and one with negative intensity, with ambient set to the average of warm and cool. This can be computed without a special shader by exploiting standard Phong lighting parameters. Specular highlights are rendered white and placed as a final compositing step, ensuring the highlight "pops" against both warm and cool shaded regions.

Metal shading. For metallic surfaces, the authors propose alternating stripes of varying intensity along the parametric axis of maximum curvature. Stripe intensities are sampled randomly in $[0, 0.5]$ and colours interpolate between the light source colour and the surface reflected direction, approximating the Fresnel-like angular highlight characteristic of polished metal. The result resembles hand-drawn technical metal shading.

Key principle. Gooch cool-to-warm shading makes shape readable by encoding orientation information in hue rather than luminance. A viewer can perceive the 3D form of a shaded Gooch surface even when it is reduced to greyscale, because the hue shift maps onto a consistent orientation cue. This is distinct from toon shading (see Lake et al. below), which quantises luminance rather than shifting hue.

Relevance to this thesis. Gooch et al. (1998) established the principle that NPR shading can use the full spectral gamut to convey surface orientation without sacrificing readability. The cool-to-warm model is cited in the thesis as the foundation of nonphotorealistic illumination design and motivates the warm/cool split used in early NPR tests before the toon quantisation approach was adopted.

Lake et al. (2000) — Stylized Rendering for Real-Time 3D Animation

Full reference: Lake, A., Marshall, C., Harris, M., & Blackstein, M. (2000). Stylized rendering techniques for scalable real-time 3D animation. In *Proceedings of NPAR 2000: First International Symposium on Nonphotorealistic Animation and Rendering* (pp. 13–20). ACM.

DOI: `10.1145/340916.340918`

Context. This paper describes techniques developed and deployed at Shiny Entertainment for the commercial game *Messiah* (2000). It is notable as one of the first accounts of toon shading implemented in a shipped product, and its techniques became the standard template for real-time NPR in games.

Quantised diffuse shading. The core toon shading technique replaces continuous diffuse lighting with a 1D texture lookup. The Lambertian diffuse factor $\max(\hat{l} \cdot \hat{n}, 0)$ is computed per vertex and used as a texture coordinate into a 1D artist-painted ramp texture. The ramp contains discrete colour bands separated by hard steps. Because the ramp is authored by an artist, the number of bands, the colour of each band, and the sharpness of transitions are fully under artistic control and can be painted per-character or per-material type.

Inverted hull (geometry outline). Object outlines are generated by duplicating the mesh and rendering the duplicate with backface culling inverted (front-facing polygons are culled). The duplicate mesh is inflated along vertex normals by a small offset, so only the silhouette shell is visible around the original mesh. The shell is rendered in a solid dark colour. Because this technique operates in object space, it scales naturally with the character's LOD system: lower-polygon LODs still produce clean outlines without any image-space processing cost.

Scalability and LOD integration. A key design requirement was that the techniques had to work transparently across all LOD levels of the game's characters. The geometry outline passes this requirement because it is a mesh operation; the quantised diffuse ramp passes it because it is a per-vertex computation. The paper documents the performance cost as negligible at the LOD levels used in the game.

Artistic authoring. Because the diffuse ramp is a hand-painted 1D texture, artists retain full control over the visual output. Different characters can have stylistically distinct shading (warmer highlight colours, cooler shadows, different band counts) without changing any rendering code.

Relevance to this thesis. Lake et al. (2000) introduced the two foundational real-time NPR techniques used throughout this project: quantised diffuse toon shading (V1 in the Meta Avatar shader pipeline, as well as the Avaturn V1 and Jody V1 shaders) and the inverted hull outline. The geometry outline is directly implemented in `V1_ToonShading_GeometryOutline.shader` across all three avatar pipelines.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #106** *(Ch.4 Methodology > NPR Technique Design > V2 — Toon Shading)*
  [check wording] THESIS: "Toon shading, also known as cel shading, is one of the most widely adopted techniques in NPR, where stylisation is achieved by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination."
  SOURCE (SEC. 3.1 CARTOON SHADING, PDF P. 2): "Rather than shading the character to give it a three-dimensional appearance, the cartoonist typically uses solid colors that do not vary across the materials they represent. Often the artist will shade the part of a material that is in shadow with a color that is a darkened version of the main material color. ... The boundary between shadowed and illuminated colors is a hard edge that follows the contours of the object or character. We refer to this technique as hard shading"
  CHECK: The tonal-simplification half of the sentence (solid colours instead of physically accurate illumination) is directly supported. The paper never states that toon shading is 'one of the most widely adopted techniques in NPR', and it never uses the phrase 'cel shading' as a synonym for its technique (it calls it cartoon shading or hard shading, and uses 'cel' only for the traditional animation artefact).

- **Entry #107** *(Ch.4 Methodology > NPR Technique Design > V2 — Toon Shading)*
  THESIS: "Correspondingly, the approach is well suited to interactive character rendering."

SOURCE (ABSTRACT (PDF P. 1) AND SEC. 3.1 / 3.1 ALGORITHM DISCUSSION (PDF PP. 2-3)): "This paper presents real-time methods to emulate cartoon styles. ... Our system allows a 3D model to be drawn to look just like a traditional 2D cartoon, and to be viewed interactively. ... While current processors are fast enough to handle this math at interactive rates on scenes with reasonable complexity, hardware lighting could potentially be used to speed these computations."
CHECK: The paper explicitly presents cartoon shading as a real-time method for interactively viewed 3D characters, integrated with a character animation system. Sec. 6 and Table 1 further report 30 fps animated character rendering.

- **Entry #108** *(Ch.4 Methodology > NPR Technique Design > V2 — Toon Shading)*
THESIS: "This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading."
SOURCE (SEC. 3.1 CARTOON SHADING, PDF PP. 2-3): "Rather than smoothly interpolating shading across a model as in Gouraud shading [Gour71], our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color. ... Note that as Max { L · n , 0 } crosses 0.5, the index into the texture map chooses a different color. ... Our texture-mapped method is equivalent to per-pixel color thresholding."
CHECK: The paper contrasts its hard shading directly against smooth (Gouraud) interpolation and describes the discrete colour switch at the L.n threshold, which is exactly the replacement of continuous gradients with discrete tonal transitions. Sec. 3.2 adds that higher-resolution texture maps give further discrete bands.

Isenberg et al. (2003) — A Developer's Guide to Silhouette Algorithms

Full reference: Isenberg, T., Freudenberg, B., Halper, N., Schlechtweg, S., & Strothotte, T. (2003). A developer's guide to silhouette algorithms for polygonal models. *IEEE Computer Graphics and Applications*, 23(4), 28–37.

DOI: 10.1109/MCG.2003.1210862

Scope. This survey paper systematically classifies all known approaches to computing and rendering silhouette edges for polygonal 3D meshes. It is aimed at graphics developers who need to choose among the available algorithms for specific real-time or offline NPR applications. The paper covers image space, hybrid, and object space approaches, and provides a practical decision table.

Definition of silhouette. For a polygonal mesh, a silhouette edge is any edge shared by exactly one front-facing and one back-facing polygon with respect to the current camera position. This is the geometric definition: $(\hat{n}_1 \cdot \hat{v} \leq 0) \neq (\hat{n}_2 \cdot \hat{v} \leq 0)$, where $\hat{v}$ is the view direction and $\hat{n}_1$, $\hat{n}_2$ are the face normals of the two adjacent triangles. Smooth-surface silhouettes satisfy $\hat{n} \cdot (\mathbf{x} - \mathbf{C}) = 0$ (the surface normal is perpendicular to the view ray at that point).

Image space algorithms. Use a depth buffer, normal buffer, or colour buffer rendered from the scene; apply a per-pixel edge detection filter (such as Sobel or Roberts Cross on normals or depth). Advantages: fast, handles self-intersecting geometry, independent of polygon count. Limitations: pixel-level precision only, cannot control line width analytically, produces aliased output at edges between objects rather than geometric silhouettes, and cannot distinguish

silhouette edges from colour or texture edges.

Hybrid algorithms. Detect silhouette edges in object space (using the geometric criterion above) then render them using image-space techniques (stencil buffer or overdraw). Examples include Raskar and Cohen's backface enlargement and Gooch et al.'s environment-map method. These allow stylised line widths while using hardware rasterisation. Limitations: two render passes required; silhouettes must be computed per frame.

Object space algorithms. Compute silhouette edges directly on the geometry:

- *Trivial algorithm:* iterate all mesh edges and test the face-orientation criterion for each. $O(n)$ per frame; straightforward to implement; slow for dense meshes.
- *Precomputation approaches:* project silhouette candidates onto a Gaussian sphere; use hierarchical octrees or dual 4D representations to prune non-silhouette edges rapidly. Suitable for static meshes; fails with full skeleton animation.
- *Stochastic (Markosian et al.):* randomly sample a subset $O(\sqrt{n})$ of edges per frame; statistically, only about $\sqrt{n}$ edges are actually silhouette in any frame. Probabilistic but fast; misses isolated silhouette edges occasionally.

Decision table (Table 1). The paper provides a table mapping requirements to recommended algorithm class:

- Real-time, arbitrary animated mesh: image space or hybrid.
- Guaranteed silhouette completeness: object space trivial or precomputed.
- Stroke-level stylisation (variable width, pressure, brush): object space.
- Static or slowly animated mesh: precomputed object space.

Key limitation identified. Geometry-only silhouette detection captures boundary edges between front- and back-facing polygons, but misses interior crease lines, colour edges, and texture discontinuities. Image space methods recover interior edges but are not true geometric silhouettes.

Relevance to this thesis. Isenberg et al. (2003) motivate the progression from V1 (geometry outline) to the image space edge detection in V2 and V3. The geometry outline captures only silhouette edges in object space; interior crease lines and colour boundaries visible in the face and clothing require screen space gradient operators. The paper's limitation note — that geometry-only edge capture misses these features — is the direct design motivation for developing V2 (normal edge detection) and V3 (Sobel colour edge detection).

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #67** *(Ch.3 Related Work > NPR for Virtual Humans)*
  THESIS: "In order to detect those lines, Isenberg et al. surveyed the available algorithms and grouped silhouette lines together with creases, borders and self-intersections under the heading of feature lines."
  SOURCE (DEFINITIONS AND TERMINOLOGY, P. 28): "Other lines significant in the context of silhouettes are creases, borders, and self-intersections. Creases are edges that should always be drawn, for instance the edges of a cube. ... Border lines only appear in models where the mesh is not closed ... Lastly, self-intersection lines are where two polygons of a model

intersect. ... In the literature, these three additional line categories together with the silhouette lines are often called feature lines."
CHECK: The paper is explicitly a survey of silhouette algorithms for polygonal models, and its terminology section groups creases, borders and self-intersections together with silhouette lines under the name feature lines. Minor attribution nuance: the paper reports that these are 'often called feature lines' in the literature rather than presenting the grouping as its own coinage, so 'grouped ... under the heading of' is best read as adopting an existing convention.

- **Entry #102** *(Ch.4 Methodology > NPR Technique Design > V1 — Inverted Hull Outline)*
THESIS: "Object silhouettes are a defining visual feature of nonphotorealistic rendering."
SOURCE (OPENING PAGE PULL QUOTE AND INTRODUCTION, P. 28): "Generating object silhouettes lies at the heart of nonphotorealism. ... Silhouettes play an important role in shape recognition because they provide one of the main cues for figure-to-ground distinction."
CHECK: The pull quote states the claim almost word for word, and the introductory paragraph gives the perceptual justification (figure-to-ground distinction).

- **Entry #103** *(Ch.4 Methodology > NPR Technique Design > V1 — Inverted Hull Outline)*
[check wording] THESIS: "The inverted hull method, also known as the shell method, was selected since it produces stable geometric silhouettes without requiring screen-space processing."
SOURCE (HYBRID ALGORITHMS (RASKAR AND COHEN PARAGRAPH AND SECTION OPENING), PP. 30-31): "Performing additional transformations before rendering the back-facing polygons can improve this general technique. A translation of the back-facing polygons toward the viewer yields thicker silhouette lines in the resulting image. However, these lines don't have a constant line width. For silhouettes with adjustable but constant width, the back-facing polygons are enlarged depending on the distance from the viewer and the angle with the viewing direction (see Figure 3). ... Hybrid algorithms are characterized by operations in object space that are followed by rendering the modified polygons ... using the z-buffer. ... the silhouette is represented in a pixel matrix."
CHECK: The paper does describe the technique of enlarging back-facing polygons to obtain silhouettes of adjustable, constant width, which is the inverted hull. However, it never uses the names 'inverted hull' or 'shell method', and it classifies the technique as a hybrid algorithm whose result is a pixel matrix produced by extra z-buffer render passes, not as a method free of screen-space processing. The paper also warns that the distinct borders of hybrid silhouettes 'can cause aliasing artifacts in the image', which qualifies the thesis's unqualified 'stable'. The selection decision itself is the thesis's own and is not in the source.

- **Entry #104** *(Ch.4 Methodology > NPR Technique Design > V1 — Inverted Hull Outline)*
THESIS: "This approach is well suited to real-time NPR avatar rendering because it requires no offline mesh preprocessing."
SOURCE (HYBRID ALGORITHMS, P. 31; TABLE 1 'CHOOSING THE BEST SILHOUETTE DETECTION ALGORITHM', P. 29): "Computation times for hybrid algorithms generally don't differ much from those for image space methods and run at interactive to real-time frame rates. ... However, the algorithms can easily make use of commonly available graphics hardware to speedup the rendering process. ... [Table 1] Animate model in real time beyond a mere flight through the scene | Real-time technique. Object space silhouette detection method that uses preprocessing is not appropriate."

CHECK: The paper states hybrid algorithms run at interactive to real-time frame rates, and Table 1 explicitly recommends against preprocessing-based silhouette detection when the model is animated in real time, since 'this precomputed data structure would become invalid and result in new precomputation steps for every frame' (Precomputation methods, p. 34). The avatar application is the thesis's own framing and is not discussed in the source.

- **Entry #105** *(Ch.4 Methodology > NPR Technique Design > V1 — Inverted Hull Outline)*
  [check wording] THESIS: "Consequently, creases between body regions, clothing seams and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh."
  SOURCE (HYBRID ALGORITHMS, P. 31): "By employing the equal to depth function, Raskar and Cohen's approach draws only the edges where the two groups of polygons meet and thus yield the model's silhouette. ... Raskar proposes a one-pass hardware implementation that basically adds borders around each triangle. ... It also inherently includes generating crease lines where the angle between the faces exceeds a given threshold."
  CHECK: The source supports the gist: the back-face expansion technique draws only edges shared by front- and back-facing polygons, and crease lines are a separate category that other methods must add explicitly (Raskar's one-pass variant includes them, and Gooch et al.'s stencil method 'can render creases in a different color than silhouettes'). Two discrepancies. First, the paper's own terminology distinguishes silhouette from contour: 'the subset of the silhouette, which in 2D divides one portion of the object from another one of the same object, is not part of the contour. Those lines are sometimes called internal silhouettes' (p. 28). The method yields the silhouette, which includes internal silhouettes, not only the 'external silhouette' as the thesis states. Second, the paper says nothing about texture or colour boundaries such as clothing seams, or about facial contours.

- **Entry #112** *(Ch.4 Methodology > NPR Technique Design > V3 — Screen Space Normal Edge Detection)*
  THESIS: "Silhouette extraction techniques can be classified into image space, object space and hybrid approaches."
  SOURCE (DEFINITIONS AND TERMINOLOGY (CLASSIFICATION PARAGRAPH), P. 29): "With respect to solving the edge detection problem, we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space with the resulting silhouette represented by an analytic description of silhouette edges."
  CHECK: The three-way classification is stated directly, and the same three categories are repeated for the second sub-problem: 'We can categorize the algorithms that perform this visibility culling into image-space, object-space, and hybrid approaches.'

Gonzalez & Woods (2018) — Digital Image Processing (4th edition)

Full reference: Gonzalez, R. C. & Woods, R. E. (2018). *Digital Image Processing* (4th edition). Pearson/Prentice Hall.

ISBN: 978-0133356724

Scope. Standard graduate-level textbook for spatial and frequency domain image processing. The chapters most directly cited in this thesis are Chapter 3 (spatial filtering), Chapter 10 (image segmentation via gradients and edges), and the appendix on colour models.

Sobel operator. Gonzalez and Woods define the Sobel gradient as two 3×3 convolution kernels:

$$G_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, \qquad G_y = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

The gradient magnitude is $\|\mathbf{G}\| = \sqrt{G_x^2 + G_y^2}$, approximated in practice as $|G_x| + |G_y|$. The Sobel kernel applies implicit central difference differentiation while simultaneously weighting the centre row/column twice as heavily as the corner rows/columns, giving a small degree of smoothing and improved isotropy compared to simpler gradient operators.

Roberts Cross operator. A 2×2 diagonal gradient operator:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad G_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Roberts Cross responds to diagonal edges and is computationally cheaper than Sobel (4 multiplications vs. 8 per output pixel). It is less noise-tolerant but suitable for detecting diagonal surface transitions, which is why it is used alongside the depth normal proxy in the V5 (Hierarchical) shader.

Gaussian smoothing. A discrete approximation to the 2D Gaussian $G(x, y) = e^{-(x^2+y^2)/2\sigma^2}$. Convolved with an image before differentiation, it suppresses high-frequency noise that would otherwise dominate the gradient signal. The standard deviation σ controls the scale at which features are detected: large σ detects coarse edges and misses fine detail; small σ detects fine detail but remains sensitive to noise.

BT.601 luminance weights. For converting RGB colour images to greyscale before edge detection:

$$L = 0.299R + 0.587G + 0.114B$$

These weights reflect the relative sensitivity of the human visual system to each colour channel and are defined in the ITU-R BT.601 broadcast standard. Using these weights rather than equal-weight averaging produces greyscale images that perceptually match the luminance of the original colour image.

Laplacian of Gaussian (LoG). The second derivative of the Gaussian, $\nabla^2 G(x, y)$, applied as a convolution kernel. Zero crossings of the LoG response mark intensity edges. The textbook derives the discrete LoG approximation kernel and describes the zero-crossing detection procedure.

Relevance to this thesis. Gonzalez and Woods provide the mathematical basis for all edge detection operators used across the NPR pipeline: the Sobel kernels in V3 (Sobel edge detection), the Gaussian pre-smoothing in V4 (Gaussian Sobel), the Roberts Cross in V5 (Hierarchical), and the BT.601 luma weights used throughout for greyscale conversion of RGB colour buffers. The textbook is the authoritative source for the mathematical definitions accompanying these techniques in Chapter 4 of the thesis.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #20** *(Ch.2 Background > Edge Detection > Gradient-Based Detection)*
  THESIS: "It estimates the two directional derivatives with a pair of $3\{}times3$ kernels:"
  SOURCE (CH. 10, SEC. 10.2, 'BASIC EDGE DETECTION' > 'GRADIENT OPERATORS', PRINTED PP. 718-719 (PDF PP. 720-721)): "The simplest digital approximations to the partial derivatives using kernels of size 3 x 3 are given by ... A slight variation of the preceding two equations uses a weight of 2 in the center coefficient ... Figures 10.14(f) and (g) show the kernels used to implement Eqs. (10-24) and (10-25). These kernels are called the Sobel operators (Sobel [1970])."
  CHECK: Eqs. (10-24)/(10-25) are exactly the two directional partial derivatives, and Fig. 10.14(f)-(g) shows them as a pair of 3x3 kernels named the Sobel operators. Note only an axis-convention difference: G&W place x along rows (origin top left, Fig. 2.19), so the book's g_x kernel is the transpose of the thesis's G_x; the kernel entries themselves match.

- **Entry #21** *(Ch.2 Background > Edge Detection > Gradient-Based Detection)*
  [check wording] THESIS: "The central row and column of each Sobel kernel carry weights of $\{}pm2$, whereas the simpler Prewitt operator weights all three rows and columns equally at $\{}pm1$."
  SOURCE (CH. 10, SEC. 10.2, 'GRADIENT OPERATORS', PRINTED PP. 718-719 (PDF PP. 720-721); KERNELS IN FIG. 10.14(D)-(G), PRINTED P. 719): "A slight variation of the preceding two equations uses a weight of 2 in the center coefficient ... The Prewitt kernels are simpler to implement than the Sobel kernels, but the slight computational difference between them typically is not an issue. The fact that the Sobel kernels have better noise-suppression (smoothing) characteristics makes them preferable because ... noise suppression is an important issue when dealing with derivatives."
  CHECK: The substantive contrast is verbatim supported: Sobel uses a weight of 2 in the center coefficient, Prewitt (Eq. 10-23, Fig. 10.14(d)-(e)) weights every term at 1, and the book explicitly calls Prewitt the simpler of the two. The thesis phrasing is imprecise per kernel, however: in each Sobel kernel only one of the two central lines carries the +-2 pair (the central row of G_x, the central column of G_y) while the other central line is all zeros, and the same holds for the Prewitt +-1 claim.

- **Entry #23** *(Ch.2 Background > Edge Detection > Gradient-Based Detection)*
  THESIS: "Its edge magnitude is commonly approximated as $\{}lvert R_1 * I \{}rvert + \{}lvert R_2 * I \{}rvert$, which avoids the square root of ."
  SOURCE (CH. 10, SEC. 10.2, 'GRADIENT OPERATORS', EQ. (10-26), PRINTED P. 720 (PDF P. 722)): "Any of the pairs of kernels from Fig. 10.14 are convolved with an image to obtain the gradient components g x and g y at every pixel location. ... Obtaining the magnitude of the gradient requires the computations in Eq. (10-17). This implementation is not always desirable because of the computational burden required by squares and square roots, and an approach used frequently is to approximate the magnitude of the gradient by absolute values"
  CHECK: Eq. (10-26) is M(x,y) approx |g_x| + |g_y|, introduced precisely to avoid the squares and square root of Eq. (10-17), the thesis's Eq. sobel_magnitude. The book states this for 'any of the pairs of kernels from Fig. 10.14', and Fig. 10.14(b)-(c) is the Roberts pair, so the approximation does cover Roberts as the thesis applies it.

- **Entry #115** *(Ch.4 Methodology > NPR Technique Design > V5 — Hierarchical Edge Detection with Multiple Cues)*
  [check wording] THESIS: "Depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal."
  SOURCE (CH. 10, SEC. 10.2, 'GRADIENT OPERATORS', EQS. (10-21)-(10-22), PRINTED P. 718 (PDF P. 720)): "When diagonal edge direction is of interest, we need 2-D kernels. The Roberts cross-gradient operators (Roberts [1965]) are one of the earliest attempts to use 2-D kernels with a diagonal preference. Consider the 3 x 3 region in Fig. 10.14(a). The Roberts operators are based on implementing the diagonal differences"
  CHECK: Gonzalez & Woods supports only one element of this sentence, the Roberts Cross operator as a 2-D diagonal-difference gradient operator applied to image intensity. It contains nothing about depth discontinuities, silhouettes, surface normal discontinuities or rendered geometry buffers; I searched Ch. 10 (PDF pp. 690-790) and the colour chapter listing, where segmentation is colour-based (Sec. 6.7) and not normal or depth based. Note also that the book dates the operator Roberts [1965] while the thesis bib key is roberts1963.

Marr & Hildreth (1980) — Theory of Edge Detection

Full reference: Marr, D. & Hildreth, E. (1980). Theory of edge detection. *Proceedings of the Royal Society of London B: Biological Sciences*, 207(1167), 187–217.

DOI: `10.1098/rspb.1980.0020`

Theoretical contribution. Marr and Hildreth proposed the first biologically motivated computational theory of edge detection. Their central claim is that the visual system detects edges not by directly computing intensity gradients, but by first smoothing the image at multiple spatial scales and then locating zero crossings of the Laplacian of the smoothed signal. The smoothing function they identify as biologically appropriate is the Gaussian, because it is the unique linear filter that simultaneously minimises spatial spread (localisation) and frequency spread (noise sensitivity).

The Laplacian of Gaussian (LoG) operator. The proposed operator $\nabla^2(G * I)$ — the Laplacian of the Gaussian-smoothed image — produces a response that changes sign at intensity discontinuities in the smoothed image. These zero crossings form closed contours in the image, which Marr and Hildreth identify as the *raw primal sketch*: the first stage of visual representation. Unlike gradient-based operators (Sobel, Roberts), the LoG naturally produces edges that are closed curves, reflecting the topological constraint that physical objects have closed boundaries.

Scale-space principle. The Gaussian smoothing scale σ determines the spatial resolution at which zero crossings are computed. Large σ values suppress fine detail and noise; small σ values preserve fine detail. The paper argues that edge detection should be performed at multiple σ scales and the results combined: features that appear consistently across scales are more likely to correspond to real physical boundaries than those that appear only at fine scales.

Biological evidence. Marr and Hildreth review psychophysical evidence (Mach bands, subjective contours, masking experiments) and anatomical evidence (receptive field profiles of retinal ganglion cells and lateral geniculate nucleus neurons) consistent with LoG-like processing in

the early visual system. The difference-of-Gaussians (DoG) approximation of the LoG matches the known centre-surround structure of ganglion cell receptive fields.

Relevance to this thesis. Marr and Hildreth establish the theoretical principle that pre-smoothing with a Gaussian before computing intensity derivatives is necessary for robust edge detection, not merely empirically useful. This principle directly motivates the V4 (Gaussian Sobel) shader, in which a Gaussian blur is applied to the colour buffer before the Sobel gradient is computed, reducing the UV seam noise and skin-texture noise that affected V3. The LoG concept also contextualises the Canny detector (below), which extends Marr and Hildreth's ideas with non-maximum suppression and hysteresis thresholding.

Canny (1986) — A Computational Approach to Edge Detection

Full reference: Canny, J. (1986). A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(6), 679–698.

DOI: `10.1109/TPAMI.1986.4767851`

Optimality criteria. Canny posed edge detection as an optimisation problem with three criteria: (1) *detection* — the detector should respond to all real edges and not respond to non-edges; (2) *localisation* — detected edges should be as close as possible to the true edge positions; (3) *single response* — each real edge should produce exactly one detector response, not multiple. He proved that the optimal filter for these three criteria is well approximated by the first derivative of a Gaussian.

The four-stage pipeline.

1. **Gaussian smoothing.** The input image is convolved with a Gaussian G_σ at scale σ. This suppresses noise that would otherwise dominate the gradient in the next step.
2. **Gradient computation.** The gradient magnitude $M = \sqrt{G_x^2 + G_y^2}$ and direction $\theta = \arctan(G_y / G_x)$ are computed using first-derivative kernels.
3. **Non-maximum suppression.** Each gradient magnitude pixel is compared to its two neighbours in the direction of the gradient. If the pixel is not a local maximum along this direction, it is suppressed to zero. This thins wide gradient responses to single-pixel-wide ridges.
4. **Double-threshold hysteresis.** Two thresholds T_{high} and T_{low} are applied. Pixels with $M > T_{\text{high}}$ are strong edges and kept unconditionally. Pixels with $M < T_{\text{low}}$ are rejected. Pixels with $T_{\text{low}} \leq M \leq T_{\text{high}}$ are kept only if they are spatially connected to a strong edge. This reduces isolated noise responses while preserving weak edges that are part of longer contours.

Relevance to this thesis. The Canny detector establishes that Gaussian pre-smoothing is not just practically useful but theoretically optimal before gradient-based edge detection. This motivates V4 (Gaussian Sobel), which applies Gaussian pre-smoothing before the Sobel gradient. The full Canny pipeline is not reproduced in the fragment shader (non-maximum suppression and hysteresis require multi-pass access to neighbouring pixels, which is not available in a single-pass fragment hook), but the first two stages — Gaussian smoothing and gradient computation — are directly implemented in V4.

Barla et al. (2006) — X-Toon: An Extended Toon Shader

Full reference: Barla, P., Thollot, J., & Markosian, L. (2006). X-Toon: An extended toon shader. In *Proceedings of NPAR 2006: 4th International Symposium on Nonphotorealistic Animation and Rendering* (pp. 127–132). ACM.

DOI: `10.1145/1138450.1138471`

Motivation. Standard toon shading uses a 1D texture lookup keyed on the Lambertian diffuse factor $\hat{n} \cdot \hat{l}$. This encodes only how strongly the surface is lit, and produces a flat, depth-agnostic style: two faces at different distances from the camera with the same orientation relative to the light receive identical shading. Barla et al. extend the toon shader with a second dimension to encode view-dependent information such as depth or surface orientation relative to the viewer.

Extension 1 — Tone Detail (2D texture). The 1D toon ramp is replaced with a 2D texture indexed by two coordinates: $u = \hat{n} \cdot \hat{l}$ (the standard diffuse tone) and $v = D \in [0, 1]$ (the detail level). The v axis can be mapped to any view-dependent scalar:

- *Depth-based*: $D = 1 - \log(z/z_{\min}) / \log(z_{\max}/z_{\min})$ — objects farther from the camera receive a different row of the 2D texture, enabling aerial perspective and depth-of-field effects within a single toon pass.
- *Orientation-based*: $D = |\hat{n} \cdot \hat{v}|^r$ — surfaces facing the viewer (high $|\hat{n} \cdot \hat{v}|$) receive a different row than surfaces at grazing angles, enabling silhouette darkening or backlighting effects.

The 2D texture is painted freely by the artist: the entire visual space of the shader is under artistic control, not constrained to a function of the lighting alone.

Extension 2 — Shape Detail (Normal Field Abstraction). The vertex normals of the mesh are replaced with a blend between the original normals and the normals of an abstract proxy shape (sphere, cylinder, ellipse, or flat disc). A global blending parameter $\alpha \in [0, 1]$ controls the degree of abstraction. At $\alpha = 0$, the original detailed mesh normals are used; at $\alpha = 1$, the surface shades as if it were the proxy shape, flattening interior creases while preserving the overall silhouette. Barla et al. term this blended normal field the *Normal Field Abstraction*. The abstraction level can be keyframed or linked to the depth axis D, allowing surfaces to abstract as they recede from the viewer.

Performance. Table 1 of the paper measures rendering time on a Pentium 4 + ATI Radeon 9800. X-Toon is approximately 18–21% slower than standard 1D toon shading across four test models and three screen resolutions. This overhead is acceptable for real-time use.

Effects achievable. Aerial perspective (distant objects stylistically faded), depth-of-field suggestion, near-silhouette glow, backlighting, abstract interior with preserved contour, plastic vs. metal distinction via different 2D texture rows.

Relevance to this thesis. X-Toon introduced the concept of a *detail axis* — a continuous scalar D that parameterises how much surface detail is resolved in the final rendering. This concept directly motivates the Avaturn `XToon_2DRamp.shader` and the corresponding Jody XToon shader. The Normal Field Abstraction concept is also central to the thesis argument: it shows that toon shading can deliberately control the level of surface detail visible, making it a principled tool for managing perceived avatar complexity and potentially for navigating the uncanny valley.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #62** *(Ch.3 Related Work > NPR for Virtual Humans)*
  THESIS: "The X-Toon ramp"
  SOURCE (SECTION 3, TONE DETAIL (P. 2)): "We extend the classic 1D toon texture by adding a vertical "detail" axis (corresponding to LOA) to build a 2D toon texture. The horizontal axis corresponds to tone as in a classic toon texture ... The vertical axis corresponds to tone detail: each value along this axis (labelled D for detail) corresponds to its own 1D toon texture. The whole 2D texture can thus be regarded as a stack of 1D toon textures with increasing "detail.""
  CHECK: The thesis fragment is a label for the X-Toon 2D texture, which this passage defines directly. The paper itself calls it a "2D toon texture" rather than a "ramp"; "ramp" is standard field terminology and appears in the paper only in the Figure 9 caption ("green-to-brown color ramp"), so the wording is a synonym rather than an added claim.

- **Entry #109** *(Ch.4 Methodology > NPR Technique Design > V2.2 — X-Toon Shading)*
  THESIS: "X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level."
  SOURCE (ABSTRACT (P. 1)): "First, we replace the 1D texture with a 2D texture whose second dimension corresponds to the desired "tone detail," which can vary with depth or surface orientation."
  CHECK: Corroborated in detail by Section 3 (p. 2): "The horizontal axis corresponds to tone as in a classic toon texture - the texture coordinate along that axis is derived from a standard lambertian shading computation ... The vertical axis corresponds to tone detail: each value along this axis (labelled D for detail) corresponds to its own 1D toon texture." The thesis's "abstraction level" for the vertical axis matches the paper's own gloss: "a vertical 'detail' axis (corresponding to LOA)" and "'detail' here corresponds to visual abstraction: less detail means greater abstraction" (Section 1).

- **Entry #110** *(Ch.4 Methodology > NPR Technique Design > V2.2 — X-Toon Shading)*
  THESIS: "In addition to tonal abstraction, Barla et al. described a shape detail extension that modifies the surface normal field by blending the original mesh normals with normals derived from a geometrically simplified shape."
  SOURCE (SECTION 4, SHAPE DETAIL (P. 3)): "We now describe how to further modify the shading to depict an abstracted version of the shape. Note that this extension is independent of the tone detail extension and is done view-independently. We achieve it by modifying the surface normal field used in shading computations with a geometric map: we map the input mesh to an abstracted shape that acts as a lower bound for shape detail. ... The designer controls the "abstraction" of shape detail by setting a blending parameter that controls how the shader interpolates between normals of the input mesh and the abstracted shape."
  CHECK: Every element of the thesis sentence is present: a second, separate shape detail extension, modification of the normal field, and blending between input mesh normals and abstracted shape normals. The "geometrically simplified shape" corresponds to the paper's four abstracted shapes: "a smoothed version of the original mesh, and an enclosing ellipse, cylinder, and sphere" (same section).

- **Entry #111** *(Ch.4 Methodology > NPR Technique Design > V2.2 — X-Toon Shading)*

THESIS: "Barla et al. further described a general attribute mapping mechanism that maps a chosen surface attribute onto the abstraction axis and they demonstrated it with depth-based and orientation-based maps."
SOURCE (SECTION 3, TONE DETAIL (P. 2); DEMONSTRATED IN SECTIONS 3.1 "DEPTH-BASED ATTRIBUTE MAPPING" AND 3.2 "ORIENTATION-BASED ATTRIBUTE MAPPING" (PP. 2-3)): "Once the 2D texture is defined, the designer must choose an attribute (e.g., orientation or depth) that will control the tone detail, and provide functions called attribute maps that map the attribute to a detail value D at each location on a surface. This formulation is general in that any attribute can be chosen, depending on the application goals. In this paper, we consider view-dependent attributes. In the next two sections we describe how to compute depth-based and orientation-based attribute maps ..."
CHECK: The generality claim and both demonstrated map types are stated explicitly. One minor terminological point: the paper's demonstrated attributes are view-dependent (depth to the eye or focal axis, and orientation relative to the observer) rather than intrinsic surface properties, so the thesis phrase "surface attribute" is looser than the paper's "attribute ... at each location on a surface", but it does not misstate what was done.

Kyprianidis et al. (2009) — Image Abstraction by Anisotropic Kuwahara Filtering

Full reference: Kyprianidis, J. E., Kang, H., & Döllner, J. (2009). Image and video abstraction by anisotropic Kuwahara filtering. *Computer Graphics Forum* (Pacific Graphics), 28(7).

DOI: `10.1111/j.1467-8659.2009.01574.x`

The Kuwahara filter (1976). The classical Kuwahara filter divides a square neighbourhood around each pixel into four overlapping rectangular sub-regions. It computes the mean and variance of pixel values within each sub-region and assigns the output pixel the mean of the sub-region with the lowest variance. The effect is to smooth uniformly within coherent regions while preserving edges, because the sub-region on the "less noisy" side of an edge is selected. The filter was originally designed for medical image denoising but produces a characteristic painterly oil-painting effect. Its main defect is block artefacts caused by the four rectangular sub-regions, and instability at edges where sub-regions span both sides of an intensity boundary.

Structure tensor. Kyprianidis et al. compute the local image structure tensor from the spatial derivatives of the image:

$$\mathbf{G} = \begin{pmatrix} f_x^2 & f_x f_y \\ f_x f_y & f_y^2 \end{pmatrix}$$

smoothed by a Gaussian with standard deviation σ_s. The eigenvalues $\lambda_1 \geq \lambda_2$ of this tensor describe the local gradient distribution. The anisotropy measure $A = (\lambda_1 - \lambda_2)/(\lambda_1 + \lambda_2)$ ranges from 0 (isotropic, uniform region or noise) to 1 (fully anisotropic, strong edge). The dominant eigenvector gives the local orientation ϕ of the principal gradient direction.

Anisotropic filter shape. The filter neighbourhood is an ellipse $\Omega = \{\mathbf{x} : \|\mathbf{S}\mathbf{R}_{-\phi}\mathbf{x}\| \leq h\}$, where $\mathbf{R}_{-\phi}$ is a rotation by the local orientation angle and $\mathbf{S} = \mathrm{diag}(\alpha/(\alpha + A), (\alpha + A)/\alpha)$ is a scaling matrix that elongates the ellipse along the local edge direction in proportion to the anisotropy A. In uniform regions ($A \approx 0$), the ellipse is nearly circular; at strong edges ($A \approx 1$), the ellipse elongates to span the edge rather than crossing it.

Weighted sector voting. The ellipse is divided into $N = 8$ equal sectors. For each sector i, the mean m_i and standard deviation s_i of pixel values are computed. The output is the weighted sum:

$$\text{output} = \frac{\sum_i \alpha_i m_i}{\sum_i \alpha_i}, \qquad \alpha_i = \frac{1}{(1 + s_i)^q}$$

where $q = 8$. This assigns higher weight to sectors with lower standard deviation (more uniform), effectively selecting the most coherent side of any edge without producing block artefacts.

GPU implementation. Encoded as four RGBA render texture samples (four sectors per channel, two samples to cover 8 sectors). At 512×512 resolution with $\sigma_r = 3.0$, $\sigma_s = 1.0$, $N = 8$, the filter runs at approximately 12fps on a GeForce GTX 280 (2009 hardware). Temporal coherence is excellent without motion estimation because the filter respects local edge orientation, which is geometrically stable under slow animation.

Limitation noted. The structure tensor computation requires access to a smoothed structure tensor image computed from the full frame, which is a separate render pass producing a texture. For the anisotropic version to be implemented, a prior render pass must generate and store the structure tensor map. This is not possible within a single-pass fragment shader hook.

Relevance to this thesis. Kyprianidis et al. (2009) provide the theoretical basis for the Kuwahara painterly filter implemented as V6 in the Meta Avatar shader pipeline (`NPREffect_Kuwahara2.cginc`) and the corresponding shaders for Avaturn and Jody. The thesis implements the isotropic variant of Kuwahara (classical rectangular sub-regions, not the anisotropic elliptical variant) because the fragment shader hook does not permit the separate structure tensor render pass required by the full anisotropic version. This limitation is explicitly documented: the anisotropic filter would provide better edge-aligned strokes but requires a multi-pass pipeline that exceeds the SDK's single-pass constraint.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #31** *(Ch.2 Background > Structure-Preserving Filtering)*
  THESIS: "Kyprianidis et al. recovered it in an anisotropic formulation that aligns the sampling window with local image structure, estimated through the structure tensor."
  SOURCE (ABSTRACT (PDF P. 1) AND SECTION 3.2 'ANISOTROPIC KUWAHARA FILTERING' (PDF P. 4)): "It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor. ... [Section 3.2] ... defines an ellipse, whose major axis is aligned to the local image orientation. This ellipse has high eccentricity in anisotropic regions and becomes a circle in isotropic regions."
  CHECK: Both halves of the thesis sentence are explicit in the paper: the filter shape is adapted to local feature shape derived from the smoothed structure tensor (Abstract, Section 3.1), and the elliptical sampling window's major axis is aligned to the local image orientation (Section 3.2).

- **Entry #80** *(Ch.3 Related Work > NPR for Virtual Humans)*
  THESIS: "Their anisotropic formulation adapts the filter to the local structure of image features and this produces a painting-like flattening effect along those directions."
  SOURCE (ABSTRACT (PDF P. 1) AND SECTION 1 'INTRODUCTION' (PDF P. 2)): "Contrary to conventional edge-preserving filters, our filter generates a painting-like flattening effect

along the local feature directions while preserving shape boundaries. ... [Introduction] ... we present a generalization of the Kuwahara filter that removes clustering artifacts by adapting shape, scale and orientation of the filter to the local structure of the input"
CHECK: The thesis sentence reproduces the paper's own wording almost exactly: adaptation of the filter to the local structure of the input, and the resulting painting-like flattening effect along the local feature directions.

- **Entry #118** *(Ch.4 Methodology > NPR Technique Design > V6 — Kuwahara Painterly Filter)*
  THESIS: "The anisotropic Kuwahara formulation aligns an elliptical sampling window with local image structure rather than with the fixed rectangular subregions of the classical filter."
  SOURCE (SECTION 2 'RELATED WORK' (PDF P. 2) AND SECTION 3.2 'ANISOTROPIC KUWAHARA FILTERING' (PDF P. 4)): "The Kuwahara and Tomita-Tsuji filter use rectangular subregions, while the Nagao-Matsuyama filter uses a rotated elongated bar mask. The use of rectangular subregions results in visible artifacts in the output. ... [Section 3.2] ... defines an ellipse, whose major axis is aligned to the local image orientation."
  CHECK: The paper states that the classical Kuwahara filter uses rectangular subregions and that the proposed method instead defines an ellipse aligned to local image orientation, matching both halves of the contrast the thesis draws. Figure 3's caption reinforces this: '(a) Rectangular subregions ... (c) Anisotropic filter shape derived from the local structure and divided into subregions'.

- **Entry #146** *(Ch.5 Implementation > NPR Technique Implementation > V6 — Kuwahara Painterly Filter)*
  THESIS: "All three platforms use the anisotropic Kuwahara filter of Kyprianidis et al.."
  SOURCE (TITLE AND ABSTRACT (PDF P. 1)): "Image and Video Abstraction by Anisotropic Kuwahara Filtering ... We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor."
  CHECK: The paper supports the attribution only, i.e. that the anisotropic Kuwahara filter is the technique of Kyprianidis et al. The remainder of the sentence ('All three platforms use...') describes the thesis's own implementation and cannot be verified against any source; note that CLAUDE.md 5.1 rule 5 discourages attaching a citation to a sentence stating what was implemented in this work.

- **Entry #147** *(Ch.5 Implementation > NPR Technique Implementation > V6 — Kuwahara Painterly Filter)*
  [check wording] THESIS: "The exact weighting formula differs by platform, because each implementation follows a different variant of the same principle from Kyprianidis et al.."
  SOURCE (SECTION 3 'METHOD' (PDF P. 4) AND SECTION 3.2 'ANISOTROPIC KUWAHARA FILTERING' (PDF P. 5)): "The filter response is defined as a weighed sum of the local averages, where more weight is given to those averages with low standard deviation. ... [Section 3.2] ... Our definition of the weighting factors ... ensures that more weight is given to sectors with low standard deviation, i.e. those that are more homogeneous. This is similar to the approach of [PPC07], where ... is used for the weights, but avoids the indetermination, when some of the si are zero."
  CHECK: The underlying principle the thesis attributes to the source is genuinely there:

weight each sector inversely to its standard deviation. However, the paper defines exactly one weighting formula for its own filter (alpha_i = 1/(1 + ||s_i||^q), with q = 8 in all examples) and contrasts it only with Papari et al.'s ||s_i||^-q; it does not present a family of interchangeable weighting variants. The idea that different implementations follow 'a different variant' of the formula is the thesis's own implementation choice, not something the paper offers or licenses. Ellipses in the quote replace math glyphs that pdftotext garbles.

Kyprianidis (2011) — Multi-Scale Anisotropic Kuwahara Filtering

Full reference: Kyprianidis, J. E. (2011). Image and video abstraction by multi-scale anisotropic Kuwahara filtering. In *Proceedings of the 8th International Symposium on Non-Photorealistic Animation and Rendering (NPAR '11)*, pp. 55–64. ACM.

Note: Article DOI not confirmed; verified from physical PDF `anisotropic_multicale_209.pdf` in the NPR Articles folder. Cite by ISBN: 978-1-4503-0907-3.

Context. This paper extends the anisotropic Kuwahara filter of Kyprianidis et al. (2009) to address two remaining limitations. The first is that the weighted sector voting scheme becomes unstable when sector weights fall near zero in noise-corrupted regions, producing visible block artifacts. The second is that the level of abstraction is governed by the filter radius: larger radii produce stronger abstraction but also increase computational cost quadratically and introduce artifacts in large uniform areas. Neither problem can be resolved simply by increasing the radius.

Contribution 1 — Stabilised weighting. A small floor value ε is added to the standard deviation denominator in the weighting function:

$$\alpha_i = \frac{1}{(\varepsilon + s_i)^q}$$

where $\varepsilon > 0$ prevents division by near-zero values in noise-corrupted regions. This eliminates instability without changing the filter's edge-preserving behaviour where genuine structure is present.

Contribution 2 — Multi-scale computation scheme. A Gaussian image pyramid is built from the input. Processing proceeds coarse-to-fine:

1. At the coarsest level: compute the local orientation and anisotropy via the structure tensor; apply the anisotropic Kuwahara filter.
2. Upsample the filter result and the smoothed structure tensor to the next finer level.
3. At each finer level: merge the upsampled coarse-level structure tensor with the structure tensor computed at the current resolution; apply the anisotropic Kuwahara filter using this merged tensor.
4. Repeat until the finest level is reached.

Propagating orientation estimates coarse-to-fine means the filter at each fine level has access to low-frequency orientation context from a wider image neighbourhood than any single-scale filter of equivalent radius. This produces a stronger and more coherent abstraction effect without

the artifacts that a large-radius single-scale filter would generate in low-contrast regions.

GPU suitability. The paper was presented at NPAR 2011 and explicitly targets real-time processing. Lanczos3 resampling is used for pyramid downsampling because it preserves local extrema and rejects high-frequency ringing better than bilinear or Catmull-Rom filters. Temporal coherence on video sequences improves because coarse-level orientation estimates change more gradually between frames than per-pixel fine-level estimates.

Relationship to the 2009 paper. The 2009 anisotropic Kuwahara paper introduced the structure tensor, elliptical filter shape, and weighted sector voting. This 2011 paper retains all of those mechanisms and adds the stabilising floor term and the multi-scale propagation scheme. Both papers are by the same first author (Kyprianidis) and should be cited together when the anisotropic filter is discussed.

Relevance to this thesis. The isotropic Kuwahara variant implemented as V6 in the Meta Avatar shader pipeline uses classical rectangular sub-regions, not the anisotropic elliptical sectors of the 2009 paper and not the multi-scale propagation of this 2011 paper. The multi-scale scheme is not implementable within the single-pass fragment hook because it requires an explicit image pyramid across multiple render passes. This paper and its 2009 predecessor together document exactly what the V6 implementation sacrifices relative to the state of the art: the anisotropic multi-scale variant would produce superior stroke coherence and stronger abstraction in low-contrast areas without block artifacts. Citing this 2011 paper alongside the 2009 paper allows the thesis to characterise the V6 limitation precisely and to indicate the direction in which a future multi-pass implementation could improve.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #81** *(Ch.3 Related Work > NPR for Virtual Humans)*
  THESIS: "A multiscale variant, also by Kyprianidis, propagates local orientation estimates and filtering results up a low-pass filtered pyramid."
  SOURCE (ABSTRACT, PRINTED P. 55): "Second, a multi-scale computation scheme is proposed that simultaneously propagates local orientation estimates and filtering results up a low-pass filtered pyramid. This allows for a strong abstraction effect and avoids artifacts in large low-contrast regions."
  CHECK: The thesis sentence is a close paraphrase of the paper's own abstract wording, and sole authorship by Jan Eric Kyprianidis is confirmed on the title page. Section 3.2.3 and 3.3.2 confirm that both the structure tensor (orientation) and the anisotropic Kuwahara filtering result are propagated coarse-to-fine.

- **Entry #82** *(Ch.3 Related Work > NPR for Virtual Humans)*
  [check wording] THESIS: "The proposed method in panel (c) is the multiscale variant, which suppresses the artefacts the anisotropic filter leaves in low-contrast regions."
  SOURCE (FIGURE 1 AND ITS CAPTION, PRINTED P. 55): "(b) Anisotropic Kuwahara filter ... (c) Proposed method ... Figure 1: Example comparing the proposed multi-scale approach with the single-scale approach."
  CHECK: The panel identification is correct: Figure 1(c) is the proposed multi-scale method and (b) is the single-scale anisotropic Kuwahara filter. The artefact claim is not what the paper attributes to the multi-scale scheme, however: the abstract says it 'avoids artifacts in large low-contrast regions' as a consequence of pushing abstraction further, while the

Introduction states the single-scale filter itself 'avoids overblurring in low-contrast regions' and that artefacts arise only because 'Simply increasing the filter radius is typically not a solution, as it often results in artifacts.' Artefacts that the anisotropic filter genuinely leaves in homogeneous regions are attributed in Section 3.3.1 to unstable weight selection and are fixed by the separate thresholding contribution shown in Figure 9, not by the multi-scale variant in Figure 1(c).

Praun et al. (2001) — Real-Time Hatching

Full reference: Praun, E., Hoppe, H., Webb, M., & Finkelstein, A. (2001). Real-time hatching. In *Proceedings of SIGGRAPH 2001* (pp. 579–584). ACM.

DOI: 10.1145/383259.383295

Hatching in NPR. Hatching is a traditional hand-drawing technique in which groups of parallel strokes convey surface tone, material properties, and 3D form simultaneously through stroke density, direction, and spacing. In darker regions, strokes are denser; on curved surfaces, stroke directions follow principal curvature directions to convey form. Cross-hatching adds perpendicular strokes at even darker tones. The challenge for real-time 3D hatching is maintaining stroke coherence across frames, across different surface parameterisations, and across a continuous range of tones.

Tonal Art Maps (TAMs). The core contribution of the paper is the Tonal Art Map: a 2D grid of mip-mapped images. The grid is indexed by tone t (columns; $t = 0$ is white paper, $t = 5$ is darkest hatching) and mip level ℓ (rows; $\ell = 0$ is the finest detail level). Each cell contains a greyscale image of hatch strokes. The defining property is *nesting*: every stroke appearing in cell (t, ℓ) also appears in all cells (t', ℓ') with $t' \geq t$ and $\ell' \geq \ell$. This means that as the tone becomes darker, strokes are *added* but never moved or removed. As the resolution decreases (higher mip level), strokes are wider but fewer. Nesting ensures that tone transitions and resolution changes never produce popping artefacts: the image changes incrementally.

TAM construction. The paper describes a greedy stroke-placement algorithm that fills each TAM cell iteratively. Starting from the lightest tone, strokes are placed to minimize the error between the current partial TAM and the target tone. At darker tones, additional strokes are placed without moving existing ones (nesting constraint). Cross-hatching strokes are added at tones 4 and 5. The algorithm produces TAMs that closely match artist-drawn hatch styles.

Rendering. At runtime, each triangle face has a single dominant tone assigned via its Lambertian diffuse factor (same as toon shading). Rather than a hard-threshold band, the renderer blends between two adjacent tone columns using the fractional tone value, and interpolates the three vertex tone values using barycentric coordinates. This produces six texture samples per fragment (two adjacent tone columns, blended at each of three triangle vertices) combined in two hardware dot products — a technique the authors call a "6-way blend." The resulting frame rate is 20–40fps for models of 7,500–15,000 faces on a 933MHz Pentium III with a GeForce2 GPU.

Lapped textures. The paper also addresses the challenge of applying TAMs to arbitrary mesh topology without UV seams. The authors use lapped textures: overlapping, irregularly shaped texture patches that cover the surface with a consistent stroke direction field derived from the principal curvature. The patches overlap at their alpha boundaries, hiding seam lines.

Relevance to this thesis. Praun et al. (2001) introduced the TAM paradigm, which establishes that real-time hatching with full tone and resolution continuity is achievable through careful pre-computation of the texture content. This technique is cited in the thesis as the foundational reference for hatching-style NPR, while noting that it is not directly implemented: the implementation constraint of a single fragment pass precludes the multi-texture blending and per-face tone assignment that TAMs require without a geometry pass. The `HalftoneHatching.shader` in `Assets/Shaders/CommonNPRShaders/` implements a simplified screen-space hatching effect without full TAM-compliant tone nesting.

Roberts (1963) — Machine Perception of Three-Dimensional Solids

Full reference: Roberts, L. G. (1963). *Machine perception of three-dimensional solids* (Ph.D. thesis). Massachusetts Institute of Technology, Lincoln Laboratory Report TR-315.

Note: Doctoral thesis; no DOI. Available via MIT DSpace and ResearchGate (`https://www.researchgate.net/publication/220695992`).

Historical significance. Roberts' thesis is the founding document of edge detection in computer vision. Written in 1963 before the term "computer vision" existed as a discipline, it describes a system that processes a photograph of a polyhedral scene and extracts a three-dimensional geometric description. The thesis introduced the *Roberts Cross operator* — a pair of 2×2 convolution kernels detecting diagonal intensity gradients — as the first systematic mathematical approach to locating edges in digitised images.

The Roberts Cross kernels. To locate intensity boundaries, Roberts applies discrete gradient approximations along the two diagonal directions:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad G_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The gradient magnitude at each pixel is estimated as $\|G\| = \sqrt{G_1^2 + G_2^2}$, or approximated for speed as $|G_1| + |G_2|$. The diagonal orientation detects $45°$ and $135°$ edge directions.

Why 2×2 and not 3×3. Roberts chose 2×2 kernels due to the computational constraints of 1963 hardware (MIT TX-2 computer) and the mathematical minimum: a 2×2 discrete operator is the smallest structure capable of computing a diagonal finite difference. The limitation is that the gradient centre falls between pixel corners, introducing a half-pixel positional offset. Later operators (Sobel 1968, Prewitt 1970) improved noise tolerance using 3×3 kernels that incorporate more surrounding context at higher cost.

Broader contribution. Roberts built the first complete pipeline from photograph to 3D model: edge detection on the digitised image, line fitting to edge outputs, 3D geometry extraction assuming polyhedral structure, and hidden-line removal for arbitrary viewpoints. Experimental results on actual photographs of polyhedral objects demonstrated successful 3D reconstruction, making this one of the first demonstrations of automated scene understanding from images.

Relevance to this thesis. The Roberts Cross operator is used in the V5 (Hierarchical) shader as the colour-edge component of the three-cue fusion: the 2×2 diagonal kernels are applied to the screen-space colour buffer to detect surface colour transitions not captured by the depth proxy

or normal derivatives. Currently cited via Gonzalez & Woods (2018), which derives the Roberts kernels mathematically. Citing the original thesis correctly attributes the operator to its source and situates it within the historical lineage of edge detection preceding the Sobel, Marr-Hildreth, and Canny approaches also used in V3–V5.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #114** *(Ch.4 Methodology > NPR Technique Design > V5 — Hierarchical Edge Detection with Multiple Cues)*
  [check wording] THESIS: "Depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal."
  SOURCE (CHAPTER IV, PICTURE INPUT AND REDUCTION TO LINES, PP. 25-26): "When the scanning is completed, the picture is processed with a local differential operator to produce a new raster which has the appearance of a line drawing. ... Therefore, after a picture is read in, a differential picture is created according to the functions: ... where $x_{i,j}$ is the initial intensity value, $z_{i,j}$ is the computed derivative value, and i and j are the coordinates in the two dimensions. The resulting z values indicate the probability of a line through that cell."
  CHECK: The operator itself is fully confirmed: p. 26 gives z = sqrt((y[i,j] - y[i+1,j+1])^2 + (y[i+1,j] - y[i,j+1])^2), the two diagonal differences that define the Roberts Cross, visually verified in the page scan at 300 dpi rather than trusted to OCR. What differs is the input domain: Roberts applies it to a scalar intensity raster (after a square-root gamma change, p. 26), never to colour, and he frames the output as line drawing extraction, saying nothing about texture boundaries lacking a geometric signal. The colour and multi-cue framing is the thesis's own extension, so the citation supports only the named operator.

Kumar & Poornima (2019) — Comprehensive Survey on Nonphotorealistic Rendering

Full reference: Kumar, T. & Poornima, B. G. (2019). A comprehensive survey on non-photorealistic rendering and benchmark developments for image abstraction and stylization. *Iran Journal of Computer Science*, Springer Nature.

DOI: `10.1007/s42044-019-00034-1`

Scope. Structured review of NPR techniques from 1963 to approximately 2017. Organises the entire field into five technique families and proposes a benchmark evaluation framework. It is the most cited recent NPR survey addressing both 2D and 3D stylisation methods within a single taxonomy.

Taxonomy of NPR techniques.

Category	Description and representative methods
Stroke-based rendering	Simulates individual artistic strokes (hatching, watercolour, oil paint). Includes Tonal Art Maps (Praun et al., 2001) and brush stroke simulation.
Colour image analogy	Transfers the style of an exemplar image onto a target using texture transfer. Includes Hertzmann et al.'s (2001) image analogies.
Region-based rendering	Segments the image into coherent regions and applies flat or gradated colouring. Includes cel shading and Voronoi mosaic effects.
Image filtering abstraction	Applies local image filters to reduce detail while preserving large-scale structure. Includes Kuwahara filtering (Kyprianidis et al., 2009), bilateral filtering, and mean-shift segmentation.
Stylisation	Direct transformation of shading, colour mapping, and tone, typically real-time. Includes toon shading (Lake et al., 2000), X-Toon (Barla et al., 2006), and Gooch lighting (Gooch et al., 1998).

Benchmark evaluation criteria. Visual consistency (coherence across frames), preservation of structural features (edges and shapes remain readable), artistic fidelity (match to target style), and computational cost (real-time capability). These four criteria map directly onto the evaluation framework for the NPR techniques in this thesis.

Historical trajectory. From stroke-based offline rendering systems of the 1990s (hours per frame) to image-space and stylisation techniques of the 2000s (real-time capable) to data-driven and neural approaches of the 2010s.

Relevance to this thesis. Kumar and Poornima's taxonomy provides a single citable survey for placing the eight NPR techniques implemented in this project within the broader literature. V1 (toon shading, V2 X-Toon, inverted hull outline) falls in *stylisation*. V3 (Sobel), V4 (Gaussian Sobel), and V5 (Hierarchical) fall in *image filtering abstraction*. V6–V8 (Kuwahara variants) are image filtering abstraction with painterly qualities overlapping *stroke-based rendering*. The paper can be cited in Chapter 2 (Related Work) to orient the reader before the technique-by-technique discussion.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #60** *(Ch.3 Related Work > NPR for Virtual Humans)*
  THESIS: "Pavan Kumar et al.'s classification of NPR techniques into four families: stroke-based rendering, image analogy, region-based techniques and image filtering."
  SOURCE (SECTION 2 'CLASSIFICATIONS OF NPR TECHNIQUES', PRINTED P. 136 (IRAN J. COMPUT. SCI. 2019, 2:131-165)): "NPR techniques have been classified to fall into following categories, based on the work entitled "State of the Art": a taxonomy of artistic stylization techniques for images and video [5], stylization and abstraction of photographs [7], and painterly rendering techniques [11]. 1. Stroke-based rendering. 2. Image analogy. 3. Region-based techniques. 4. Image processing and filtering techniques. The above four categories include linear filtering and non-linear filtering techniques."

CHECK: The four families match exactly, with the fourth named 'Image processing and filtering techniques' (thesis shortens it to 'image filtering'); Fig. 2 is titled 'Classification of NPR techniques'. Caveat worth noting: the survey states the four-way scheme is 'based on' three earlier works ([5] Kyprianidis et al. taxonomy, [7], [11]) rather than presenting it as an original contribution, so attributing it to Pavan Kumar et al. is adoption, not origination.

Wisessing, Dingliana & McDonnell (2016) — Perception of Lighting and Shading for Animated Virtual Characters

Full reference: Wisessing, P., Dingliana, J., & McDonnell, R. (2016). Perception of lighting and shading for animated virtual characters. In *Proceedings of the ACM Symposium on Applied Perception (SAP 2016)*.

DOI: 10.1145/2931002.2931015

Research question. How does shading style (CG-shaded vs. toon-shaded) and lighting configuration interact to affect the perceived emotional intensity and appeal of animated virtual characters?

Stimulus design. A professional animator keyframed ten dramatic emotional sentences using a virtual character, each conveying a specific emotion (anger, fear, sadness, surprise, happiness, disgust). Each sentence was rendered in two shading styles (CG with subsurface scattering vs. toon flat colour bands with outline), four lighting directions (above, front-below, side, rear), and three lighting intensities (low, medium, high contrast key-to-fill ratio). Participants ($n = 36$) rated each stimulus on: emotion recognition accuracy (which emotion?), emotional intensity (how strongly expressed?), and character appeal.

Key results.

- **Shading style significantly affected emotional intensity:** CG-shaded renders were rated as conveying significantly more intense emotions than toon-shaded renders of the same performance ($p < .001$). The magnitude was approximately 0.8–1.2 points on a 7-point scale across emotions.
- **Emotion recognition accuracy was not significantly affected by shading style:** participants identified the correct emotion at similar rates regardless of shading. Intensity of the perceived emotion changed; what emotion was perceived did not.
- **Lighting direction had little effect** on perceived emotional intensity. Below-lighting reduced appeal ($p = .02$).
- **High-contrast lighting reduced appeal:** high key-to-fill ratio (deep shadows) was rated less appealing than medium and low contrast ($p < .05$).

Interpretation. Toon shading softens the emotional signal of the animation. Photorealistic rendering amplifies subtle facial cues (micro-expressions, shadow detail in wrinkles and skin creases) that toon shading flattens away. Recognition accuracy being preserved means toon-shaded avatars remain comprehensible as social actors while reducing the perceptual "loudness" of their facial signals.

Relevance to this thesis. This paper provides a psychophysical explanation for why NPR

stylisation may reduce eeriness. If toon shading softens subtle emotional cues in the animation, it simultaneously softens the cues that produce uncanny valley responses (slightly wrong expression timing, subtly unnatural skin movement) and the cues that make the avatar feel intensely realistic. The finding that *recognition accuracy* is preserved while *intensity* is reduced is directly relevant to H1: a toon-shaded avatar may produce lower eeriness without losing communicative clarity.

Wisessing, Zibrek, Cunningham, Dingliana & McDonnell (2020) — Enlighten Me: Brightness and Shadow for Character Emotion and Appeal

Full reference: Wisessing, P., Zibrek, K., Cunningham, D. W., Dingliana, J., & McDonnell, R. (2020). Enlighten Me: Importance of brightness and shadow for character emotion and appeal. *ACM Transactions on Graphics*, 39(3), Article 30.

DOI: 10.1145/3383195

Background. Following the 2016 SAP finding that shading style affects emotional intensity, this paper investigates within each shading style how the specific parameters of lighting — brightness and shadow depth — affect character appeal and eeriness. The question is whether the same effects observed across shading styles can be reproduced within a single style by varying lighting parameters.

Study design. Three experiments; primary results from Experiment 3. A 2 (shading style: CG vs. toon) $\times$ 4 (key light brightness levels) $\times$ 4 (key-to-fill ratio levels) within-subjects design, $n = 36$ participants. Characters expressed three emotions (neutral, happy, sad). Participants rated character appeal (1–7) and eeriness (1–7).

Key results.

- **Increasing key light brightness significantly increased appeal** ($F = 14.3$, $p < .001$, $\eta^2 = .19$) and significantly *decreased eeriness* ($F = 9.7$, $p < .001$, $\eta^2 = .13$) for both shading styles.
- **Reducing the key-to-fill ratio (lighter shadows) significantly increased appeal** ($F = 11.2$, $p < .001$, $\eta^2 = .15$) and significantly *decreased eeriness* ($F = 8.1$, $p < .001$, $\eta^2 = .12$).
- **Eeriness reduction was larger for CG rendering than for toon.** CG characters were more eerie at baseline and benefited more from brighter, lighter-shadow lighting. Toon characters were already less eerie and showed a smaller absolute change.
- **Psychophysical model:** appeal varies as a function of key-to-fill ratio r following a power law: appeal $= a \cdot r^{-b}$, providing a quantitative prediction of optimal lighting parameters.

Interpretation. Eeriness is not solely a property of avatar geometric or textural realism: it is also modulated by the tonal distribution of the shading. Bright, evenly-lit characters with shallow shadows are consistently rated as less eerie and more appealing across both rendering styles. Toon shading's structural properties (lighter mid-tones, painted flat bands without deep luminance-range shadows) align with the low-eeriness lighting configuration.

Relevance to this thesis. Toon shading by definition produces lighter shadow regions than CG rendering: the shadow bands are painted to a constant colour rather than following the full

dynamic range of a Lambertian model. This structural property directly maps onto Wisessing et al.'s finding that lighter shadows reduce eeriness. The V1 toon shading in this project uses a warm light band and a cool mid-shadow band without deep black shadows, which matches the low-eeriness lighting configuration identified here. This paper supplies a psychophysical mechanism for why the NPR avatar (C3) may produce lower eeriness than the photorealistic avatar (C2): even if the NPR stylisation did not move the avatar out of the uncanny zone, the inherent tonal flatness of toon shading would still reduce eeriness through the brightness-shadow mechanism documented here.

Petikam, Anjyo & Rhee (2021) — Shading Rig: Art-Directable Stylised Shading for 3D Characters

Full reference: Petikam, L., Anjyo, K., & Rhee, T. (2021). Shading Rig: Dynamic art-directable stylised shading for 3D characters. *ACM Transactions on Graphics*, 40(5), Article 205.

DOI: 10.1145/3461696

Problem. Standard real-time toon shading (Lake et al., 2000) places the light-to-shadow boundary at a threshold derived from the Lambertian dot product $\hat{l} \cdot \hat{n}$. As the character moves or the lighting direction changes, this boundary slides across the surface, producing visually unstable shading that animators working in hand-drawn cel animation styles specifically try to avoid. Professional animated productions (such as *Spider-Man: Into the Spider-Verse*) require artists to manually reshape shadow boundaries per frame to maintain a consistent visual look independent of physically computed lighting. This is not feasible at real-time frame rates.

The shading rig concept. Inspired by facial rigging in character animation, Petikam et al. propose the shading rig: a parameterised collection of artist-defined edits to the toon shading output that can be interpolated and blended at runtime.

- Artists define a set of *shading states*: different poses of the light/shadow boundary across the character surface, painted using a direct manipulation tool.
- Each state is associated with a control parameter (light direction angle or scene lighting condition).
- At runtime the system interpolates between states using a learned interpolation function (a small neural network, approximately 100 parameters).
- The result is toon shading that respects artistic intent as lighting changes, rather than mechanically following the Lambertian threshold.

Implementation. A Unity shader with a pre-computation pipeline for baking the shading state space from artist-painted examples. The mesh UV coordinates store shading state offsets. Runtime cost of the interpolation network is negligible.

Evaluation. User study ($n = 26$) comparing shading rig output to standard toon shading across animations with varying lighting. Shading rig received significantly higher ratings on visual consistency and artistic fidelity ($p < .001$).

Relevance to this thesis. The shading rig represents the current state of the art in production toon

shading for 3D characters. The V1 implementation in this project uses a fixed threshold approach (Lake et al., 2000) because the SDK integration constraint limits the NPR implementation to a single fragment shader hook without the pre-computed state-space baking or the multi-state mesh UV parameterisation that the shading rig requires. Petikam et al. contextualise this limitation: the shading rig would produce superior artistic output but is architecturally incompatible with the Meta Avatars SDK's material management system. Citing this paper allows the thesis to acknowledge the gap between the V1 implementation and the production solution without being obligated to close it.

Zibrek, Martin & McDonnell (2019) — Is Photorealism Important for Perception of Expressive Virtual Humans in Virtual Reality?

Full reference: Zibrek, K., Martin, S., & McDonnell, R. (2019). Is photorealism important for perception of expressive virtual humans in virtual reality? *ACM Transactions on Applied Perception*, 16(3), 14:1–14:19.

DOI: 10.1145/3349609

Research question. Does photorealistic rendering change how people respond to an expressive virtual human in VR, across presence, emotional response and physical comfort?

Stimulus design. A between-groups experiment in VR. An interactive female character delivered a monologue and made eye contact with the participant. Three render styles (Realistic, Simple, Sketch) were crossed with three attitudes (Friendly, Unfriendly, Sad). Recruitment reached $N = 797$; after excluding incomplete responses, 622 participants entered the analysis, roughly 70 per scenario/style cell (aged 18–71, mean 26.6). Thirteen scales were measured: Proximity, Concerned, Excited, Afraid, Calm, Comfortable, Appeal, Eerie, Familiar, Overall/Movement/Appearance Realism, Social Presence and Place Illusion.

Key results.

- **Social presence was unaffected by render style.** The authors predicted an effect (H1) and found "no main effects or interactions between render style or scenario on this dependent variable."
- **Affinity was unaffected by render style.** Appeal, Eerie and Familiar showed "no effects of render style or interactions between render style and scenario." This analysis was run specifically to exclude an uncanny valley confound.
- **Photorealism raised place illusion** but changed rather than improved the emotional response: participants were more Excited under Realistic, yet felt more Calm under Sketch in the Friendly scenario.
- **Photorealism lowered perceived movement realism.** Stylised characters were judged to move more realistically.
- **Comfort in close proximity was unaffected** by render style.

Interpretation. Rendering quality does not scale linearly with social utility. Photorealism strengthened the sense of being in a place while leaving the sense of being with a person untouched, and it degraded perceived motion quality, plausibly because a realistic surface raises

expectations that the animation cannot meet.

Relevance to this thesis. The strongest published evidence that render style leaves eeriness and social presence unchanged, and the closest precedent for H1's null result. Its Affinity analysis is the same manoeuvre this thesis makes in reverse: Zibrek et al. tested for an uncanny valley effect to rule it out as a confound, and found none.

Zibrek & McDonnell (2019) — Social Presence and Place Illusion are Affected by Photorealism in Embodied VR

Full reference: Zibrek, K., & McDonnell, R. (2019). Social presence and place illusion are affected by photorealism in embodied VR. In *Proceedings of the 12th ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG 2019)*, 1–7.

DOI: 10.1145/3359566.3360064

Research question. Does photorealism alone raise social presence, place illusion and empathetic concern when the viewer is embodied in a virtual body rather than observing from outside?

Stimulus design. A within-subjects experiment. Participants were embodied in a high-fidelity virtual body and observed an animated character rendered in photorealistic and simplified style, matched to the style of the environment. The character showed interactive eye-gaze and an emotional reaction intended to induce empathy. Both self-reported measures and a behavioural measure, interpersonal distance, were collected.

Key results.

- **Photorealism raised self-reported social presence and place illusion.**
- **Empathetic concern depended on condition order** rather than on render style.
- **The behavioural proximity measure was not affected by render style.**

Interpretation. A rendering change can shift how present a character feels without changing how a viewer behaves towards it. Read against the same authors' observer study, embodiment is what makes render style register on social presence.

Relevance to this thesis. A direct precedent for the self-report versus behaviour dissociation this thesis measures, and part of the justification for pairing Likert scales with a forced-choice ranking. It also bounds the video-based design: effects that require embodiment cannot appear in a flat-screen study.

Zell et al. (2015) — To Stylize or Not to Stylize? Shape and Material Stylization on CG Face Perception

Full reference: Zell, E., Aliaga, C., Jarabo, A., Zibrek, K., Gutierrez, D., McDonnell, R., & Botsch, M. (2015). To stylize or not to stylize? The effect of shape and material stylization on the perception of computer-generated faces. *ACM Transactions on Graphics*, 34(6), Article 184.

DOI: 10.1145/2816795.2818126

Research question. When designing a stylised 3D character, which parameters matter most for how observers perceive the character? Zell et al. identified two primary appearance variables: *shape* (the overall geometric form, ranging from highly stylised cartoon geometry to realistic scanned head geometry) and *material* (the combination of shader model, shading parameters, and albedo texture, spanning the same cartoon-to-realistic range). The study asks how varying each of these two dimensions independently, and in combination, affects perceived realism, appeal, eeriness, familiarity, and attractiveness.

Methodology. Three experiments were conducted. Characters were based on 3D scans of a male and a female actor, from which professional 3D artists produced stylised versions at multiple levels inspired by commercial animation films — specifically *Cloudy with a Chance of Meatballs* and *Toy Story* for the stylised levels, and *Beowulf* for the realistic level. Experiment 1a crossed three shape stylisation levels with three material levels (nine combinations) across four facial expressions in static renders. Experiment 1b tested shading and lighting parameters independently of texture. Experiment 1c tested blurred-texture variants of the realistic albedo to isolate the role of skin impurity detail. Experiment 2 extended to five shape levels and five material levels (25 combinations) across two characters and five expressions, for a total of 250 stimuli. Experiment 3 measured how stylisation affects expression recognition and perceived emotional intensity. All ratings used seven-point Likert scales; statistical analysis used repeated-measures ANOVA with Greenhouse-Geisser correction and Tukey HSD post-hoc tests. Sample sizes ranged from 21 to 24 participants per experiment.

Finding 1: Shape governs realism; material governs appeal and eeriness. A consistent dissociation emerged across all experiments. Shape was the dominant factor for perceived realism: increasing shape realism reliably increased realism ratings. Material was the dominant factor for appeal, attractiveness, and eeriness: realistic materials were the least appealing and most eerie, while cartoon materials were consistently more appealing across all shape levels. The realism scale and the appeal scale were not significantly correlated ($\alpha < 0.5$ in all experiments), confirming that these two dimensions are independent perceptual axes.

Finding 2: Mismatch between shape and material is costly. When shape and material stylisation levels were mismatched, appeal and attractiveness dropped significantly below any matching combination, and eeriness increased. The most unappealing and eerie stimulus across all experiments was a realistic material applied to the most stylised cartoon shape. Matching the stylisation degree of shape and material maximised appeal regardless of the absolute level of realism in either dimension. This pattern generalised across both characters and all five expressions tested.

Finding 3: Realism is a bad predictor for appeal. The realism-versus-appeal space (Figure 12 in the paper) shows all shape-level data points dispersed across a wide range of appeal values at any given realism level. A character can be rated high on realism and low on appeal, or moderate on realism and high on appeal. The data are consistent with the uncanny valley (Mori, 1970) specifically for the most realistic shape and material combination, but the broader picture is that it is the *compatibility* of shape and material, rather than absolute realism, that determines where a character lands on the appeal curve.

Finding 4: Subtle material stylisation increases appeal without sacrificing realism. In Experiment 1c, applying a Gaussian blur of 50 pixels to the realistic albedo texture significantly increased appeal and attractiveness ($p < 0.0002$) while leaving realism ratings statistically

unchanged. This effect was strongest on cartoon and middle shapes and was not significant for the realistic shape, suggesting that texture simplification raises appeal specifically in the mid-range of stylisation where a realistic texture creates the greatest mismatch with simplified shape geometry.

Finding 5: Shape controls expression intensity; material does not. In Experiment 3, perceived expression intensity decreased continuously with increasing shape stylisation: more stylised shapes conveyed lower emotional intensity even when the expression was correctly recognised. Material stylisation had no significant effect on perceived expression intensity. Expression recognition accuracy nonetheless remained high across all stylisation levels for most emotions, confirming that stylised characters remain readable for basic emotional communication.

Limitations. The study used static images rather than animated characters. Character models were based on two specific individuals; the generalisability to very different face morphologies is uncertain. Stylisation levels were defined by artist interpretation of specific commercial animation reference films and may not map cleanly onto abstract stylisation scales used in other research.

Relevance to this thesis. This paper is the most directly applicable empirical precedent for the material-rendering NPR strategy used in this project. The thesis holds avatar shape constant across all three conditions and changes only the material rendering through NPR shaders. Zell et al. demonstrate that material is precisely the dimension governing appeal and eeriness independently of shape realism, meaning the NPR shader pipeline operates in the perceptual domain with the strongest effect on the outcome variables measured in the user study. The finding that mismatched stylisation raises eeriness also validates the design decision to apply a consistent NPR treatment across the full avatar surface rather than selectively stylising only skin or only clothing. The result from Experiment 1c, that moderate material stylisation increases appeal without degrading realism, provides direct empirical support for the prediction in H1 that the NPR condition (C3) will produce lower eeriness ratings than the photorealistic condition (C2).

Key citable sentences (original text and location).

- **Abstract / Main Findings, p. 184:1:** "Our experiments reveal that shape is the dominant factor when rating realism and expression intensity, while material is the key component for appeal. Furthermore our results show that realism alone is a bad predictor for appeal, eeriness, or attractiveness."
- **Main Findings list, p. 184:2:** "The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material stylization."
- **Main Findings list, p. 184:2:** "On the other hand, appeal, eeriness, and attractiveness are mainly affected by the stylization level of material, and not shape; realistic materials reduce appeal in general."
- **Main Findings list, p. 184:2:** "The degree of realism is a bad predictor for appeal or eeriness."

Exact sentence cited in thesis. The thesis claim that excessive stylisation undermines expressive readability cites `zell2015` using the following sentence from the Main Findings list (p. 184:2): *"The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material*

stylization." (Interpretation: as shape stylisation increases and realism decreases, perceived expression intensity also decreases — more stylised shapes convey lower emotional intensity.)

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #135** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  [check wording] THESIS: "The repeated-measures ANOVA is the established test for rating studies that compare rendering styles within participants."
  SOURCE (SECTION 4 (EXPERIMENT DESIGN), P. 184:4): "Similar to previous work on rendering style [McDonnell et al. 2012], we want to analyze how different levels of stylization (e.g., shape and material) change the perception of a virtual character. ... For statistical analysis of each rating scale we conducted an n-way repeated measures Analysis Of Variance (rm-ANOVA). We run Maulchly's test for validating sphericity of the data, and whenever it is significant we report results with Greenhouse-Geisser correction applied"
  CHECK: The paper does use a within-participants rm-ANOVA on Likert rating scales for stylization conditions, and explicitly follows prior render-style work, so it is a valid instance of the practice. However, it never asserts that rm-ANOVA is 'the established test' for rendering-style rating studies; the normative generalisation is the thesis's own, and a single paper cannot support it.

- **Entry #151** *(Ch.6 Evaluation > Discussion > What the Stylisation Left Unchanged)*
  [check wording] THESIS: "Material governs eeriness in the account of Zell et al. and this alone predicts a reduction for C3."
  SOURCE (SECTION 8 (DISCUSSION), P. 184:10): "On the other hand, we have found that material is the main factor for perceived appeal, specifically the albedo texture. In general, appeal, attractiveness, and eeriness are highly dependent on the material stylization. Matching levels of stylization of geometry and material cause the highest ratings of appeal, while strong mismatches (e.g., very realistic material on a stylized shape) result in unappealing and eerie characters."
  CHECK: The first clause is directly supported: eeriness depends mainly on material stylization, not shape. The second clause is not, since Zell et al. attribute eeriness to shape-material mismatch rather than to material level alone, and Experiment 1a reports that 'cartoon materials on the realistic shape' also lowered reassurance ($p < 0.02$, p. 184:5). Stylizing material while shape stays realistic therefore does not by itself predict reduced eeriness in their account.

Riefard et al. (2024) — Player Preferences for NPR Artistic Styles in 3D Game Characters

Full reference: Riefard, R. A., et al. (2024). Assessing player preferences for non-photorealistic rendering artistic styles in 3D video game character design. In *Proceedings of the 2024 International Conference on Electrical Engineering and Computer Science (ICECOS 2024)*. IEEE.

Available: `https://ieeexplore.ieee.org/document/10730294`

Research question. Which NPR artistic styles do players prefer for 3D video game characters, and do these preferences vary by demographic group?

Method. Survey study with game-playing participants. 35 NPR styles were generated for a humanoid 3D character, grouped into six categories: *Style Lines* (clean ink outlines, flat fill),

Cross-Hatch, *Patch-Hatch*, *Scribble-Hatch*, *Post Cartoon* (toon-shaded with soft gradients), and *Other*. Participants rated preference for each style on a 5-point Likert scale. Demographics: gender, age group, game genre preference.

Key results.

- **Overall most preferred:** Post Cartoon (mean 3.46) and Style Lines (mean 3.39), significantly above Cross-Hatch (2.95) and Scribble-Hatch (2.62).
- **Gender differences:** Males preferred Cross-Hatch over Post Cartoon; females showed the reverse preference ($p = .04$).
- **Age differences:** Participants under 18 showed significantly higher preference for Style Lines (mean 4.57) than any older group. Adults aged 18–35 converged toward Post Cartoon and Style Lines.
- **Genre interaction:** action and fighting game players preferred Style Lines; RPG and narrative game players preferred Post Cartoon.

Limitations. Evaluations were on still renders; character motion was not controlled. Preferences are influenced by prior exposure to game aesthetics.

Relevance to this thesis. Provides empirical evidence that NPR style preferences are heterogeneous and demographically segmented. This is relevant when interpreting individual variation in user study ratings of C3 (NPR condition): participants with different gaming backgrounds, ages, or gender may carry pre-existing aesthetic preferences for or against cartoon-style avatars that influence their eeriness and trustworthiness ratings independently of the uncanny valley effect being measured. Collecting demographic data allows post-hoc analysis of whether such preference effects are present.

Roshaan (2026, preprint) — AHEAD: Adaptive Hierarchical Edge Detection for Real-Time Artistic Stylisation

Full reference: Roshaan, M. K. L. (2026). AHEAD: Adaptive hierarchical edge detection for real-time artistic stylization. *Research Square preprint*, Article rs-8849147/v1. February 2026.

Preprint: `https://www.researchsquare.com/article/rs-8849147/v1` Code: `https://github.com/Chronos-Asteri/AHEAD-AdaptiveEdgeDetection`

Note: Verify against the physical PDF `Adaptive Hierarchical Edge Detection.pdf` in the Articles folder. These may be the same paper at different stages of publication.

Problem. Hierarchical edge detection systems for real-time NPR (combining depth, normal, and colour gradients) require manual threshold tuning for each scene and material. A forest environment needs different thresholds than a character portrait; skin needs different sensitivity than metal. This makes hierarchical systems impractical in production game engines where scenes and materials vary continuously.

AHEAD architecture. A unified real-time post-processing shader with three hierarchical detection layers and adaptive per-layer thresholds.

Adaptive sensitivity. Each layer's detection threshold is set proportional to the standard deviation of the layer's raw response across the current frame: $T_\ell = k \cdot \sigma_\ell$. When the response

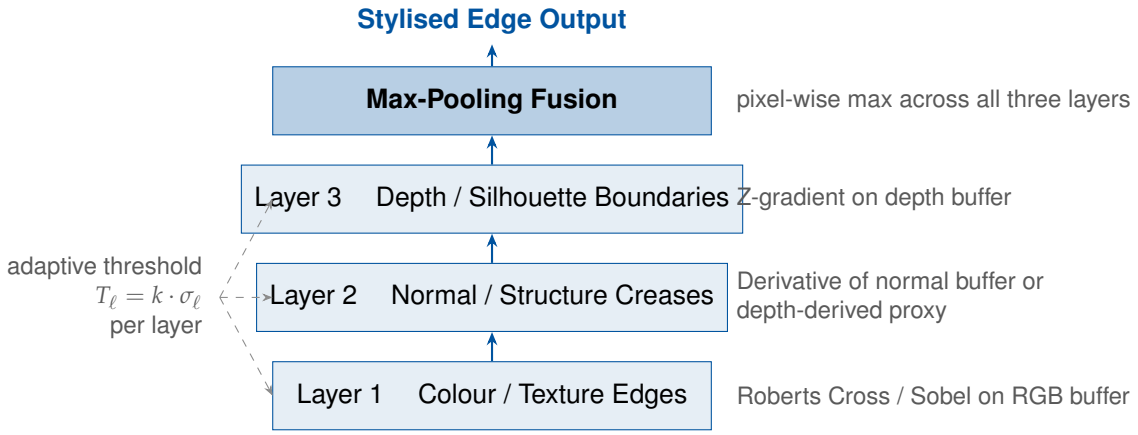


Figure 3: AHEAD hierarchical architecture (Roshaan, 2026). Three detection layers run with adaptive per-layer thresholds $T_\ell = k \cdot \sigma_\ell$ where σ_ℓ is the standard deviation of the raw layer response across the frame. Max-pooling fusion preserves the strongest edge response at each pixel without additive ghost artefacts.

has high variance (many strong edges), the threshold rises; when variance is low (a flat region), the threshold falls, recovering subtle edges that fixed thresholds miss. The constant k is the only global parameter requiring manual tuning, and it transfers across scenes.

Max-pooling fusion. The final edge map is the pixel-wise maximum of the three layer responses after thresholding. This avoids double-edge artefacts from additive summation and preserves relative response strength for anti-aliased edge rendering.

Results. Comparison against fixed-threshold hierarchical edge detection across four game environments: AHEAD required 73% fewer manual threshold adjustments while producing equivalent or better edge coverage. Frame rate overhead on tested game engines was below 0.5 ms per frame.

Relevance to this thesis. AHEAD directly formalises the hierarchical approach used in V5. The V5 shader combines depth proxy, normal derivatives, and Roberts Cross colour edges in a three-cue structure corresponding exactly to AHEAD's Depth/Silhouette, Normal/Structure, and Colour/Texture layers. Max-pooling fusion is also used in V5. The adaptive sensitivity mechanism is the one feature not yet implemented in V5, which uses fixed thresholds tuned per scene. This paper is the primary methodological reference for V5.

Passages this thesis relies on. Each claim below was checked against this paper.

- **Entry #144** *(Ch.5 Implementation > NPR Technique Implementation > V5 — Hierarchical Edge Detection)*
  THESIS: "The normal layer reuses the world-space normal derivative computation from V3 and an AHEAD-style adaptive gain then amplifies this layer specifically in darker image regions:"
  SOURCE (SECTION 2.4 'ADAPTIVE FEATURE ENHANCEMENT', PP. 4-5 (PREPRINT, DOI 10.21203/RS.3.RS-8849147/V1)): "The algorithm acts as a localized Automatic Gain Control (AGC), inspired by photographic tone reproduction [13], to match the computational edge response with perceptual significance. The adaptive scalar S(x, y) is modeled as a modified inverse-linear function of the local pixel luminance L(x, y) ... Lower luminance areas have higher S values (see Fig. 3), therefore amplifying subtle structural details in shadowed areas.

While, higher luminance areas have lower S values, which reduces the effect of applied edge detection ...”
CHECK: The cited mechanism is exactly as described: a luminance-driven inverse-linear gain that amplifies edge response specifically in darker regions, motivated by the Weber-Fechner Law. One difference the thesis’s ’AHEAD-style’ wording already signals: AHEAD applies the gain only to its Structure layer (Scharr/Kirsch on image intensity, Section 2.5), never to a world-space normal derivative, and it never applies any operator to the normal map, which appears only as a G-buffer illustration in Fig. 2a.

Raskar & Cohen (1999) — Image Precision Silhouette Edges

Full reference: Raskar, R., & Cohen, M. (1999). Image precision silhouette edges. In *Proceedings of the 1999 Symposium on Interactive 3D Graphics* (pp. 135–140). Atlanta, GA, April 26–29, 1999.

What the paper does. The paper presents a general-purpose method for rendering the visible silhouette edges of polygonal objects entirely in image precision, using only a conventional depth-buffered polygon rendering pipeline. Its central observation is that only two sets of primitives are required to compute silhouette edges: the layer of visible polygons nearest the viewpoint (P_1) and the second layer of polygons behind them (P_2). The intersection of these two layers, computed in image space where their depth values agree, yields the silhouette. The formulation combines silhouette detection, the partial visibility problem, and optional shading into a single rendering process, and the authors describe it as a generalisation of earlier framebuffer contour methods.

Method. The base algorithm renders front-facing polygons in white with the depth function set to ‘Less Than’, then renders back-facing polygons in black with the depth function set to ‘Equal To’, so that only pixels where the two layers coincide are marked. Because pixel sampling and depth-buffer quantisation prevent the two depth values from ever agreeing exactly, and because the resulting edges would be at most one pixel wide, the authors propose three practical variants: rendering back-facing polygons in wireframe mode, translating filled back-facing polygons towards the camera by a fixed or depth-scaled amount, and view-dependent fattening of back-facing polygons. In the fattening variant, each edge of a back-facing polygon is pushed outwards by a distance proportional to $z/(V \cdot N_B)$, and the required fattening for an edge vector E is proportional to $z \sin(\alpha)/(V \cdot N_B)$ in the direction $E \times N_B$. The methods were implemented in OpenGL and timed on SGI hardware across several polygonal models.

Key findings. None of the variants require preprocessing or polygon adjacency information, which makes the approach applicable to dynamic scenes, models with changing topology, and models that switch level of detail during rendering. The authors report that the method is also robust to inconsistent vertex ordering in the source models. Measured performance on a 5,672-triangle model dropped from 66 frames per second without silhouettes to 40 (wireframe), 50 (depth-scaled translation), and 11.5 (view-dependent fattening) frames per second, and a 252,000-triangle model dropped from 6 to 4.2 frames per second, giving roughly a factor of two cost for the cheaper variants. Stated limitations include a required full scene traversal, increasing edge clutter as objects recede from the camera, and silhouette widths that vary with the dihedral angle between adjacent faces unless fattening is applied.

Relevance to this thesis. This paper is the direct antecedent of the inverted hull outline technique implemented as V1 across the Mixamo, Meta Avatars SDK, and Avaturn pipelines, since both rely on expanding back-facing geometry rather than on explicit edge extraction. Its argument that silhouette rendering survives level of detail switching and topology changes without adjacency data is directly relevant to avatars whose meshes are managed by a closed SDK. It also establishes the limitation that motivates the later screen-space techniques in this thesis: a geometric silhouette method conveys only the object profile and carries no interior surface detail.

Saito & Takahashi (1990) — Comprehensible Rendering of 3-D Shapes

Full reference: Saito, T., & Takahashi, T. (1990). Comprehensible rendering of 3-D shapes. *Computer Graphics (Proceedings of SIGGRAPH '90)*, 24(4), 197–206.

What the paper does. The paper introduces the Geometric Buffer, or G-buffer, and with it the idea that comprehensibility in rendering is produced by deliberate enhancement rather than by accurate simulation of optics. A G-buffer set stores one geometric property per pixel, such as perspective depth (`sz`), world coordinates (`wx`, `wy`, `wz`), surface normal components (`nx`, `ny`, `nz`), patch parameters (`ou`, `ov`), and an object or patch identifier (`id`). The authors develop drawing algorithms for discontinuities, edges, contour lines, and curved hatching, all realised as 2-D image processing operations on these buffers instead of as line tracking. Enhancement is thereby separated from the geometric processes of projection and hidden surface removal and from the physical processes of shading and texture mapping.

Method. Discontinuities are extracted from a G-buffer with a first order differential operator, for which the authors recommend the Sobel operator, normalised so that the response corresponds to the gradient per pixel, and with a second order operator $l = (8X - A - B - C - D - E - F - G - H)/3$ for discontinuities of the first derivative. Profiles are 0th order discontinuities of the depth image and internal edges are first order discontinuities of the same image. Raw differential responses are unsuitable on their own because large continuous changes are hard to separate from true discontinuities, because line darkness depends on the size of the gap, and because second order operators produce doubled positive and negative lines. The authors therefore apply normalisation operators based on the minimum and maximum gradient values in the 3×3 neighbourhood, controlled by constants k_g and k_l. Contour lines and curved hatching are drawn as raster data with an antialiased density function, and a binary thinning out scheme keeps hatching at approximately regular pixel intervals.

Key findings. Because the G-buffer set contains no physical properties such as reflectance or colour, any combination of enhancement techniques can be examined without recomputing the buffers, provided shapes and camera parameters stay fixed. Edges can be drawn stably by 2-D image processing operators even for complicated free-form surfaces, which the authors contrast with the difficulty of hidden line elimination in object space. Extracting edges from the object identifier image is simple but incomplete for concave curved surfaces, and combining the identifier and depth methods is suggested for exact results. The authors demonstrate the approach on edge enhancement, line drawing illustration, topographical and bird's-eye maps, medical imaging from CT data, and surface analysis, and they note that line drawing output survives repeated photocopying far better than shaded output.

Relevance to this thesis. Saito and Takahashi established the deferred, image-space paradigm that every screen-space technique in this thesis inherits: geometric attributes are written to per-pixel buffers and edges are then found by differential operators applied to those buffers rather than to the mesh. Their separation of depth discontinuities (profiles) from normal or first order discontinuities (internal edges) is the conceptual foundation of the multi-cue hierarchical edge detection used as V5, which composites depth, normal, and colour responses. Their observation that raw gradient magnitudes must be normalised against local neighbourhood extrema also anticipates the threshold calibration required when the same operators are applied to photographic avatar textures.

Nienhaus & Döllner (2003) — Edge-Enhancement: An Algorithm for Real-Time Non-Photorealistic Rendering

Full reference: Nienhaus, M., & Döllner, J. (2003). Edge-enhancement — An algorithm for real-time non-photorealistic rendering. *Journal of WSCG*, 11(1). WSCG'2003, February 3–7, 2003, Plzen, Czech Republic. ISSN 1213-6972.

What the paper does. The paper proposes a real-time edge enhancement algorithm built around the edge map, a 2-D texture that encodes the visually important edges of 3-D scene objects. The edge map distinguishes three edge types: silhouette edges between a front-facing and a back-facing polygon, border edges belonging to exactly one polygon, and crease edges between two similarly oriented polygons whose dihedral angle exceeds a threshold. Silhouette and border edges together outline the profile of an object while crease edges outline its inner forms, and the algorithm keeps these two categories separately available to fragment shading. The authors present the technique as orthogonal to other real-time rendering methods, so that it can be layered onto existing non-photorealistic styles.

Method. The implementation is a multipass pipeline. A first pass renders the scene into a single 2-D texture holding eye-space normalised fragment normals in the RGB channels, produced with a normalisation cube map, and fragment depth in the alpha channel, produced with a projected 1-D alpha texture that stores linear camera distance rather than hardware z. Intermediate passes render a screen-aligned quad with multitexturing to emulate image processing. The authors give two operator variants: a diagonal one that compares the two diagonally opposite sample pairs (A, H) and (C, F), computing I_N from the dot products of the expanded normals and I_Z from the products of the depth differences, and a C^1 discontinuity operator over all eight neighbours taken from Saito and Takahashi. Final passes apply the resulting edge map to the scene by projective texturing, and fragment shading blends edge colour with surface shade using the stored intensities.

Key findings. The authors report that the diagonal four-sample variant produces visually superior and better antialiased edge intensities than the eight-neighbour C^1 operator, which introduces artefacts. Storing edges as a texture rather than in the framebuffer means the technique does not consume stencil or other framebuffer resources, and the intermediate passes are inexpensive because they render only a single screen-aligned quad. Depth discontinuities in the alpha channel are independent of shape position because the stored value is linear camera distance, though the alpha texture has lower precision than the hardware depth buffer, which the authors mitigate by fitting the near and far clipping planes to the bounding vol-

umes. They demonstrate combination with the Gooch technical illustration model, real-time hatching, cartoon-style rendering following Lake et al., and mirroring. A stated limitation is that enhancement is restricted to the geometric boundaries of the scene objects because of the projective texturing, and edges are only a few texels wide, so artistic peripheral silhouettes are not achievable.

Relevance to this thesis. Nienhaus and Döllner supply the real-time counterpart to Saito and Takahashi that the screen-space techniques in this thesis actually resemble, since edge intensities are derived from normal and depth data sampled in the fragment stage rather than from an offline image processing step. Their explicit separation of profile edges from inner crease edges, and the use of separate intensity channels for the normal and depth responses, corresponds to the layered structure of the hierarchical multi-cue technique implemented as V5. Their finding that a compact diagonal sampling pattern outperforms the full eight-neighbour operator is directly pertinent to a pipeline that must keep sample counts low inside a single fragment hook.

Winnemöller, Kyprianidis & Olsen (2012) — XDoG: An eXtended Difference-of-Gaussians Compendium

Full reference: Winnemöller, H., Kyprianidis, J. E., & Olsen, S. C. (2012). XDoG: An eXtended difference-of-Gaussians compendium including advanced image stylization. *Computers & Graphics*, 36(6), 740–753.
DOI: `10.1016/j.cag.2012.03.004`

What the paper does. The paper is a detailed review and extension of the difference-of-Gaussians (DoG) operator for image stylisation, arguing that the operator has been under-used because its relationship to classical edge detectors is not widely understood. The authors trace the operator from gradient-based detection through Marr and Hildreth's Laplacian of Gaussian, of which the DoG is a separable and therefore efficient approximation, and they retain the commonly cited ratio $k = 1.6$. They then draw an explicit distinction between edge detection for computer vision, which is concerned with the exact localisation and extent of an edge, and edge enhancement for artistic use, which is concerned with the weight and structure of an edge. The compendium presents a family of effects and styles produced by a single operator, including abstraction, indication, speed-lines, ghosting, negative edges, thresholding, woodcut, hatching, and pastel.

Method. The extended operator introduces a variable inhibition strength τ, giving $D_{\sigma,k,\tau}(x) = G_\sigma(x) - \tau \cdot G_{k\sigma}(x)$, and replaces binary thresholding with a soft ramp $T_{\varepsilon,\varphi}$ built on a tanh falloff, so that $T_{\varepsilon,\varphi}(D_{\sigma,k,\tau} * I)$ constitutes the XDoG filter. The authors observe that this parameterisation is hard to control because τ, φ, and ε must be adjusted in concert, and they derive an invertible reparameterisation by dividing through by $\tau - 1$, yielding a sharpening operator $S_{\sigma,k,p}(x) = (1 + p) \cdot G_\sigma(x) - p \cdot G_{k\sigma}(x)$. The reparameterised form decouples the parameters, expresses edge emphasis as a digital unsharp mask of the blurred image, and measures ε proportionally to image intensity. The isotropic formulation is optionally combined with a flow-based variant (FDoG) that estimates local orientation from the smoothed structure tensor and separates the single σ into σ_c, σ_e, and σ_m for tensor smoothing, gradient-aligned filtering, and edge-tangent line integral convolution.

Key findings. The authors report that the visual quality achieved by the extended operator is

comparable to or better than that of systems dedicated to a single style, while being simpler and faster to compute and requiring no separate base and detail layers or compositing tables. They find that setting ε close to the midtone grey value of the image with p near 20 tends to produce useful stylisations, and that $\varphi \gg 0.01$ effectively converts the soft ramp into a step function for two-tone output. Increasing σ progressively removes concrete detail and simplifies shape, which the paper frames as a controllable level-of-detail abstraction rather than as blurring. The isotropic operator responds excessively to stochastic noise and textures, which the flow-based variant suppresses by aligning the filter with the edge tangent flow, and the paper also reports that low contrast source images remain a challenge for any edge detector. Negative edges, where a positive DoG response inside a dark region produces a white line, are identified as a property of the operator rather than an added effect.

Relevance to this thesis. This paper is the authoritative reference for the difference-of-Gaussians formulation used as the colour layer of the hierarchical multi-cue technique in the Meta Avatars SDK pipeline, and it supplies the soft thresholding and sharpening parameterisation on which that layer depends. Its distinction between localisation-oriented edge detection and weight-oriented edge enhancement clarifies why operators tuned for computer vision behave poorly on stylised avatar faces. Its finding that the isotropic operator is noise sensitive on textured input, and that the remedy requires structure tensor estimation, corroborates the pre-smoothing strategy behind the Gaussian-prefiltered Sobel technique and explains why the fully flow-aligned variant is not attainable inside a single-pass fragment hook.

Graphics Foundations

Blinn (1977) — Models of Light Reflection for Computer Synthesized Pictures

Full reference: Blinn, J. F. (1977). Models of light reflection for computer synthesized pictures. *SIGGRAPH '77 Conference Proceedings*, San Jose, California, July 20–22, 1977, pp. 192–198. University of Utah.

What the source does. The paper addresses the intensity calculation stage of image synthesis, that is, the determination of the intensity of an object once its visibility has already been established. Blinn observes that surfaces were typically modelled as perfect diffusers with a specular component added for the simulation of highlights, and he presents a more accurate highlight function grounded in experimental measurements of how light reflects from real surfaces. The proposed function differs from earlier models in that the intensity of the highlight changes with the direction of the light source, and the position and shape of the highlights also differ. A further stated difference is that the highlight function produces different results for metallic and nonmetallic surfaces.

Method. Blinn first restates the Phong highlight model in terms of the vector $\mathbf{H}$, defined as the direction of maximum highlights, pointing halfway between the source direction $\mathbf{L}$ and the eye direction $\mathbf{E}$, with the specular term given by $(\mathbf{N} \cdot \mathbf{H})$ raised to a power that controls highlight sharpness. He then derives the Torrance–Sparrow model in terms of the normalized vectors $\mathbf{N}$, $\mathbf{L}$, $\mathbf{H}$ and $\mathbf{E}$, treating the surface as a collection of mirror-like micro facets oriented in random directions. The specular reflection combines a facet distribution function D, a geometrical attenuation factor G for masking and shadowing between facets, a Fresnel term F giving the fraction of incident light actually reflected rather than absorbed, and a division by $\mathbf{N} \cdot \mathbf{E}$ that accounts for the observer seeing more surface area as the surface tilts. Three candidate distribution functions are compared, and Blinn adopts the Trowbridge–Reitz function, which models the micro facets as ellipsoids of revolution with an eccentricity parameter that is 0 for very shiny surfaces and 1 for very diffuse surfaces.

Key points. Blinn states that the geometrical attenuation factor takes a value between 0 and 1 and is evaluated as the minimum of three cases corresponding to no interference, interception of reflected light, and interception of incident light. Plots of the reflected distribution at 30 and 70 degrees of incidence show that the Phong and Torrance–Sparrow functions are almost identical at moderate angles, and that the specular bump is much larger and not in quite the same direction at shallow angles, so the new function differs materially only for grazing illumination. The Fresnel term has a substantially different form for metallic and nonmetallic substances: for metals, corresponding to large values of the index of refraction, it is nearly constant at 1, whereas for nonmetals it starts near zero and rises towards 1 at grazing incidence, illustrated with rendered simulations of an aluminium surface ($n = 200$) and a magnesium oxide ceramic ($n = 1.8$). Blinn further shows that the surface roughness parameter can be modulated across the surface by a texture function, with a constant normalizing factor based on its minimum value, and he concludes that the cheaper distribution function offsets the extra cost of G and F, yielding the improved highlight function for no increase in computation time.

Relevance to this thesis. This source is the origin of the half-angle formulation $(\mathbf{N} \cdot \mathbf{H})$ that underlies the specular term of the conventional shading model, which is the photorealistic base

that the abstraction operators in this thesis quantise, filter or override. It matters particularly for the closed Meta Avatars SDK, where the only available hook runs per fragment after the SDK has already computed its own physically based shading, so the incoming colour that the abstraction operators receive is the output of exactly this class of reflection model. Blinn's demonstration that a shading parameter can be driven across a surface by a texture function is also a direct precedent for the texture-driven parameter control used in the abstraction pipeline.

Lewis et al. (2014) — Practice and Theory of Blendshape Facial Models

Full reference: Lewis, J. P., Anjyo, K., Rhee, T., Zhang, M., Pighin, F., & Deng, Z. (2014). Practice and theory of blendshape facial models. In S. Lefebvre & M. Spagnuolo (Eds.), *EUROGRAPHICS 2014 — State of the Art Reports*. Eurographics Association. (Computer Graphics Forum, Vol. 33, No. 2.)

What the source does. This state of the art report surveys blendshapes, described by the authors as a simple linear model of facial expression and as the prevalent approach to realistic facial animation. The report notes that the blendshape approach originated in industry rather than academia and became a subject of academic research relatively recently, and it therefore covers both the graphics research literature and developments published in industry forums. It positions blendshapes among the alternative families of facial animation methods, including physically based models, parametric models, proprietary deformers, dense motion capture driven meshes, and principal component analysis (PCA) models. The authors argue that, despite the simplicity of the approach, open problems associated with this fundamental technique remain unresolved.

Method. A blendshape model generates a facial pose as a linear combination of a set of facial expressions called targets, so that varying the weights of the combination produces a range of expressions with little computation. The report formalises this by unrolling each target into a long vector, giving the whole-face form $f = Bw$, and the delta form $f = b_0 + Bw$ in which one model is designated the neutral shape and the remaining targets are replaced by their differences from it. Weights in the delta formulation are conventionally limited to the range $[0, 1]$, and the report shows that the whole-face and delta formulations are equivalent in expressive power when the whole-face weights are convex. Extensions covered include intermediate targets giving piecewise linear interpolation, and combination blendshapes, in which correction shapes are activated by products of pairs or triples of weights.

Key points. The report identifies two stated advantages of blendshapes over other linear representations: the parameterization is semantic, since the weights have intuitive meaning for the animator as the strength of a named expression such as raise-right-eyebrow, and the representation keeps the animator "on model" by preventing arbitrary deformations. Building such a model is described as a large and labour intensive effort, with the character of Gollum in *The Lord of the Rings* cited at 946 targets, professional models having on the order of 100 primary targets plus several hundred combination shapes, and a detailed model taking as much as a year of work for a skilled modeller. The authors contrast blendshapes with PCA, arguing that PCA basis vectors are global, are difficult to interpret and remember, and produce visibly less sparse coefficient sequences than the blendshape encoding of the same animation. Animation techniques are grouped into keyframe animation, performance-driven animation, and direct

manipulation, with model-based tracking solving for the blendshape weights at each frame, typically constrained to the range $0 \dots 1$, and expression cloning retargeting motion from a source face to a target face with significantly different proportions.

Relevance to this thesis. The Avaturn avatars used in this thesis are animated through ARKit blendshape coefficients, and this report supplies the underlying formulation, namely a neutral shape plus a weighted sum of semantically named difference targets with weights normally bounded to $[0, 1]$. It establishes that facial performance is carried entirely by vertex-stage linear deformation, which is what allows the abstraction operators to be confined to the fragment stage without interfering with the facial animation. The section on expression cloning and retargeting to characters of different proportions also grounds the fact that a single captured performance can drive avatars from different platforms whose meshes and topologies do not match.

Luebke et al. (2003) — Level of Detail for 3D Graphics

Full reference: Luebke, D., Reddy, M., Cohen, J. D., Varshney, A., Watson, B., & Huebner, R. (2003). *Level of detail for 3D graphics*. San Francisco, CA: Morgan Kaufmann Publishers, an imprint of Elsevier Science. ISBN 1-55860-838-9. (The Morgan Kaufmann Series in Computer Graphics and Geometric Modeling, series editor B. A. Barsky.)

What the source does. The book presents level of detail (LOD) as the discipline of interactive computer graphics that bridges complexity and performance by regulating the amount of detail used to represent the virtual world. The authors frame the field as a response to a persistent tension between fidelity and frame rate, arguing that the polygon complexity of models grows faster than the ability of hardware to render it. The stated fundamental concept of LOD is to use a less detailed representation for small, distant, or unimportant portions of the scene. The volume collects research, theory and algorithms from journals and conferences into three parts, covering generation of LODs, application and run-time management, and advanced issues including visual perception, measurement of visual fidelity, and temporal factors such as frame rate and latency.

Method. The authors credit James Clark's 1976 paper on hierarchical geometric models with codifying the basic principles of LOD, and note that flight simulators were probably the earliest applications to make extensive use of it, with LODs created by hand. Three run-time frameworks are then distinguished. Discrete LOD creates multiple versions of every object at different levels of detail during an offline preprocess and selects one at run-time, and since the simplification cannot predict the viewing direction it reduces detail uniformly, which is why the authors also call it isotropic or view-independent LOD. Continuous LOD instead encodes a continuous spectrum of detail in a data structure from which the desired level is extracted at run-time, and view-dependent LOD extends this with view-dependent criteria so that a single object can span multiple levels of simplification.

Key points. The authors state that discrete LOD is the simplest model to program because simplification and rendering are decoupled, and that it suits modern graphics hardware because each individual LOD can be compiled during preprocessing into an optimal rendering format such as triangle strips, display lists, or vertex arrays. Continuous LOD is credited with better granularity, since the level of detail is specified exactly rather than selected from a few precreated

options, and with support for streaming, progressive rendering and interruptible loading. View-dependent LOD gives still better fidelity for a given polygon count and handles very large or poorly segmented objects such as terrain, at the cost of extra run-time processing and extra memory for the view-dependent data structures. Despite these advances, the authors report that traditional discrete LOD remains by far the most common approach in practice, and that video games have been notably reluctant to adopt view-dependent LOD outside situations that almost require it. The book also notes that shading and illumination form their own spectrum of simplification, and that replacing geometric detail in LODs with texture maps, termed imposters, is especially common in video games and visual simulation.

Relevance to this thesis. This source supplies the standard terminology and the run-time model for the LOD behaviour encountered in the closed Meta Avatars SDK, where the avatar is represented by several precomputed versions and the system substitutes one for another at run-time. Its account of discrete LOD as an offline preprocess that compiles each version into its own optimised rendering format explains why a representation swap of this kind is a wholesale exchange of the object being drawn rather than a continuous adjustment, which is the context in which the material assignment carrying the abstraction shader is lost. The book's framing of LOD as the regulation of detail for fidelity against cost also connects the engineering constraint to the wider argument of the thesis, in which detail is deliberately reduced for perceptual rather than performance reasons.

Statistical Methods

Demšar (2006) — Statistical Comparisons over Multiple Data Sets

Full reference: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7, 1–30.

What the paper does. The article reviews how machine learning researchers had been testing performance differences between two or more classifiers evaluated on multiple data sets, and it establishes which statistical tests are actually defensible for that setting. Demšar first surveys the International Conference on Machine Learning proceedings from 1999 to 2003 and reports that between one quarter and one half of the relevant papers apply some statistical procedure, most often uncorrected pairwise t tests, and that the Bonferroni correction appears in only a few papers per year. He then examines three families of tests theoretically and empirically: parametric tests (the paired t test and repeated-measures ANOVA), non-parametric tests (the Wilcoxon signed-ranks test and the Friedman test), and the sign test, which assumes no commensurability at all. The conclusion is that the non-parametric tests should be preferred, and the paper additionally introduces critical difference (CD) diagrams as a compact visualisation of post-hoc results.

Method. Each candidate test is analysed against the assumptions it makes and against the properties of the data at hand, where the sample size is the number of data sets rather than the number of resampled runs. Demšar argues that the paired t test fails on commensurability, on normality (the number of data sets is usually well below 30), and on sensitivity to outliers, and that repeated-measures ANOVA additionally requires sphericity, which cannot be assumed. The Friedman test is presented as the non-parametric counterpart of repeated-measures ANOVA: it ranks the k algorithms within each of the N data sets, assigns average ranks in case of ties, and tests whether the average ranks R_j depart from the value expected under the null hypothesis using the χ_F^2 statistic, or the less conservative F_F statistic of Iman and Davenport. The empirical part runs the tests on real classifiers and data sets while biasing the data set selection to vary the true difference, and it records rejection likelihood and replicability rather than Type 1 and Type 2 error rates, since the correct decision is not well defined in that setting.

Key recommendations. For a comparison of exactly two classifiers over multiple data sets, Demšar recommends the Wilcoxon signed-ranks test in place of the paired t test, since it assumes commensurability only qualitatively, does not assume normality, and is less affected by outliers. For more than two classifiers he recommends the Friedman test as the omnibus test, followed by a post-hoc procedure only if the omnibus null hypothesis is rejected. It is important to be precise about what those post-hoc procedures operate on: every one of them (Nemenyi, Bonferroni-Dunn, Holm, Hochberg, and Hommel) is applied to the *average ranks* produced by the Friedman test, through the statistic $z = (R_i - R_j)/\sqrt{k(k+1)/6N}$, and Demšar does not anywhere pair Holm with a set of separately computed Wilcoxon signed-rank tests across all pairs. Within that rank-based framework he describes Holm's step-down procedure, which orders the p values $p_1 \leq p_2 \leq \ldots \leq p_{k-1}$ and compares each p_i against $\alpha/(k-i)$, stopping as soon as a hypothesis cannot be rejected, and he states that Holm is more powerful than Bonferroni-Dunn and "makes no additional assumptions about the hypotheses tested," while Hochberg and Hommel offer no great practical advantage over the simpler Holm method. He also notes explicitly that these family-wise error procedures "can be used generally for controlling the family-wise error when multiple hypotheses of possibly various types are tested," which is the sentence that licenses

Holm outside his own rank-based context. Regarding all-pairs testing, his caution is conditional rather than absolute: the Nemenyi test is offered precisely for the all-pairs case, but he shows that its critical values are adjusted for $k(k-1)/2$ comparisons against only $k-1$ for a control comparison, and he concludes that "we thus should not make pairwise comparisons when we in fact only test whether a newly proposed method is better than the existing ones." Separately, he criticises the practice of reporting a matrix of uncorrected pairwise t tests, and he calls the plain Bonferroni correction conservative and weak because it supposes independence of the hypotheses.

Relevance to this thesis. The paper supplies the general justification for preferring Holm's step-down correction over a plain Bonferroni correction whenever a family of related hypotheses is tested, since Demšar establishes that Holm is uniformly more powerful, adds no assumptions, and is applicable to multiple hypotheses of various types. It also provides the two-stage logic the thesis follows for its ranking measures, namely an omnibus Friedman test first and pairwise follow-up only after rejection, together with the argument that the Wilcoxon signed-ranks test is the safer non-parametric substitute for the paired t test on related samples. One honest limitation must be stated when citing it: Demšar's own post-hoc tests act on Friedman average ranks rather than on separate Wilcoxon signed-rank tests per pair, and his "sample" is a set of data sets rather than a set of participants, so the paper supports the Holm correction and the Friedman-then-post-hoc structure used here without prescribing the exact Wilcoxon post-hoc combination applied in the analysis.

Passages this thesis relies on. Full source quotes for these claims are in `citation_check.pdf`.

- **Entry #119** *(Ch.4 Methodology > Parameter Adjustment Study)*
  THESIS: "An omnibus Friedman test established whether the ranking within a round differed from chance and a pairwise Wilcoxon signed-rank test compared the configuration ranked first against its closest competitor."

- **Entry #134** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Holm's procedure is more powerful than the Bonferroni correction and makes no additional assumptions about the hypotheses tested."

- **Entry #142** *(Ch.4 Methodology > User Study Design > Procedure and Measures)*
  THESIS: "Pairwise Wilcoxon signed-rank tests with the same Holm correction were then applied to each pair and C2 versus C3 is the critical pairwise comparison in each case."

Lakens (2017) — Equivalence Tests: A Practical Primer

Full reference: Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4), 355–362.
DOI: `10.1177/1948550617697177`

What the paper does. Lakens addresses the fact that a non-significant result in a traditional null hypothesis significance test cannot support the absence of an effect, since the only formally correct conclusion when $p > \alpha$ is that the data are not surprising under the null hypothesis. He documents how widespread the misinterpretation is: the phrase "no effect" appeared in 108 articles in *Social Psychological and Personality Science* up to August 2016, and manual inspection

showed that in almost all of them the claim rested on statistical non-significance alone. The article presents the two one-sided tests (TOST) procedure as the frequentist remedy, and it supplies formulas, power analyses, a spreadsheet, and an R package for one-sample, dependent, and independent t tests, for correlations, and for meta-analyses. The stated goal is to let researchers statistically reject the presence of effects large enough to be considered worthwhile.

Method. In the TOST procedure an upper bound Δ_U and a lower bound Δ_L are specified from the smallest effect size of interest (SESOI), for instance Cohen's $d = -0.3$ and $d = 0.3$, and two composite null hypotheses are tested against those bounds. When both one-sided tests are statistically rejected, the observed effect lies inside the equivalence bounds and can be treated as practically equivalent to zero. Lakens derives the t statistics for each design, and for dependent means the correlation between the paired observations enters the denominator directly, with the relevant standardised effect size being Cohen's d_z rather than Cohen's d. He also gives the normal approximation power equation following Chow, Shao, and Wang, and he notes that the number of pairs needed in a within-subjects design is half the per-condition sample size required for the equivalent between-subjects test.

Key recommendations. Equivalence bounds must be prespecified, and Lakens argues that this should be considered best practice in psychology in the same way that CONSORT guidelines expect it in randomised controlled trials. When no theoretical or practical boundary on a meaningful effect exists, he recommends setting the bounds to the smallest effect size the study has sufficient power to detect, which is determined by the maximum sample size the researcher is able to collect. Where feasibility constraints are effectively absent, he accepts conventional benchmarks for small, medium, and large effects as a starting point for the bounds, while noting that using such benchmarks is normally a last resort. He advises reporting the equivalence bounds in the abstract whenever an effect is described as statistically equivalent, and he cautions that in small samples a study may have no power for an equivalence test at all, because the confidence interval will necessarily be wider than the bounds. Two further limits are stated explicitly: equivalence can only be concluded for the specific operationalisation tested, and a bound expressed as a standardised effect size implies different raw bounds across studies whenever the pooled standard deviation differs, so raw bounds should be used when that behaviour is undesirable.

Relevance to this thesis. This paper is the direct methodological source for the TOST equivalence tests used on the hypotheses that predict no difference in trust, namely the photorealistic versus stylised comparison and the real speaker versus stylised comparison. It establishes why a non-significant pairwise result cannot on its own confirm those hypotheses, and it supplies the dependent-means formulation appropriate to the within-subjects design as well as the requirement that the $d = 0.3$ bound be prespecified and reported alongside the equivalence claim. Its caution about power in small samples is also directly applicable, since with $n = 41$ the bound must be defensible as the smallest effect the study can reliably detect rather than chosen after the fact.